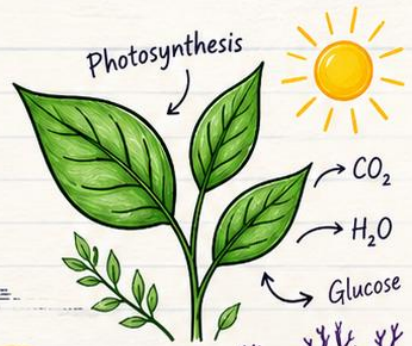
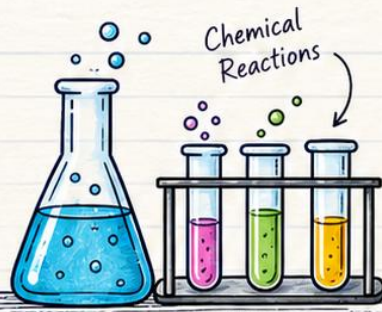
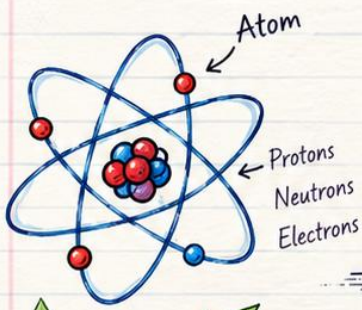
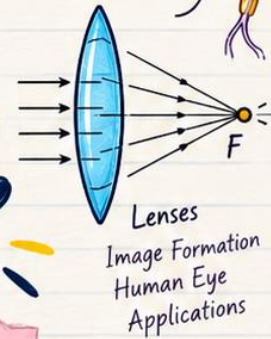
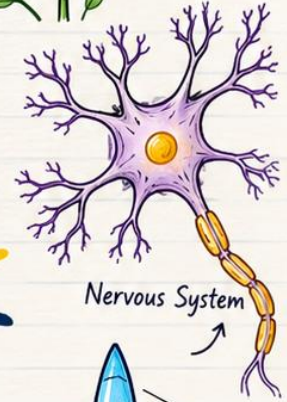
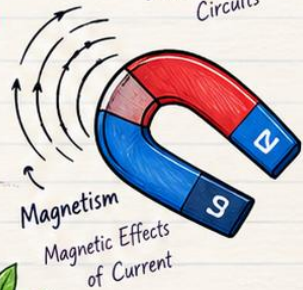
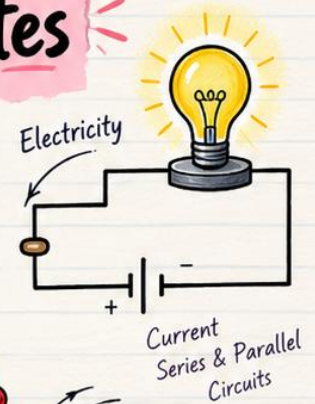
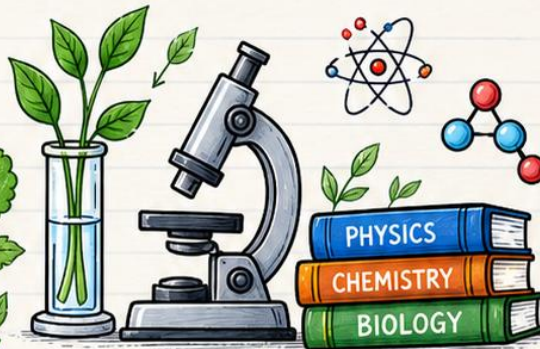
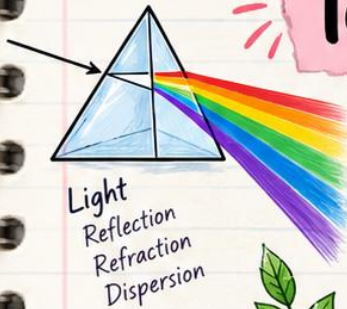


CBSE Class 10 Science



Topper Revision Notes

Chapter 1 to Chapter 13



sjmaths.com

CBSE CLASS 10 SCIENCE

Topper Revision Textbook

Handwritten-style notes • diagrams • board-focused revision

Chapters 1–13 | 2026–27 Edition

About this edition

- Designed for fast revision with clear concepts, exam keywords and labelled diagrams.
- Built around the Class 10 Science chapter structure used throughout this book.
- Best used together with NCERT reading, school notes, PYQs and regular question practice.

Publication note

- Educational revision resource prepared for students and teachers.
- Syllabus, sample-paper patterns and assessment guidance may change; always check the latest official CBSE/NCERT updates.
- CBSE and NCERT names identify the curriculum context only; this is an independent revision resource.

For PYQs, notes and practice: sjmaths.com

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Preface

This book is designed as a compact, student-friendly Science revision textbook for CBSE Class 10. It combines chapter-wise concept notes, exam-focused points, neat handwritten diagrams and quick explanations in one place. The aim is simple: help students revise faster without losing conceptual clarity.

The chapter pages are intentionally notebook-like rather than textbook-heavy. Important definitions, laws, reactions, examples, formulae and diagrams are placed where students need them during revision.

Use this book with NCERT, class notes and question practice. Read actively, redraw important diagrams, solve numericals on paper and revisit weak areas repeatedly.

A simple 3-pass revision method

- Pass 1 — Understand: read the concept and follow the diagram or example.
- Pass 2 — Recall: close the page and reproduce the key points, equation or labelled figure.
- Pass 3 — Apply: solve NCERT/PYQ/competency questions and correct your mistakes.

Use the visual cues

- Highlighted words = examiner keywords.
- Tables = fast comparison / recall.
- Handwritten diagrams = figures to practise.
- Caution boxes = common mark-losing errors.

Before moving on

- ☐ Can I explain the concept aloud?
- ☐ Can I write the formula / reaction correctly?
- ☐ Can I draw the key diagram without copying?
- ☐ Can I solve one question without help?

How to Use This Book

Step	Action	What to do
1	Preview	Read the chapter title, scope and headings before detailed revision.
2	Learn	Revise definitions, concepts, equations, laws and important reasoning.
3	Draw	Practise every board-drawable diagram without looking at the page.
4	Test	Answer short questions and numericals without checking the solution first.
5	Repair	Mark mistakes and return to the exact concept that caused them.
6	Repeat	Use the final revision grids for quick recall before tests and boards.

Answer-writing guide by marks

Marks	Best structure	Typical length
1 mark	Direct definition / fact / formula	1 precise point
2 marks	Two distinct points or formula + result	2 clear points
3–5 marks	Stepwise explanation / labelled diagram / calculation	Pointwise, complete and relevant

Before the board exam

- Reactions, formulae, laws and diagrams are easy marks when remembered accurately.
- For numericals: formula → substitution with units → calculation → final answer with unit.
- For diagrams: keep labels horizontal, use clean leader lines and avoid unnecessary artistic detail.
- Write scientific keywords and distinct points instead of long vague paragraphs.

Chapter-wise Index

Use this page as your revision checklist

Ch.	Chapter	Area	Page	✓
1	Chemical Reactions and Equations	Chemistry	6	☐
2	Acids, Bases and Salts	Chemistry	14	☐
3	Metals and Non-metals	Chemistry	25	☐
4	Carbon and its Compounds	Chemistry	33	☐
5	Life Processes	Biology	40	☐
6	Control and Coordination	Biology	49	☐
7	How do Organisms Reproduce?	Biology	62	☐
8	Heredity	Biology	70	☐
9	Light - Reflection and Refraction	Physics	77	☐
10	The Human Eye and the Colourful World	Physics	84	☐
11	Electricity	Physics	94	☐
12	Magnetic Effects of Electric Current	Physics	101	☐
13	Our Environment	Environment	107	☐

Tip: tick a chapter only after you can explain its key concepts without looking at the notes.

CHAPTER 1 — CHEMICAL REACTIONS & EQUATIONS

Topper-style revision • board-answer language • 2026–27 scope audited

BOARD SCOPE — what you must be able to do

- ★ Recognise a chemical change from evidence and distinguish it from a physical change.
- ★ Write word, skeletal and balanced chemical equations; use correct chemical formulae, states and conditions.
- ★ Identify combination, decomposition, displacement, double displacement and precipitation reactions.
- ★ Explain exothermic/endothermic reactions and oxidation/reduction (redox) using correct reasoning.
- ★ Answer observation-based and error-correction questions, not just memorise definitions.

1. Physical Change vs Chemical Change — DO NOT SKIP

[1M] What is a physical change?

→ A change in which no new substance is formed; only physical properties such as shape, size, state or arrangement may change.

Underline: no new substance

[1M] What is a chemical change?

→ A change in which one or more new substances with different properties are formed.

Underline: new substance(s) • different properties

PHYSICAL	CHEMICAL
Physical change	Chemical change
No new substance is formed.	New substance(s) are formed.
Usually only state/shape/size/arrangement changes.	Chemical composition changes.
Example: melting ice.	Example: burning Mg to form MgO.

2026–27 Sample-paper thinking

- ★ A colour change alone does NOT prove a chemical reaction. Ask: “Was a new substance formed?”
- ★ CBSE can give an unfamiliar situation and ask you to infer physical vs chemical change.

2. Chemical Reaction — meaning & evidence

[1M] Define a chemical reaction.

→ A process in which reactant(s) undergo chemical change to form new product(s) having properties different from the reactants.

Underline: reactants • chemical change • products • different properties

Common indicators / observations of a reaction

- Change in state
- Change in colour
- Evolution of a gas
- Change in temperature / heat change
- Formation of an insoluble precipitate

Exam rule: OBSERVATION $\neq$ INFERENCE

- ★ Observation: what you see — colour change, gas, precipitate, temperature change.
- ★ Inference: what it means — new substance formed, displacement occurred, etc.

3. Chemical Equations — word $\rightarrow$ skeletal $\rightarrow$ balanced

[1M] What is a chemical equation?

$\rightarrow$ A chemical equation is the symbolic representation of a chemical reaction using the chemical formulae of reactants and products.

$\rightarrow$ **Magnesium + Oxygen $\rightarrow$ Magnesium oxide** *word equation*

$\rightarrow$ **Mg + O₂ $\rightarrow$ MgO** *skeletal / unbalanced*

$\rightarrow$ **2Mg + O₂ $\rightarrow$ 2MgO** *balanced*

[1M] What is a balanced chemical equation?

$\rightarrow$ An equation in which the number of atoms of each element is equal on the reactant and product sides.

[2M] Why must an equation be balanced?

$\rightarrow$ The Law of Conservation of Mass states that mass can neither be created nor destroyed in a chemical reaction.

- Hence atoms of each element must be equal on both sides of the equation.

Underline: Law of Conservation of Mass • equal atoms

Golden rules for balancing

- ✓ Never change a chemical formula or subscript while balancing.
- ✓ Change only the numerical coefficients placed before formulae.
- ✓ Use the smallest whole-number coefficients.
- ✓ Recount every element after balancing.

[3M] Balance: **Fe + H₂O $\rightarrow$ Fe₃O₄ + H₂**

$\rightarrow$ Balanced equation: **3Fe + 4H₂O $\rightarrow$ Fe₃O₄ + 4H₂**

- Check: Fe = 3, O = 4, H = 8 on both sides.

Making an equation informative

- (s) = solid, (l) = liquid, (g) = gas, (aq) = aqueous solution.
- Conditions are written over/near the arrow: Δ (heat), sunlight, electricity, catalyst.
- Use correct formulae first; balancing cannot repair a wrong formula.

$\rightarrow$ **CaCO₃(s) $\xrightarrow{\Delta}$ CaO(s) + CO₂(g)**

$\rightarrow$ **2AgCl(s) $\xrightarrow{\text{sunlight}}$ 2Ag(s) + Cl₂(g)**

What coefficients tell you

- ★ Coefficients give the relative number of particles / moles participating in the reaction.
- ★ Example: $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$ means 2 units of Mg react with 1 unit of O_2 to form 2 units of MgO.
- ★ They do NOT directly give a mass ratio or density ratio.

Equation-correction trap — current CBSE style

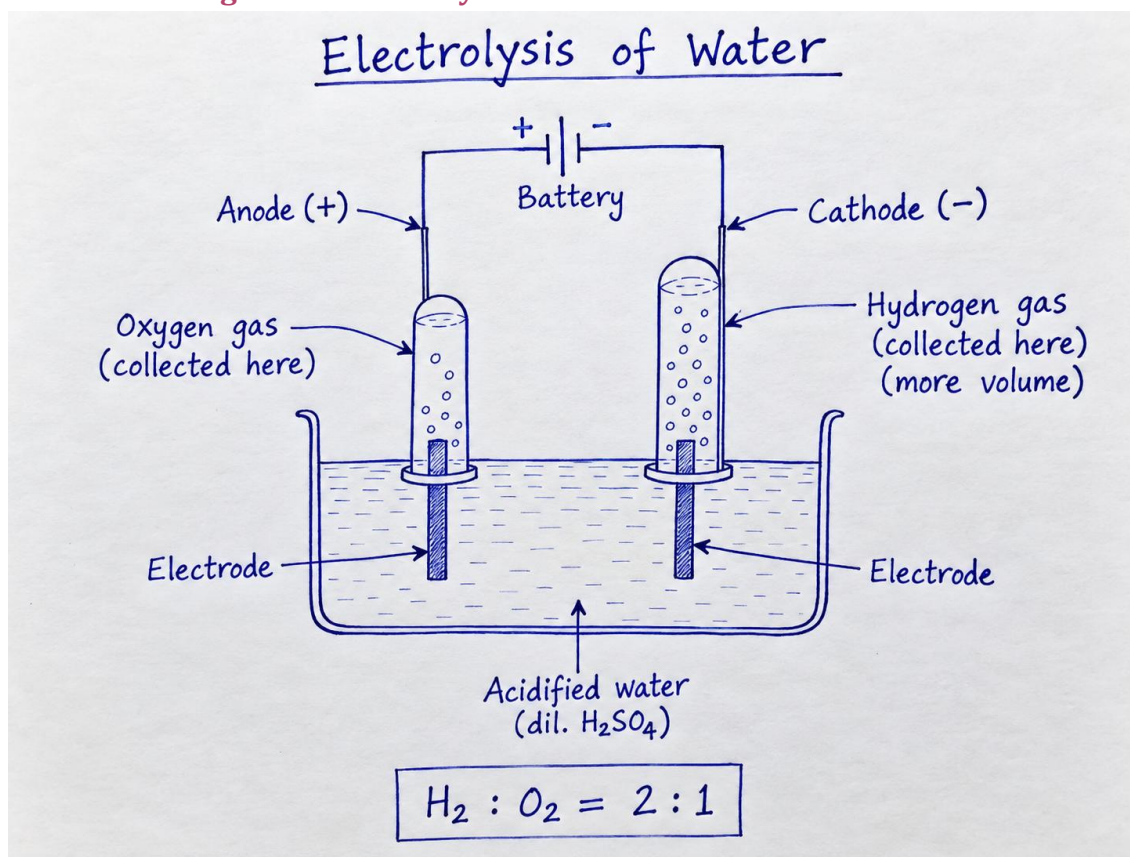
[2M] Correct the formula/equation: $\text{MgCl}_2 + \text{AgNO}_3 \rightarrow \text{AgCl}_2 + \text{MgNO}_3$

- Correct product formulae are AgCl and $\text{Mg}(\text{NO}_3)_2$.
- Balanced equation: $\text{MgCl}_2 + 2\text{AgNO}_3 \rightarrow 2\text{AgCl} + \text{Mg}(\text{NO}_3)_2$.

Underline: correct formula first • then balance

4. Types of Chemical Reactions — pattern first

Handwritten Board Diagram — Electrolysis of Water



Useful for decomposition reaction and activity-based questions: remember that hydrogen is collected in double volume as compared to oxygen.

Master map

- ★ Combination: two or more reactants → one product.
- ★ Decomposition: one compound → two or more simpler products.
- ★ Displacement: a more reactive element replaces a less reactive one.
- ★ Double displacement: two compounds exchange ions.
- ★ Precipitation: an insoluble solid forms from solutions.
- ★ Energy classification: exothermic / endothermic; redox classification: oxidation + reduction.

4.1 Combination Reaction

[2M] Define combination reaction with an example.

→ A reaction in which two or more reactants combine to form a single product is called a combination reaction.

- Example: $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{heat}$.

→ $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$ *dazzling white flame; white MgO powder*

Topper cue

- ★ Pattern = MANY → ONE. Quicklime + water is combination AND exothermic.

4.2 Decomposition Reaction

[2M] Define decomposition reaction.

→ A reaction in which one compound breaks down into two or more simpler substances is called a decomposition reaction.

- It is the opposite pattern of a combination reaction.

A. Thermal decomposition — by heat

→ $\text{CaCO}_3 \xrightarrow{\Delta} \text{CaO} + \text{CO}_2$

→ $2\text{FeSO}_4 \xrightarrow{\Delta} \text{Fe}_2\text{O}_3 + \text{SO}_2 + \text{SO}_3$ *green crystals → brown solid; pungent gases*

→ $2\text{Pb(NO}_3)_2 \xrightarrow{\Delta} 2\text{PbO} + 4\text{NO}_2 + \text{O}_2$ *brown NO₂ fumes; yellow PbO residue*

B. Electrolytic decomposition — by electricity

→ $2\text{H}_2\text{O} \xrightarrow{\text{electricity}} 2\text{H}_2 + \text{O}_2$ *H₂ volume : O₂ volume = 2 : 1*

C. Photolytic decomposition — by light

→ $2\text{AgCl} \xrightarrow{\text{sunlight}} 2\text{Ag} + \text{Cl}_2$ *white AgCl turns grey*

→ $2\text{AgBr} \xrightarrow{\text{sunlight}} 2\text{Ag} + \text{Br}_2$ *light-sensitive silver salt*

Concept trap

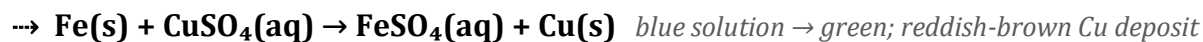
- ★ A decomposition product need not always be two elements; it may be elements and/or compounds.
- ★ The same reaction may have more than one label: thermal decomposition describes the pattern + condition.

5. Displacement, Double Displacement & Precipitation

5.1 Displacement Reaction

[2M] Define displacement reaction.

→ A reaction in which a more reactive element displaces a less reactive element from its compound is called a displacement reaction.



Reactivity logic — board favourite

- ★ A displacement occurs only if the free element is MORE reactive than the element in the compound.
- ★ $\text{Fe} + \text{MgSO}_4 \rightarrow$ no reaction, because Fe is less reactive than Mg.
- ★ $\text{Cu} + \text{ZnSO}_4 \rightarrow$ no reaction, because Cu is less reactive than Zn.

[2M] Iron filings are added to MgSO_4 solution. Will the solution turn green?

→ No. Iron cannot displace magnesium from MgSO_4 because magnesium is more reactive than iron.

- Hence no FeSO_4 is formed and the stated colour change is incorrect.

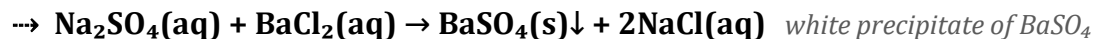
Underline: compare reactivity before predicting reaction

5.2 Double Displacement Reaction

[2M] Define double displacement reaction.

→ A reaction in which two compounds exchange ions to form two new compounds is called a double displacement reaction.

Underline: exchange of ions



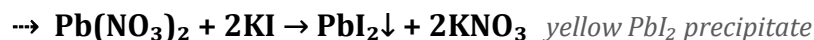
5.3 Precipitation Reaction

[1M] What is a precipitate?

→ An insoluble solid formed during a reaction in solution is called a precipitate.

[2M] Why is the above reaction also called a precipitation reaction?

→ Because insoluble barium sulphate (BaSO_4) is formed as a white solid precipitate from two aqueous solutions.



Fast distinction

- ★ Displacement: ONE element replaces another — $\text{A} + \text{BC} \rightarrow \text{AC} + \text{B}$.
- ★ Double displacement: TWO compounds exchange ions — $\text{AB} + \text{CD} \rightarrow \text{AD} + \text{CB}$.
- ★ Precipitation is a common outcome of double displacement when an insoluble product forms.

6. Energy Changes — Exothermic & Endothermic

[1M] Define an exothermic reaction.

→ A reaction in which heat/energy is released to the surroundings is called an exothermic reaction.

[1M] Define an endothermic reaction.

→ A reaction in which heat/energy is absorbed from the surroundings is called an endothermic reaction.

- $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{heat}$ *exothermic*
- $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$ *combustion; exothermic*
- $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{energy}$ *respiration; exothermic*

[2M] Why is respiration considered exothermic?

→ Glucose is oxidised in cells in the presence of oxygen and releases energy required for life processes.

Do not confuse categories

- ★ “Decomposition” tells how reactants/products are arranged.
- ★ “Endothermic/exothermic” tells the energy change.
- ★ A reaction can therefore belong to more than one category.

7. Oxidation, Reduction & Redox — high priority

[1M] What is oxidation?

→ Addition of oxygen OR removal of hydrogen from a substance is oxidation.

[1M] What is reduction?

→ Removal of oxygen OR addition of hydrogen to a substance is reduction.

- $2\text{Cu} + \text{O}_2 \rightarrow 2\text{CuO}$ *Cu gains O → oxidised*
- $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$ *CuO loses O → reduced; H₂ gains O → oxidised*

[3M] Explain a redox reaction with an example.

- Oxidation and reduction occur simultaneously in a redox reaction.
- In $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$, CuO loses oxygen and is reduced, while H₂ gains oxygen and is oxidised.

Underline: simultaneously • identify both substances

CBSE reasoning examples

- $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$ *N gains H → reduction; H is oxidised → redox*

Rusting is also redox

- ★ Iron is oxidised while oxygen is reduced during rust formation.
- ★ If asked “oxidation OR redox?”: rusting contains oxidation, but the complete process is redox because reduction occurs simultaneously.

8. Oxidation in Everyday Life — board-linked applications

8.1 Corrosion

[1M] Define corrosion.

→ The gradual deterioration of a metal due to chemical reaction with substances in its environment is called corrosion.

- Iron: reddish-brown rust forms in the presence of oxygen and moisture.
- Silver: black coating of silver sulphide may form.
- Copper: green coating develops on prolonged exposure to moist air.

[2M] Why does an iron bridge near the sea rust faster?

- Air near the sea contains more moisture (humidity), which speeds up rusting.
- A dry region has less moisture, so rusting is slower.

Note

- ★ Detailed prevention of corrosion belongs mainly with “Metals and Non-metals”, but the oxidation/redox idea can be integrated in board questions.

8.2 Rancidity

[1M] Define rancidity.

→ Oxidation of fats and oils in food causing unpleasant smell and taste is called rancidity.

[2M] How can rancidity be prevented?

- Reduce contact with oxygen by airtight packing or nitrogen flushing.
- Use antioxidants and keep food at lower temperature/refrigerated where appropriate.

Why nitrogen in chips packets?

- ★ Nitrogen is relatively unreactive and reduces contact of fats/oils with oxygen, slowing oxidation and rancidity.
- ★ The official 2026–27 sample paper tests this idea through a food-label/antioxidant question.

9. NCERT / Board Observations — memorise exactly

ACTIVITY / REACTION	OBSERVATION TO WRITE
Mg ribbon burns in air	Dazzling white flame; white MgO powder.
Fe nail + blue CuSO_4	Blue → green solution; reddish-brown Cu deposit.
$\text{Pb}(\text{NO}_3)_2 + \text{KI}$	Yellow PbI_2 precipitate.
$\text{Na}_2\text{SO}_4 + \text{BaCl}_2$	White BaSO_4 precipitate.
Heat FeSO_4 crystals	Green → brown solid; SO_2/SO_3 gases.
Heat $\text{Pb}(\text{NO}_3)_2$	Brown NO_2 fumes; yellow PbO residue.
AgCl in sunlight	White → grey due to metallic Ag.
$\text{CaO} + \text{H}_2\text{O}$	Mixture becomes hot; slaked lime forms.

10. Must-Write Equation Bank — learn with type + clue

EQUATION	TYPE / OBSERVATION
$2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$	Combination • dazzling white flame
$\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca}(\text{OH})_2 + \text{heat}$	Combination + exothermic
$\text{CaCO}_3 \xrightarrow{\Delta} \text{CaO} + \text{CO}_2$	Thermal decomposition
$2\text{FeSO}_4 \xrightarrow{\Delta} \text{Fe}_2\text{O}_3 + \text{SO}_2 + \text{SO}_3$	Thermal decomposition • green→brown
$2\text{Pb}(\text{NO}_3)_2 \xrightarrow{\Delta} 2\text{PbO} + 4\text{NO}_2 + \text{O}_2$	Thermal decomposition • brown NO_2
$2\text{AgCl} \xrightarrow{\text{sunlight}} 2\text{Ag} + \text{Cl}_2$	Photolytic decomposition • white→grey
$2\text{H}_2\text{O} \xrightarrow{\text{electricity}} 2\text{H}_2 + \text{O}_2$	Electrolytic decomposition • 2:1 gases

$\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$	Displacement • blue→green
$\text{Na}_2\text{SO}_4 + \text{BaCl}_2 \rightarrow \text{BaSO}_4\downarrow + 2\text{NaCl}$	Double displacement + precipitation
$\text{Pb}(\text{NO}_3)_2 + 2\text{KI} \rightarrow \text{PbI}_2\downarrow + 2\text{KNO}_3$	Yellow precipitate
$\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$	Redox
$\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$	Redox reasoning example

11. Mark-wise Answer Bank — write to the marks

[1M] Why is magnesium ribbon cleaned before burning?

→ To remove the thin magnesium oxide layer from its surface so that magnesium can burn readily.

[2M] Differentiate combination and decomposition reactions.

→ Combination: two or more reactants form one product; e.g. $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca}(\text{OH})_2$.

• Decomposition: one compound breaks into two or more simpler substances; e.g. $\text{CaCO}_3 \xrightarrow{\Delta} \text{CaO} + \text{CO}_2$.

[2M] A white salt turns grey in sunlight. Identify the reaction.

→ The salt may be AgCl: $2\text{AgCl} \xrightarrow{\text{sunlight}} 2\text{Ag} + \text{Cl}_2$.

• It is photolytic decomposition; grey colour is due to metallic silver.

[3M] Explain the three types of decomposition reactions.

→ Thermal: caused by heat — $\text{CaCO}_3 \xrightarrow{\Delta} \text{CaO} + \text{CO}_2$.

• Electrolytic: caused by electricity — $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$.

• Photolytic: caused by light — $2\text{AgCl} \rightarrow 2\text{Ag} + \text{Cl}_2$.

12. Competency-Based Questions — topper method

When a question gives an experiment / unfamiliar situation, write in this order

- ★ 1) Observation — what changed?
- ★ 2) Scientific reason / inference — why did it happen?
- ★ 3) Balanced equation — with correct formulae and states/conditions where relevant.
- ★ 4) Reaction type — only after checking the reaction pattern.

[3M] Case: iron nail is placed in blue CuSO_4 solution. Explain.

→ Observation: blue solution becomes green and reddish-brown copper deposits on the nail.

• Reason: iron is more reactive than copper and displaces it from CuSO_4 .

• Equation: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$; displacement reaction.

[3M] Case: aluminium is added to dilute H_2SO_4 and brisk effervescence occurs. Explain.

→ The bubbles are due to hydrogen gas evolved when a reactive metal reacts with dilute acid.

• Balanced equation: $2\text{Al} + 3\text{H}_2\text{SO}_4 \rightarrow \text{Al}_2(\text{SO}_4)_3 + 3\text{H}_2\uparrow$.

• The observation is consistent with a displacement reaction.

13. Common Mistakes a Topper Avoids

- ✗ Assuming every colour change is chemical — first check whether a new substance forms.
- ✗ Changing subscripts while balancing — never alter a correct chemical formula.
- ✗ Balancing an equation that contains wrong product formulae — correct formulae first.
- ✗ Calling every decomposition reaction endothermic without considering the energy statement.

- ✗ Predicting displacement without comparing reactivity.
- ✗ Confusing displacement (one element replaces another) with double displacement (ion exchange).
- ✗ Writing only oxidation in a redox answer — identify reduction too.
- ✗ Giving only an equation when the question asks for observation/reason.

14. 5-Minute Final Revision Grid

MEMORY LINES

- ★ Physical change → no new substance | Chemical change → new substance.
- ★ Balance by coefficients only; atoms equal on both sides.
- ★ Combination: many→one | Decomposition: one→many.
- ★ Displacement: more reactive kicks out less reactive.
- ★ Double displacement: exchange of ions | Precipitation: insoluble solid forms.
- ★ Exothermic releases energy | Endothermic absorbs energy.
- ★ Oxidation = +O / -H | Reduction = -O / +H | both together = redox.
- ★ AgCl white→grey | CuSO₄ blue→green with Fe | PbI₂ yellow ppt.
- ★ For 2–3 marks: definition + keyword + balanced equation + observation/reason.

Self-check before you close the chapter

- ☐ 1. Can I distinguish physical and chemical change in an unfamiliar situation?
- ☐ 2. Can I correct a wrong chemical formula before balancing?
- ☐ 3. Can I write states/conditions correctly?
- ☐ 4. Can I give one example each of all reaction types?
- ☐ 5. Can I predict “reaction / no reaction” from reactivity?
- ☐ 6. Can I identify both oxidation and reduction in a redox reaction?
- ☐ 7. Can I write the observations of the key NCERT activities?
- ☐ 8. Can I write the 12 equation-bank reactions from memory?

CHAPTER 2 — ACIDS, BASES & SALTS

Topper-style revision • answer-ready board language • syllabus + NCERT exercise angles audited

BOARD SCOPE — what you must be able to do

- ★ Define acids and bases in terms of H^+ / OH^- ions and use indicators to identify acidic/basic solutions.
- ★ Write balanced equations for reactions of acids and bases with metals, carbonates, metal oxides and each other.
- ★ Use the pH scale to compare solutions and explain pH in digestion, tooth decay, soil and self-defence by animals/plants.
- ★ Explain preparation, properties and uses of sodium hydroxide, bleaching powder, baking soda, washing soda and Plaster of Paris.
- ★ Solve competency questions involving observations, pH order, dilution, gas tests, neutralisation and salt identification.
- ★ Explain conductivity of acid/base solutions, glucose/alcohol and distilled/rainwater; distinguish base from alkali and acidic/basic oxides.

1. Acids & Bases — exact board definitions

[1M] What is an acid?

→ An acid is a substance which furnishes H^+ ions (actually H_3O^+ ions) when dissolved in water.

Underline: H^+ / H_3O^+ ions in aqueous solution

[1M] What is a base?

→ A base is a substance which furnishes OH^- ions when dissolved in water.

Underline: OH^- ions in aqueous solution

Base vs alkali — common 1-mark distinction

[1M] What is an alkali?

→ An alkali is a base that is soluble in water. All alkalis are bases, but all bases are not alkalis.

• Examples of alkalis: NaOH, KOH. Example of a base that is not appreciably soluble in water: CuO.

Natural acids — useful for MCQ / matching

- Vinegar → acetic acid • Curd → lactic acid • Lemon/orange → citric acid
- Tamarind → tartaric acid • Tomato → oxalic acid • Ant/nettle sting → methanoic acid

Topper note: water is essential

- ★ Acids show acidic behaviour only in aqueous solution because H^+ ions are produced in the presence of water.
- ★ Dry HCl gas does not change dry blue litmus red; moist blue litmus turns red.
- ★ Likewise, bases furnish OH^- ions in water.

Common examples

ACIDS

BASES

Hydrochloric acid — HCl

Sodium hydroxide — NaOH

Sulphuric acid — H₂SO₄

Potassium hydroxide — KOH

Nitric acid — HNO₃

Calcium hydroxide — Ca(OH)₂

Acetic acid — CH₃COOH

Magnesium hydroxide — Mg(OH)₂

2. Indicators — colour changes you must memorise

IN ACID

Blue litmus → red

Methyl orange → red

Phenolphthalein → colourless

Turmeric → no change

IN BASE

Red litmus → blue

Methyl orange → yellow

Phenolphthalein → pink

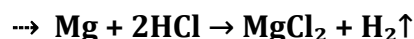
Turmeric → reddish-brown

Olfactory indicators

- ★ Onion, vanilla and clove can be used as olfactory indicators.
- ★ Their smell is retained in acidic medium but is not detected / changes in basic medium in the usual school activity.

3. Chemical Properties of Acids — equations + gas tests

3.1 Acid + metal → salt + hydrogen gas



[2M] How will you test the gas evolved when a metal reacts with dilute acid?

→ Bring a burning splint near the gas.

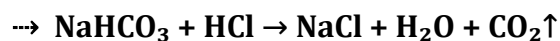
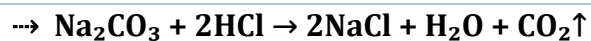
- Hydrogen burns with a characteristic “pop” sound.

Underline: hydrogen • pop sound

Important exception / reasoning

- ★ Acids generally react with metals above hydrogen in the reactivity series to liberate H₂.
- ★ Do not blindly write H₂ for every metal–acid pair; reactivity matters.

3.2 Acid + metal carbonate / hydrogencarbonate → salt + CO₂ + H₂O



[2M] How is carbon dioxide identified?

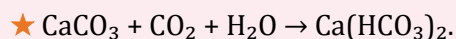
→ Pass the gas through lime water.

- Lime water turns milky because insoluble calcium carbonate is formed: $\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3\downarrow + \text{H}_2\text{O}$.

Underline: lime water turns milky • CaCO₃ precipitate

Excess CO₂ — common competency trap

★ On passing excess CO₂, milkiness may disappear because insoluble CaCO₃ changes into soluble calcium hydrogencarbonate.



3.3 Acid + metal oxide → salt + water



[2M] Why are metal oxides called basic oxides?

→ Metal oxides react with acids to form salt and water, showing basic character.

4. Chemical Properties of Bases

4.1 Base + acid → salt + water (neutralisation)



[1M] What is neutralisation?

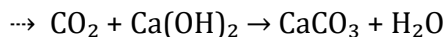
→ The reaction between an acid and a base to form salt and water is called a neutralisation reaction.

Underline: acid + base → salt + water

4.2 Base + certain metals → salt + hydrogen



4.3 Non-metallic oxide + base → salt + water



[2M] Why are non-metallic oxides generally called acidic oxides?

→ They react with bases to form salt and water. Therefore, oxides such as CO₂ show acidic character.

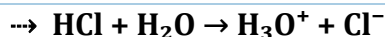
Exam caution

★ Bases do not react with all metals. Amphoteric metals such as zinc and aluminium can react with strong bases.

★ For Class 10, learn the zinc + NaOH example and its H₂ evolution.

5. What actually happens in water?

Ionisation of acids



[2M] Why can H⁺ ions not exist independently in water?

→ H⁺ ions are highly reactive and associate with water molecules to form hydronium ions, H₃O⁺.

Dissociation of bases



Electrical conductivity — board reasoning

[2M] Why do aqueous acids and bases conduct electricity?

→ Their aqueous solutions contain mobile ions that carry electric current.

[2M] Why do glucose and alcohol solutions not conduct electricity like acids?

→ Although glucose and alcohol contain hydrogen, they do not ionise in water to furnish H^+ / H_3O^+ ions; hence their solutions have too few ions to conduct like acid solutions.

[2M] Why does rainwater conduct electricity better than distilled water?

→ Distilled water contains very few ions and is a poor conductor. Rainwater contains dissolved salts and gases that produce ions, so it conducts electricity better.

What really happens in neutralisation?

→ $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$

★ Basic solutions also contain some H^+ ions, but they are basic because OH^- ions are present in greater concentration than H^+ ions.

Key concept

- ★ Acidic nature is due to H_3O^+ / H^+ ions; basic nature is due to OH^- ions.
- ★ Aqueous acidic/basic solutions conduct electricity because mobile ions are present.

6. Dilution — small concept, frequent reasoning question

[2M] What happens when an acid is diluted with water?

→ The concentration of H_3O^+ ions per unit volume decreases, so the solution becomes less acidic and its pH increases.

[2M] What happens when a base is diluted with water?

→ The concentration of OH^- ions per unit volume decreases, so the solution becomes less basic and its pH moves towards 7.

SAFETY RULE — ALWAYS REMEMBER

- ★ While diluting a concentrated acid, add acid slowly to water with constant stirring — never add water to concentrated acid.
- ★ Dilution is highly exothermic; adding water to acid can cause sudden boiling and splashing.

7. Strength vs Concentration — do not confuse

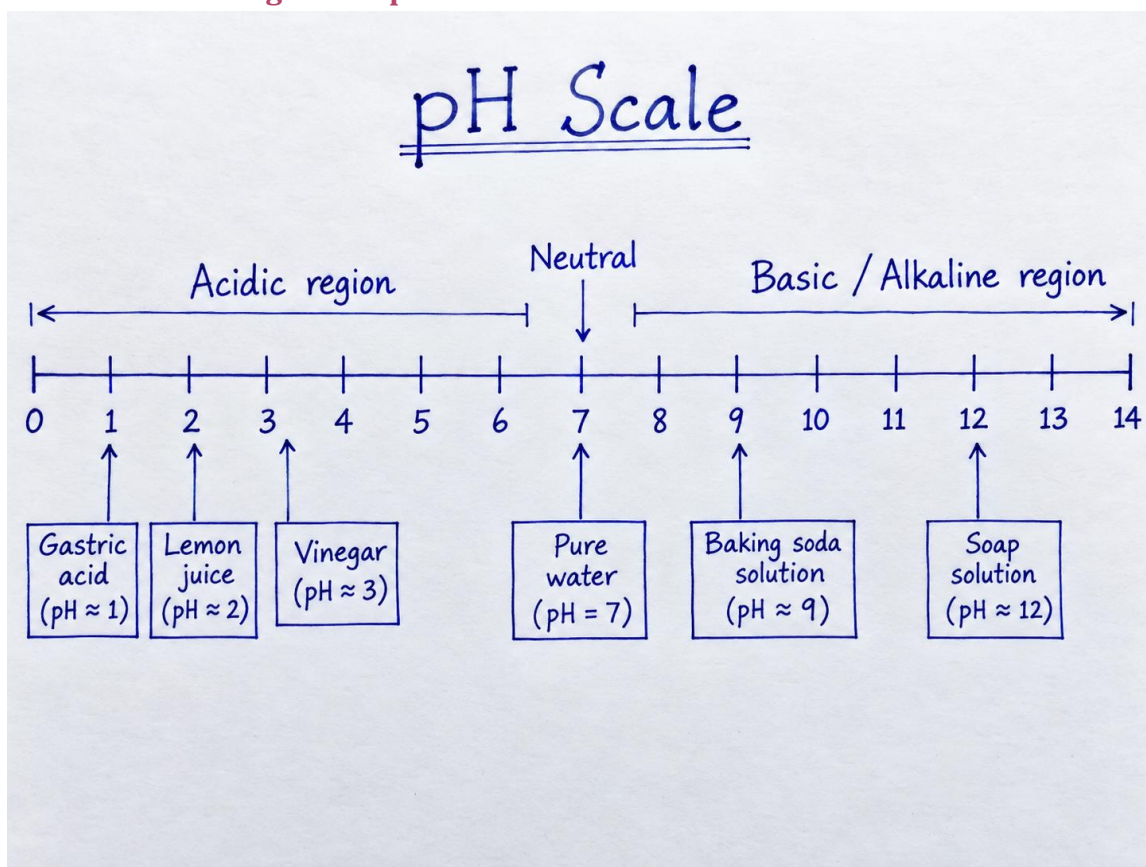
STRENGTH	CONCENTRATION
Strength depends on extent of ionisation in water.	Concentration depends on amount of acid/base present per volume of solution.
Strong acid: ionises almost completely.	A strong acid can still be dilute.
Weak acid: ionises only partially.	A weak acid can be concentrated.

Examples

- ★ Strong acids commonly used at this level: HCl , HNO_3 , H_2SO_4 .
- ★ Weak acid example: CH_3COOH (acetic acid).
- ★ Strong base examples: NaOH , KOH .

8. pH Scale — board favourite

Handwritten Revision Diagram — pH Scale



Memorise the position of 7 as neutral and learn a few everyday examples on both acidic and basic sides.

[1M] What does pH indicate?

- pH indicates how acidic or basic an aqueous solution is by relating to the concentration of H^+ / H_3O^+ ions.
- The logarithmic definition/formula is not required in the Class 10 syllabus.

High-yield pH reasoning

- Adding more base to a basic solution increases OH^- concentration and generally increases its basic character / pH.
- When milk turns into curd, acids are produced; therefore pH decreases.
- Adding a small amount of baking soda to milk makes it slightly basic, so more acid must form before curdling occurs; curdling is delayed.

pH IDEA

$\text{pH} < 7$

MEANING

Acidic

pH = 7	Neutral
pH > 7	Basic / alkaline
Lower pH	Higher H ⁺ concentration / stronger acidic character
Higher pH	Lower H ⁺ concentration / stronger basic character

pH ordering shortcut

- ★ For acidic solutions: smaller pH $\Rightarrow$ more H⁺ ions.
- ★ For basic solutions: larger pH $\Rightarrow$ fewer H⁺ ions / more basic character.
- ★ If pH values are 6, 8 and 9.5, maximum H⁺ is at pH 6. Increasing H⁺ concentration: $9.5 < 8 < 6$.

Universal indicator / pH paper

- Universal indicator gives different colours over a range of pH values and helps estimate pH, unlike a single indicator that mainly tells acid/base.

9. Importance of pH in Everyday Life

9.1 Our digestive system

[2M] Why are antacids used during acidity?

- $\rightarrow$ The stomach contains HCl for digestion. Excess acid causes pain/indigestion.
- A mild base such as magnesium hydroxide neutralises excess acid.

Underline: neutralisation • excess stomach acid

9.2 Tooth decay

[2M] Why does tooth decay start when mouth pH falls below about 5.5?

- $\rightarrow$ Tooth enamel is attacked by acids below about pH 5.5.
- Basic toothpaste helps neutralise acids in the mouth.

9.3 Soil pH

[2M] How is unsuitable soil pH corrected?

- $\rightarrow$ If soil is too acidic, farmers add quicklime (CaO), slaked lime [Ca(OH)₂] or limestone (CaCO₃).
- Treatment depends on the crop and soil condition.

9.4 Self-defence by animals/plants

[2M] Why can baking soda help in a bee sting?

- $\rightarrow$ A bee sting injects acidic material; a mild base such as baking soda can help neutralise it.

[2M] Nettle sting — what is the neutralisation idea?

$\rightarrow$ Nettle hairs can inject methanoic acid. Traditionally, dock leaves are rubbed on the affected area; the treatment is explained through acid–base neutralisation.

Competency link

- ★ CBSE may describe a sting, soil, indigestion or dental situation and ask you to choose an acid/base treatment with reason.
- ★ Write the neutralisation logic, not just the product name.

10. Salts — formation, families & nature

[1M] What is a salt?

→ A salt is an ionic compound formed when the replaceable hydrogen ion(s) of an acid are replaced by a metal or ammonium ion; salts are commonly obtained in neutralisation reactions.

Families of salts

- Salts with the same positive ion belong to one family, e.g. NaCl, Na₂SO₄, NaNO₃.
- Salts with the same negative ion belong to one family, e.g. NaCl, KCl, CaCl₂.

Specific salt examples — practise classification

- Neutral examples: NaCl, KNO₃. • Acidic example: NH₄Cl.
- Basic examples: CH₃COONa, Na₂CO₃, NaHCO₃.
- In NCERT-type activities, salts such as CuSO₄, ZnSO₄ and AlCl₃ may also be compared using pH/indicator observations; use the observed pH rather than assuming every salt is neutral.

pH of salt solutions — reasoning

PARENT ACID/BASE	NATURE OF SALT SOLUTION
Strong acid + strong base	Usually neutral; e.g. NaCl
Strong acid + weak base	Acidic; e.g. NH ₄ Cl
Weak acid + strong base	Basic; e.g. CH ₃ COONa

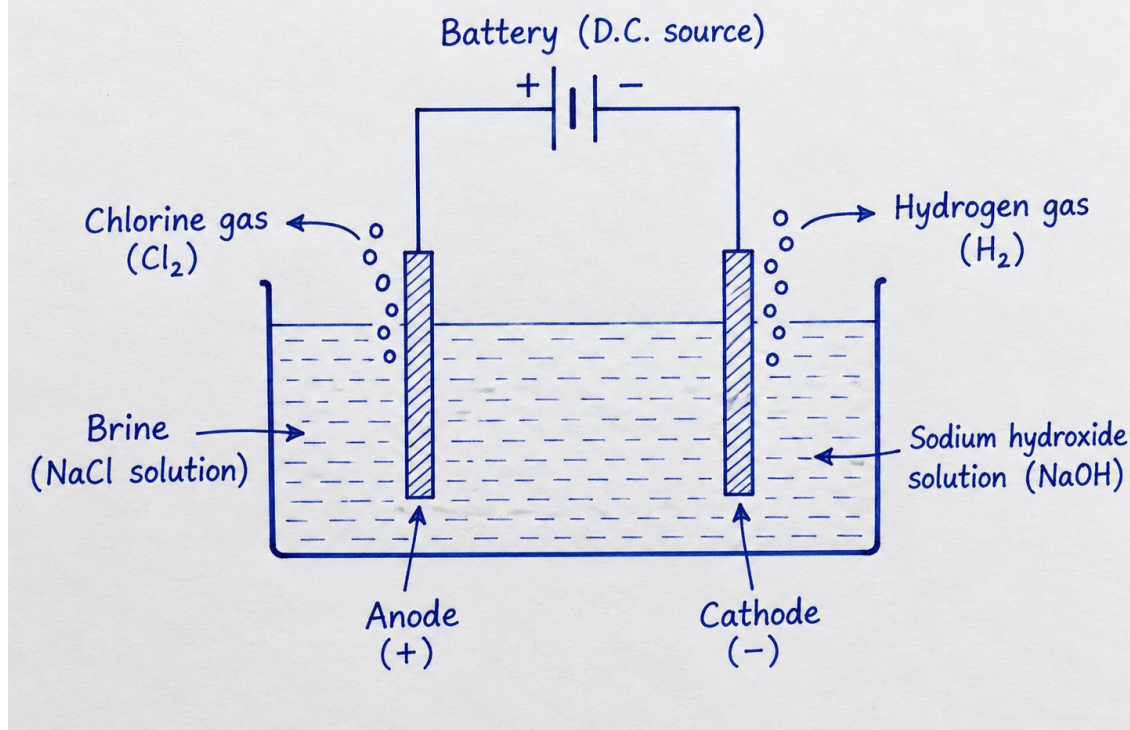
Do not memorise “all salts are neutral”

★ A salt solution may be acidic, basic or neutral depending on the acid and base from which it is derived.

11. Common Salt (NaCl) — raw material for useful chemicals

Handwritten Board Diagram — Chlor-Alkali Process

Chlor-Alkali Process



This is a very useful board diagram: brine gives chlorine at the anode, hydrogen at the cathode and sodium hydroxide remains in solution.

[1M] What is brine?

→ A concentrated aqueous solution of sodium chloride is called brine.

[1M] What is rock salt?

→ Large natural deposits of common salt formed from ancient seas are called rock salt; they are mined from the earth.

11.1 Chlor-alkali process: preparation of NaOH



Why “chlor-alkali”?

- ★ Chlor = chlorine is produced; alkali = sodium hydroxide is produced.
- ★ Products: chlorine at anode, hydrogen at cathode, NaOH remains in solution.

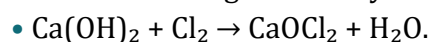
Uses of NaOH

- making soaps and detergents
- paper manufacture
- artificial fibres / degreasing metals

11.2 Bleaching powder — CaOCl₂

[2M] How is bleaching powder prepared?

→ Pass chlorine gas over dry slaked lime.



Uses

- bleaching cotton/linen and wood pulp
- disinfecting drinking water
- as an oxidising agent

12. Baking Soda — NaHCO_3

[2M] How is baking soda obtained?

- In the Solvay process, brine reacts with ammonia and carbon dioxide to form sodium hydrogencarbonate.
- $\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2 + \text{NH}_3 \rightarrow \text{NH}_4\text{Cl} + \text{NaHCO}_3$.

Heating baking soda



Uses

- ingredient of baking powder; CO_2 makes bread/cakes soft and spongy
- mild antacid to neutralise excess stomach acid
- used in soda-acid type fire extinguishers

Baking powder $\neq$ baking soda

- ★ Baking soda is NaHCO_3 .
- ★ Baking powder is a mixture containing baking soda and a mild edible acid (such as tartaric acid).
- ★ Tartaric acid neutralises bitter Na_2CO_3 that may form on heating.

13. Washing Soda — $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$

[2M] How is washing soda prepared?

- Heat baking soda to get sodium carbonate.
- Recrystallisation of sodium carbonate with water gives washing soda, $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$.
- $$\rightarrow 2\text{NaHCO}_3 \xrightarrow{\Delta} \text{Na}_2\text{CO}_3 + \text{CO}_2 + \text{H}_2\text{O}$$
- $$\rightarrow \text{Na}_2\text{CO}_3 + 10\text{H}_2\text{O} \rightarrow \text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$$

Uses

- cleaning agent for domestic purposes
- removing permanent hardness of water
- manufacture of glass, soap and paper

14. Water of Crystallisation

[1M] What is water of crystallisation?

- The fixed number of water molecules chemically associated with one formula unit of a crystalline salt is called water of crystallisation.

Examples

- ★ $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ — blue copper sulphate crystals.
- ★ $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ — washing soda.
- ★ $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ — gypsum.

15. Plaster of Paris (POP) & Gypsum

[1M] Write the formula of gypsum.

→ $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$.

[1M] Write the formula of Plaster of Paris.

→ $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$ (calcium sulphate hemihydrate).

Preparation of POP

→ $\text{CaSO}_4 \cdot 2\text{H}_2\text{O} \xrightarrow{373\text{ K}} \text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O} + 1\frac{1}{2}\text{H}_2\text{O}$ *gypsum* → *POP*

Setting of POP

→ $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O} + 1\frac{1}{2}\text{H}_2\text{O} \rightarrow \text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ *POP + water* → *gypsum*

[2M] Why should POP be stored in moisture-proof containers?

→ POP reacts with moisture to form hard gypsum, so it loses its setting property if exposed to damp air.

Uses of POP

- making casts for fractured bones
- making toys, decorative materials and statues
- making smooth surfaces / moulds

16. Must-Memorise Equations — rapid board bank

EQUATION	EXAM TAG
$\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\uparrow$	acid + metal; H_2 pop test
$\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2\uparrow$	carbonate + acid
$\text{Ca}(\text{OH})_2 + \text{CO}_2 \rightarrow \text{CaCO}_3\downarrow + \text{H}_2\text{O}$	lime water turns milky
$\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$	neutralisation
$\text{CuO} + 2\text{HCl} \rightarrow \text{CuCl}_2 + \text{H}_2\text{O}$	basic oxide + acid
$2\text{NaCl} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{Cl}_2 + \text{H}_2$	chlor-alkali
$\text{Ca}(\text{OH})_2 + \text{Cl}_2 \rightarrow \text{CaOCl}_2 + \text{H}_2\text{O}$	bleaching powder
$2\text{NaHCO}_3 \rightarrow \text{Na}_2\text{CO}_3 + \text{CO}_2 + \text{H}_2\text{O}$	heating baking soda
$\text{Na}_2\text{CO}_3 + 10\text{H}_2\text{O} \rightarrow \text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$	washing soda
$\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O} + 1\frac{1}{2}\text{H}_2\text{O} \rightarrow \text{CaSO}_4 \cdot 2\text{H}_2\text{O}$	POP sets to gypsum
$\text{CO}_2 + \text{Ca}(\text{OH})_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$	non-metal oxide + base; acidic oxide concept
$\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$	net ionic equation of neutralisation

17. Mark-wise Answer Bank — write like this

[1M] Why does dry HCl gas not change dry blue litmus?

→ HCl produces H^+ / H_3O^+ ions only in the presence of water; hence dry HCl does not show acidic behaviour on dry litmus.

[2M] Why should curd and sour substances not be stored in brass/copper vessels?

→ Acids present in sour food may react with the metal to form salts.

- These metal salts can contaminate the food and may be harmful.

[2M] A solution has pH 3 and another has pH 6. Which has more H^+ ions?

→ The pH 3 solution has a higher H^+ concentration because lower pH means greater acidity.

[2M] Equal concentrations of HCl and CH_3COOH react with magnesium. Which gives more vigorous effervescence and why?

→ HCl gives more vigorous effervescence because it is a stronger acid and furnishes a greater concentration of H^+ ions than acetic acid of the same concentration. More H^+ ions react with Mg to produce H_2 faster.

[2M] Why does milk become more acidic when it changes into curd?

→ During curd formation, acidic substances such as lactic acid are produced. H^+ concentration increases, so the pH decreases.

- The pH scale is logarithmic, so a difference of 3 pH units represents a large difference in H^+ concentration; exact logarithmic calculation is not required.

[3M] Explain why baking soda is used in baking powder.

→ On heating, NaHCO_3 releases CO_2 gas.

- CO_2 bubbles make dough rise and make cakes/bread soft and spongy.
- A mild edible acid such as tartaric acid is added to neutralise bitter sodium carbonate formed during heating.

[3M] Explain the chlor-alkali process.

→ Electrolysis of brine produces sodium hydroxide, chlorine and hydrogen.

- $2\text{NaCl} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{Cl}_2 + \text{H}_2$.
- Chlorine is obtained at the anode and hydrogen at the cathode; NaOH remains in solution.

18. Competency / Case-based Thinking

If CBSE gives a pH table or graph

- ★ Step 1: locate smallest/largest pH.
- ★ Step 2: translate pH into H^+ concentration — lower pH means more H^+ .
- ★ Step 3: decide acid / neutral / base and explain in one line.

If CBSE gives an unknown white powder

- ★ Use clues: reaction with acid, gas evolved, pH, heating behaviour and stated use.
- ★ NaHCO_3 + acid gives CO_2 ; washing soda contains water of crystallisation; POP sets with water; bleaching powder is used for disinfection/bleaching.

If CBSE asks “why this treatment works?”

- ★ Identify whether the harmful substance is acidic/basic.
- ★ Choose a mild opposite reagent.
- ★ Use the word neutralisation and connect it to pH.

19. Common Mistakes a Topper Avoids

- ✗ Writing “acid contains H^+ ” without saying in aqueous solution / furnishes ions in water.
- ✗ Confusing strength with concentration.
- ✗ Saying lower pH means fewer H^+ ions — it is the opposite.
- ✗ Forgetting that H^+ exists as hydronium, H_3O^+ , in water.
- ✗ Calling every salt neutral.
- ✗ Writing Na_2CO_3 as baking soda — baking soda is $NaHCO_3$; washing soda is $Na_2CO_3 \cdot 10H_2O$.
- ✗ Confusing bleaching powder $CaOCl_2$ with baking powder.
- ✗ Writing POP formula as $CaSO_4 \cdot 2H_2O$ — that is gypsum; POP is $CaSO_4 \cdot \frac{1}{2}H_2O$.
- ✗ Adding water directly to concentrated acid during dilution.
- ✗ Giving only a product name in application questions without neutralisation / pH reasoning.
- ✗ Assuming a substance is an acid merely because its formula contains hydrogen — it must furnish H^+ / H_3O^+ ions in water.
- ✗ Saying all bases are alkalis — only water-soluble bases are alkalis.
- ✗ Forgetting that non-metallic oxides such as CO_2 can react with bases and show acidic character.

20. 5-Minute Final Revision Grid

MEMORY CUE	ONE-LINE RECALL
Acid	furnishes H^+ / H_3O^+ in water
Base	furnishes OH^- in water
Acid + metal	salt + H_2
Acid + carbonate	salt + H_2O + CO_2
Neutralisation	acid + base $\rightarrow$ salt + water
$pH < 7$ / $=7$ / >7	acid / neutral / base
Lower pH	more H^+ ions
NaOH	chlor-alkali from brine
$CaOCl_2$	bleaching powder
$NaHCO_3$	baking soda
$Na_2CO_3 \cdot 10H_2O$	washing soda
$CaSO_4 \cdot \frac{1}{2}H_2O$	Plaster of Paris
Alkali	water-soluble base
Non-metal oxide + base	salt + water; non-metal oxide shows acidic character
Glucose/alcohol	contain H but do not furnish H^+ in water

Rainwater vs distilled
Basic solution

rainwater has more ions → better conductor
contains H^+ too, but $\text{OH}^- > \text{H}^+$

21. Self-check — can you answer without looking?

- ☐ Define acid and base in terms of ions furnished in water.
- ☐ Write all indicator colour changes.
- ☐ Write and explain acid + metal, carbonate and metal oxide reactions.
- ☐ Explain why dry HCl does not affect dry litmus.
- ☐ Explain dilution safety and pH change on dilution.
- ☐ Arrange solutions by H^+ concentration from given pH values.
- ☐ Explain pH in digestion, teeth, soil and stings.
- ☐ Write preparation + 2 uses each of NaOH, bleaching powder, baking soda, washing soda and POP.
- ☐ Define water of crystallisation with two examples.
- ☐ Write the 12 must-memorise equations from memory.
- ☐ Distinguish base and alkali; give one example of each.
- ☐ Explain why glucose/alcohol do not behave as acids and why distilled water is a poor conductor.
- ☐ Explain non-metallic oxide + base with equation and acidic-oxide conclusion.
- ☐ Explain why a basic solution can still contain H^+ ions.
- ☐ Explain milk → curd pH change and why baking soda can delay curdling.
- ☐ Compare equal-concentration HCl and CH_3COOH reactions using acid strength.

TOPPER RULE FOR THIS CHAPTER

- ★ For a board answer, aim for: definition / principle + correct equation + observation or reason + application when relevant.
- ★ In competency questions, do not merely name the substance — explain why it works.

22. Final Board Trigger → Must-Write Drill

QUESTION TRIGGER	MUST-WRITE IDEA
Dry HCl + dry litmus	No $\text{H}^+/\text{H}_3\text{O}^+$ without water; no acidic behaviour.
Glucose contains H but is not acid	Does not ionise in water to furnish $\text{H}^+/\text{H}_3\text{O}^+$.
$\text{CO}_2 + \text{Ca}(\text{OH})_2$	Salt + water; CO_2 is a non-metallic acidic oxide.
Distilled vs rainwater	Rainwater has dissolved ions; distilled water has very few ions.
Milk → curd	Acid formation increases H^+ ; pH falls.
Baking soda added to milk	Makes milk slightly basic; more acid needed before curdling.
HCl vs CH_3COOH with Mg	HCl stronger → more H^+ → faster H_2 evolution.
Basic solution contains H^+ ?	Yes; it is basic because OH^- concentration is greater than H^+ .
NaCl / NH_4Cl / CH_3COONa	Neutral / acidic / basic salt solution.
POP left in moist air	Hydrates to gypsum and loses useful setting property.

★ If a competency question gives an unfamiliar situation, identify the acid/base clue, write the relevant ion/equation, and then explain the observation in one sentence.

CHAPTER 3 — METALS & NON-METALS

Topper-style revision • board-answer language • syllabus + NCERT + competency angles audited
BOARD SCOPE — what you must be able to do

- ★ Compare physical and chemical properties of metals and non-metals, including standard exceptions.
- ★ Use the reactivity series to predict reactions with oxygen, water, acids and salt solutions.
- ★ Explain formation and properties of ionic compounds using electron transfer.
- ★ Explain occurrence of metals and the basic metallurgy route for low, medium and highly reactive metals.
- ★ Explain electrolytic refining, corrosion and methods of corrosion prevention.
- ★ Solve competency questions using observations, graphs/experiments, reactivity logic and correct balanced equations.

1. Metals & Non-metals — physical properties

PROPERTY	METALS	NON-METALS
Appearance	Generally lustrous (shiny).	Generally dull.
State	Mostly solids at room temperature.	May be solid, liquid or gas.
Hardness	Generally hard.	Generally soft/brittle as solids.
Malleability	Can be beaten into sheets.	Usually non-malleable.
Ductility	Can be drawn into wires.	Usually non-ductile.
Conductivity	Good conductors of heat/electricity.	Generally poor conductors.
Sound	Sonorous.	Generally non-sonorous.

Exceptions CBSE loves to test

- ★ Mercury (Hg) is a liquid metal at room temperature.
- ★ Sodium (Na) and potassium (K) are very soft; they can be cut with a knife.
- ★ Graphite, a non-metal form of carbon, conducts electricity.
- ★ Iodine is a lustrous non-metal; bromine is a liquid non-metal.
- ★ Diamond (carbon) is a non-metal but is extremely hard.

⚠ EXAM TRAP Do not answer “all metals are hard/conductive solids” or “all non-metals are dull insulators.” Use “generally” and remember the exceptions.

2. Chemical Properties of Metals — learn pattern + observation

2.1 Reaction with oxygen

[2M] What happens when metals react with oxygen?

- Metals generally react with oxygen to form metal oxides.
 - Most metal oxides are basic in nature.
 - Some metal oxides, especially Al_2O_3 and ZnO , are amphoteric — they react with both acids and bases.
 - $2\text{Cu} + \text{O}_2 \rightarrow 2\text{CuO}$ *copper forms black copper(II) oxide*
 - $4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$ *aluminium oxide forms a protective coating*

Amphoteric oxides — high-yield

- $\text{ZnO} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\text{O}$
- $\text{ZnO} + 2\text{NaOH} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2\text{O}$
- $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$
- $\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2 + \text{H}_2\text{O}$

★ **COMPETENCY LINK** 2026–27 sample-paper angle: aluminium is reactive, yet aluminium wires do not corrode easily because a thin, adherent Al_2O_3 layer protects the metal underneath.

2.2 Reaction with water — order matters

METAL	REACTION WITH WATER	KEY PRODUCT / OBSERVATION
K, Na	Very violent with cold water	Hydroxide + H_2 ; enough heat may ignite H_2 .
Ca	Less violent with cold water	$\text{Ca}(\text{OH})_2$ + H_2 ; floats due to H_2 bubbles.
Mg	Slow with hot water; reacts with steam	$\text{Mg}(\text{OH})_2$ / MgO + H_2 depending on condition.
Al, Zn, Fe	React with steam	Metal oxide + H_2 .
Pb, Cu, Ag, Au	No reaction with water	Too low in reactivity for this reaction.



★ **MEMORY LINE** Sodium and potassium are stored under kerosene because they react vigorously with air/moisture and can catch fire.

2.3 Reaction with dilute acids

[2M] State the general reaction of a metal with dilute acid.

→ A metal above hydrogen in the reactivity series usually displaces hydrogen from dilute acids to form a salt and hydrogen gas.



[1M] How is hydrogen gas identified?

→ Bring a burning splint near the gas; hydrogen burns with a characteristic pop sound.

⚠ **EXAM TRAP** Nitric acid is a strong oxidising agent, so metals generally do NOT evolve H_2 with HNO_3 because the hydrogen formed is oxidised to water. NCERT exception: Mg and Mn can evolve H_2 with very dilute HNO_3 .

2.4 Reaction with salt solutions — displacement

[2M] When can one metal displace another from its salt solution?

→ A more reactive metal displaces a less reactive metal from the solution of its salt.

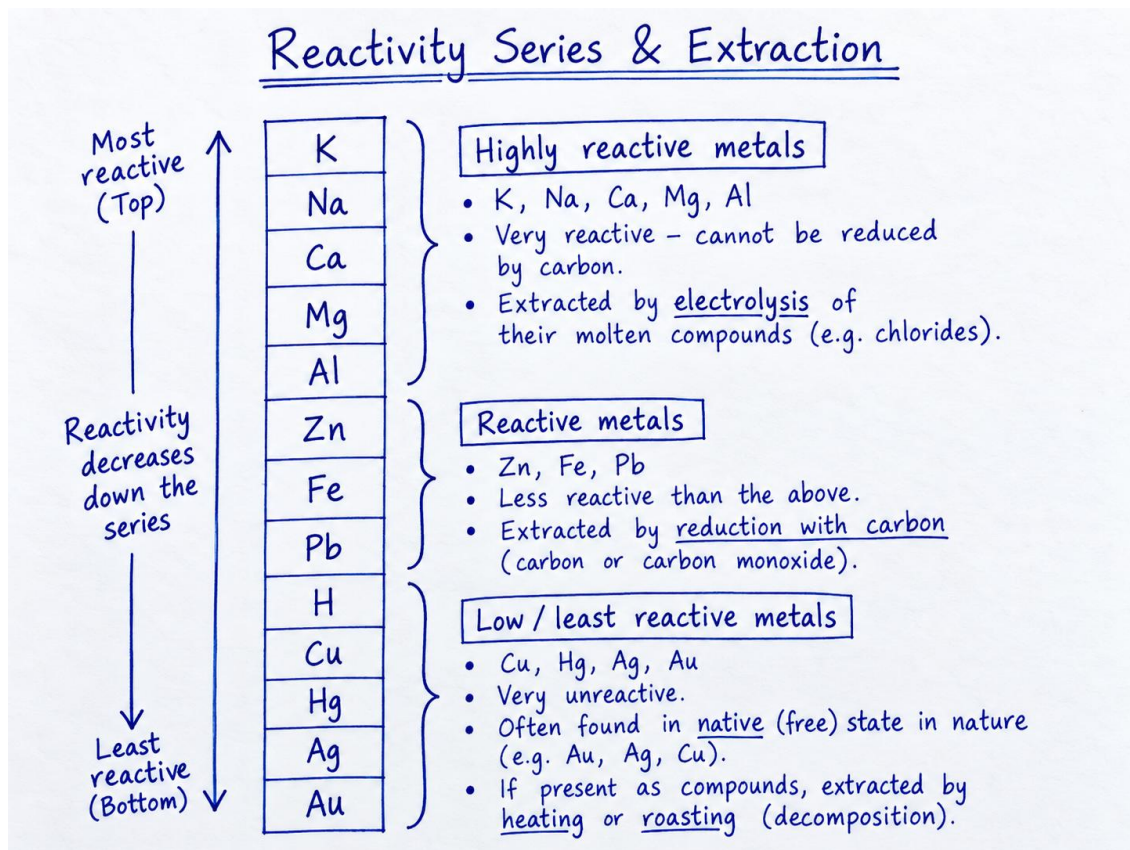


★ **COMPETENCY LINK** Current competency pattern: “ $\text{Fe} + \text{MgSO}_4$ ” gives NO reaction because iron is less reactive than magnesium. Do not invent a colour change if reactivity does not permit displacement.

3. Reactivity Series — the chapter’s decision tool

Handwritten Revision Chart — Reactivity Series & Extraction

Reactivity Series & Extraction



Use this chart to quickly decide whether a metal is extracted by electrolysis, carbon reduction or simply by heating/roasting.

MORE REACTIVE →	SERIES	QUICK USE
Highest	K Na Ca Mg Al	Very reactive; extraction by electrolysis.
Middle	Zn Fe Pb	Moderately reactive; oxides can be chemically reduced.
Reference	H	Metals above H usually liberate H ₂ from dilute acids.
Lower	Cu Hg Ag Au	Less reactive; some occur free / easier extraction.

★ **TOPPER NOTE** Mnemonic idea: learn the order, not only the words. Most prediction questions reduce to: "Which element is higher in the reactivity series?"

Reaction prediction in 10 seconds

- ✓ Metal + salt solution: reaction only if free metal is more reactive than the metal in the salt.
- ✓ Metal + dilute acid: metals above H generally release H₂; metals below H generally do not.
- ✓ Gold/silver are found relatively uncombined because of very low reactivity.

4. How Metals & Non-metals Form Ionic Compounds

[2M] What is an ionic/electrovalent compound?

→ A compound formed by transfer of electron(s) from one atom to another, producing oppositely charged ions held together by strong electrostatic forces.

Example 1: Formation of NaCl

- Na electronic configuration = 2,8,1 → loses 1 e⁻ → Na⁺ (2,8).
- Cl electronic configuration = 2,8,7 → gains 1 e⁻ → Cl⁻ (2,8,8).
- Na⁺ and Cl⁻ attract electrostatically → NaCl.
- Na → Na⁺ + e⁻
- Cl + e⁻ → Cl⁻

Example 2: Formation of MgCl_2

- $\text{Mg} = 2,8,2 \rightarrow \text{loses } 2 \text{ e}^- \rightarrow \text{Mg}^{2+}$.
- Two chlorine atoms each gain one electron $\rightarrow 2\text{Cl}^-$.
- Mg^{2+} combines with $2\text{Cl}^- \rightarrow \text{MgCl}_2$.

5. Properties of Ionic Compounds — explain WHY

PROPERTY	WHAT TO WRITE IN BOARD ANSWER
Physical nature	Usually hard, crystalline solids; brittle because shifting layers can bring like charges close, causing repulsion.
Melting / boiling point	High, because strong electrostatic attraction between oppositely charged ions requires much energy to overcome.
Solubility	Generally soluble in water; generally insoluble in organic solvents such as kerosene/petrol.
Electrical conduction	Do not conduct in solid state because ions are fixed. Conduct when molten or dissolved in water because ions become mobile.

[2M] Why does solid NaCl not conduct electricity but molten NaCl does?

- In solid NaCl the ions are held at fixed positions and cannot move. On melting, the ions become free to move and carry electric current.

⚠ **EXAM TRAP** Never write “ionic compounds contain no charges in solid state.” They contain ions; the ions simply cannot move in the solid.

6. Occurrence of Metals — mineral, ore & gangue

[1M] What is a mineral?

- A naturally occurring substance in the earth's crust containing a metal or its compounds is called a mineral.

[1M] What is an ore?

- A mineral from which a metal can be extracted conveniently and economically is called an ore.

[1M] What is gangue?

- Unwanted earthy impurities such as sand, clay and rock associated with an ore are called gangue.

★ **MEMORY LINE** All ores are minerals, but all minerals are not ores.

Why reactivity controls occurrence

- Highly reactive metals (K, Na, Ca, Mg, Al) are found only in combined form.
- Moderately reactive metals (Zn, Fe, Pb) usually occur as sulphides, carbonates or oxides.
- Less reactive metals (Cu, Hg, Ag, Au) may occur in free/native state or as compounds.
- Gold can occur native because it has very low reactivity — a common competency reasoning point.

7. Metallurgy — the complete board flow

STEP	PURPOSE
1. Enrichment / concentration of ore	Remove gangue and increase percentage of desired metal compound.
2. Convert ore to oxide	Oxides are generally easier to reduce than sulphides/carbonates.
3. Reduction to metal	Method depends on position in reactivity series.
4. Refining	Purify the crude metal; electrolytic refining is important.

7.1 Converting ores to oxides

[2M] Differentiate roasting and calcination.

- Both convert ores into oxides, but the conditions and ore types differ.

PROCESS	USED FOR	CONDITION / EXAMPLE
Roasting	Sulphide ores	Strong heating in excess air. $2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2$
Calcination	Carbonate ores	Strong heating in limited/no air. $\text{ZnCO}_3 \rightarrow \text{ZnO} + \text{CO}_2$

8. Extraction According to Reactivity

8.1 Low-reactivity metals — heating / self-reduction

- Low-reactivity metal compounds can often be reduced by heating.
- Mercury is obtained from cinnabar (HgS) by roasting to HgO, followed by heating HgO.



8.2 Middle-reactivity metals — chemical reduction

- Sulphide/carbonate ores are first converted to oxides by roasting/calcination.
- Metal oxides are then reduced using carbon, carbon monoxide or a more reactive metal.



Thermite reaction — very important application



[2M] Why is aluminium used in the thermite process?

- Aluminium is more reactive than iron, so it removes oxygen from iron(III) oxide. The reaction is highly exothermic and produces molten iron.
- Application: welding railway tracks and repairing heavy iron parts.

8.3 Highly reactive metals — electrolytic reduction

[2M] Why can sodium, calcium and magnesium not be obtained by reducing their oxides with carbon?

- These metals have greater affinity for oxygen and are more reactive than carbon; therefore their molten compounds are reduced by electrolysis.



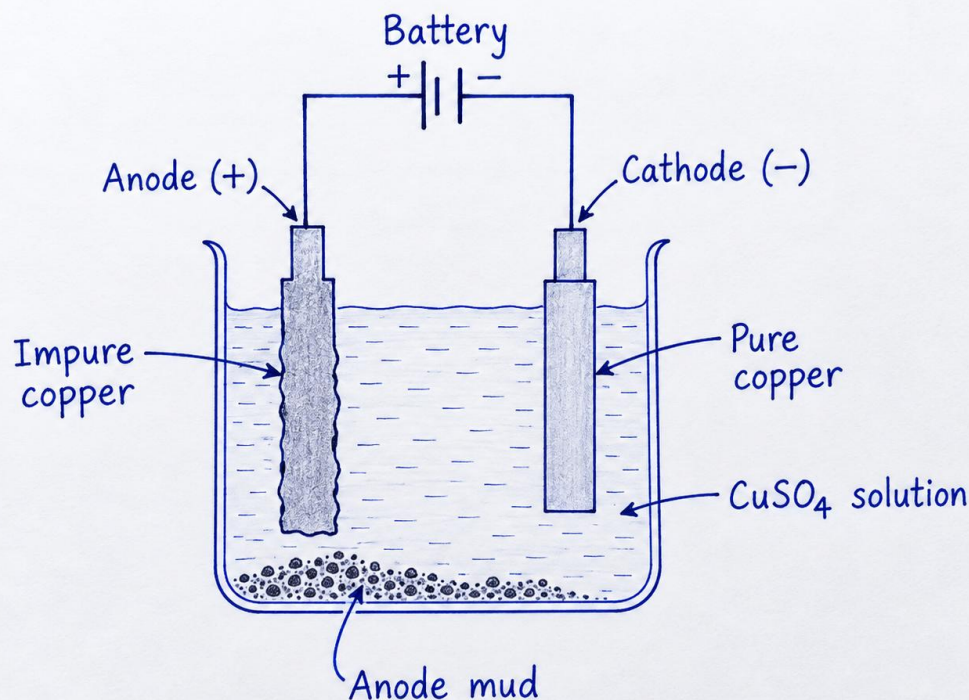
- At cathode: positive metal ions gain electrons → metal is deposited.
- At anode: negative ions lose electrons; e.g. chloride ions give chlorine gas.

★ **COMPETENCY LINK** Choose the extraction method from reactivity. If the question gives an unfamiliar metal, first place it conceptually as high / middle / low reactivity.

9. Refining of Metals — electrolytic refining

Handwritten Board Diagram — Electrolytic Refining of Copper

Electrolytic Refining of Copper



A common labelled diagram: impure copper is the anode, pure copper is the cathode and anode mud settles at the bottom.

[3M] Explain electrolytic refining of copper.

→ Electrolytic refining purifies impure copper using electric current.

- Anode: thick slab of impure copper.
- Cathode: thin sheet of pure copper.
- Electrolyte: acidified copper sulphate solution.
- On passing current, copper from anode dissolves as Cu^{2+} and pure copper deposits on cathode.
- Insoluble impurities settle below the anode as anode mud.

★ **MEMORY LINE** Board diagram labels: impure metal = anode (+), pure metal sheet = cathode (-), solution of metal salt = electrolyte, insoluble impurities = anode mud.

10. Corrosion — definition, examples, prevention

[1M] Define corrosion.

→ The gradual deterioration of a metal due to chemical reaction with substances in the environment such as oxygen, moisture or other chemicals is called corrosion.

Examples to memorise

METAL	CORROSION PRODUCT / OBSERVATION
Iron	Reddish-brown rust (hydrated iron(III) oxide).
Silver	Black coating mainly due to silver sulphide.
Copper	Green coating of basic copper carbonate.
Aluminium	Thin Al_2O_3 layer adheres strongly and protects further attack.

[2M] Why does an iron bridge near the sea rust faster than one in a dry hilly region?

→ Rusting needs oxygen and moisture. Sea air contains more moisture and dissolved salts, which make the corrosion process faster.

Methods of preventing corrosion

- ✓ Painting, oiling or greasing: blocks contact with air and moisture.
- ✓ Galvanisation: coating iron/steel with zinc.
- ✓ Electroplating: coating with a protective metal layer.
- ✓ Alloying: mixing with suitable elements to improve corrosion resistance.
- ✓ Anodising (for aluminium): deliberately thickens the protective oxide layer.

11. Alloys — why metals are mixed

[1M] What is an alloy?

- An alloy is a homogeneous mixture of two or more metals, or of a metal with a non-metal, prepared to obtain improved properties.

Why alloying is useful

- Can increase hardness/strength.
- Can improve resistance to corrosion.
- Can change melting point or electrical resistance.

ALLOY	MAIN COMPONENTS	COMMON USE / POINT
Brass	Cu + Zn	Utensils, fittings, decorative items.
Bronze	Cu + Sn	Statues, medals, bearings.
Solder	Pb + Sn (traditional)	Low melting point; joining wires/metals.
Stainless steel	Fe + Cr + Ni (with C)	Hard and corrosion resistant.
Amalgam	Mercury + another metal	Alloy containing mercury.

★ **TOPPER NOTE** Pure gold is soft; jewellery is commonly made from gold alloyed with other metals to increase hardness.

12. Current CBSE Competency Patterns — think, don't memorise

A. Heat-conduction experiment

- If rods have equal length and thickness and one end is heated, the candle/sensor responding first corresponds to the better heat conductor.
- To compare materials fairly, control variables such as rod length, thickness and heating conditions.
- Increasing rod length generally increases the time for heat to reach the far end.
- If thicknesses differ, a direct comparison of material conductivity becomes unreliable because geometry also affects heat flow.

★ **COMPETENCY LINK** The 2026–27 CBSE sample paper explicitly tests interpreting a heat-conduction graph and judging whether rod length/thickness makes the experiment a fair comparison.

B. Protective oxide reasoning

[1M] Why do aluminium electrical wires not corrode easily despite aluminium being reactive?

- Aluminium rapidly forms a thin, adherent aluminium oxide layer that prevents oxygen/moisture from reaching fresh aluminium.

C. Observation validation

- ✓ Iron + MgSO_4 : no reaction; Fe cannot displace more reactive Mg.
- ✓ Aluminium + dilute H_2SO_4 : hydrogen can evolve because Al is above H in the reactivity series (after the oxide layer is overcome).

13. Must-Memorise Equations — board bank

EQUATION	EXAM TAG
$4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$	Protective oxide; metal + oxygen
$2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2\uparrow$	Very reactive metal + cold water
$\text{Ca} + 2\text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{H}_2\uparrow$	Calcium + water
$3\text{Fe} + 4\text{H}_2\text{O(steam)} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2\uparrow$	Iron + steam
$\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\uparrow$	Metal + dilute acid
$\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$	Displacement + observation
$\text{ZnO} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\text{O}$	Amphoteric oxide + acid
$\text{ZnO} + 2\text{NaOH} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2\text{O}$	Amphoteric oxide + base
$2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2$	Roasting
$\text{ZnCO}_3 \xrightarrow{\Delta} \text{ZnO} + \text{CO}_2$	Calcination
$\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$	Reduction of middle-reactivity metal oxide
$\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow \text{Al}_2\text{O}_3 + 2\text{Fe} + \text{heat}$	Thermite reaction
$2\text{NaCl(l)} \xrightarrow{\text{electrolysis}} 2\text{Na} + \text{Cl}_2$	Extraction of highly reactive metal

14. Mark-wise Answer Bank — write like this

[1M] Why is sodium kept immersed in kerosene?

→ Sodium reacts vigorously with oxygen/moisture and may catch fire; kerosene isolates it from air and water.

[1M] Name one amphoteric oxide.

→ Zinc oxide (ZnO) or aluminium oxide (Al_2O_3).

[2M] Why do ionic compounds have high melting points?

→ Oppositely charged ions are held by strong electrostatic forces. A large amount of heat energy is required to overcome these attractions.

[2M] Why does copper not displace zinc from ZnSO_4 ?

→ Copper is less reactive than zinc and lies below zinc in the reactivity series; therefore the displacement is not feasible.

[3M] Differentiate roasting and calcination.

→ Roasting is strong heating of sulphide ores in excess air, whereas calcination is strong heating of carbonate ores in limited/no air. Both convert ores into oxides before reduction.

- Roasting: $2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2$.
- Calcination: $\text{ZnCO}_3 \rightarrow \text{ZnO} + \text{CO}_2$.

15. Board Answers for 3–5 Marks

[3M] Explain why different metals need different extraction methods.

→ The method of extraction depends on metal reactivity.

- Highly reactive metals cannot be chemically reduced easily → electrolysis of molten compounds.
- Middle-reactivity metals: ores are converted to oxides and reduced with carbon/CO or a more reactive metal.
- Low-reactivity metals: compounds can often be reduced by heating or self-reduction.

[3M] State three properties of ionic compounds with reasons.

→ Ionic compounds show characteristic properties because they consist of strongly attracted positive and negative ions.

- High melting/boiling points — strong electrostatic attractions.
- Conduct electricity only in molten/aqueous state — ions are mobile only then.

- Generally soluble in water — polar water stabilises separated ions.

[3M] What is corrosion? Give two examples and two prevention methods.

→ Corrosion is gradual deterioration of metals by environmental reactions.

- Examples: iron rusts; silver blackens; copper develops a green coating.
- Prevention: painting/oiling/greasing; galvanising; alloying/electroplating.

[5M] Explain extraction of metals from low, medium and high positions in the reactivity series.

→ Classify the metal first, then state the correct process and one equation for each category.

- Low reactivity: heating/roasting followed by thermal or self-reduction; e.g. mercury from HgS/HgO .
- Middle reactivity: sulphide/carbonate → oxide by roasting/calcination; oxide → metal by carbon/CO or aluminium reduction.
- High reactivity: electrolytic reduction of molten salts; e.g. molten NaCl gives Na at cathode and Cl_2 at anode.
- After extraction, crude metal may require refining.

16. Common Mistakes a Topper Avoids

- ✗ Writing every metal + acid reaction as H_2 evolution — check reactivity and remember the HNO_3 exception.
- ✗ Saying aluminium “does not react” because it is unreactive — it is reactive but passivated by Al_2O_3 .
- ✗ Confusing mineral and ore — economic extraction distinguishes an ore.
- ✗ Writing “ionic solids have no ions” — ions are present but fixed.
- ✗ Using carbon reduction for Na/K/Ca/Mg/Al — highly reactive metals need electrolysis.
- ✗ Confusing roasting (sulphide + excess air) with calcination (carbonate + limited/no air).
- ✗ Ignoring observation in displacement questions: state colour/deposit only when the reaction actually occurs.

17. 5-Minute Final Revision Grid

MEMORY CUE	ONE-LINE RECALL
Physical properties	Metals: lustrous, malleable, ductile, sonorous, conductors — with exceptions.
Amphoteric oxides	Al_2O_3 , ZnO react with acids AND bases.
Water	Na/K cold & violent → Ca → Mg hot → Al/Zn/Fe steam → Pb/Cu/Ag/Au no reaction.
Acid	Above H usually gives H_2 ; HNO_3 is the major oxidising exception.
Displacement	More reactive metal displaces less reactive one from salt solution.
Ionic compounds	Electron transfer; hard; high mp; conduct molten/aqueous, not solid.
Ore	Mineral from which metal can be extracted economically.
Roasting / calcination	Sulphide + excess air / carbonate + limited or no air.
High reactivity extraction	Electrolysis.
Thermite	Al reduces Fe_2O_3 ; huge heat; railway welding.
Refining	Impure anode, pure cathode, salt solution electrolyte, anode mud.
Corrosion	Prevent by barrier coatings, galvanising, electroplating, alloying.

18. Final Board Trigger → Must-Write Drill

“Aluminium wire resists corrosion” → protective, adherent Al_2O_3 layer.

“Fe placed in MgSO_4 ” → no reaction; $\text{Fe} < \text{Mg}$ in reactivity.

“Solid ionic compound does not conduct” → ions fixed; no mobile charge carriers.

“Molten ionic compound conducts” → ions become mobile.

“Sulphide ore → oxide” → roasting in excess air.

“Carbonate ore → oxide” → calcination.

“Na extraction” → electrolysis of molten compound; cannot use carbon.

“Iron bridge near sea rusts faster” → more moisture + dissolved salts speed corrosion.

“Fair heat-conduction test” → same rod length/thickness + same heating conditions.

“Railway track welding” → thermite: $\text{Fe}_2\text{O}_3 + \text{Al} \rightarrow \text{molten Fe} + \text{heat}$.

19. Self-check — can you answer without looking?

- ☐ Differentiate metals and non-metals with four properties + exceptions.
- ☐ Write reactivity series and use it to predict three reactions.
- ☐ Explain $\text{NaCl}/\text{MgCl}_2$ formation by electron transfer.
- ☐ Give four properties of ionic compounds with reasons.
- ☐ Define mineral, ore and gangue.
- ☐ Differentiate roasting and calcination with equations.
- ☐ Explain extraction of low/middle/high-reactivity metals.
- ☐ Explain electrolytic refining of copper with labels.
- ☐ Explain corrosion + prevention + aluminium passivation.
- ☐ Solve a heat-conduction fair-test question using variables.

★ **MEMORY LINE** TOPPER RULE: For metals questions, combine the property/reaction + reactivity reason + balanced equation + observation/application whenever the mark value allows.

CHAPTER 4 — CARBON AND ITS COMPOUNDS

Topper-style revision • board-answer language • syllabus + NCERT/competency angles audited

BOARD SCOPE — what you must be able to do

- ★ Explain covalent bonding in carbon and why carbon forms such a huge number of compounds.
- ★ Classify compounds as saturated/unsaturated, straight-chain/branched/cyclic; identify homologous series and functional groups.
- ★ Name simple alkanes, alkenes, alkynes and compounds containing halo, alcohol, aldehyde and ketone groups.
- ★ Write and reason through combustion, oxidation, addition and substitution reactions.
- ★ Explain properties/reactions of ethanol and ethanoic acid, esterification/saponification, and cleansing action of soaps/detergents.
- ★ Draw structures/electron-dot diagrams and solve unfamiliar competency questions from molecular formulae and observations.

1. Why Carbon Forms Covalent Bonds

[2M] Why does carbon not form C^{4+} or C^{4-} ions easily?

- Carbon has 4 valence electrons. Losing 4 electrons requires very large energy, while gaining 4 electrons would make a 10-electron ion on a 6-proton nucleus, causing strong repulsion and instability.
- Carbon therefore shares electrons and forms covalent bonds.
- Tetravalency = carbon forms four covalent bonds.

[1M] What is a covalent bond?

→ A bond formed by mutual sharing of one or more pairs of electrons between atoms is called a covalent bond.

Electron-dot / structural examples

- $H-H$ (H_2): one shared pair
- $O=O$ (O_2): two shared pairs
- $N \equiv N$ (N_2): three shared pairs
- CH_4 : carbon shares one electron with each of four H atoms

EXAM TRAP

- ★ In a single bond, 1 pair is shared; double bond = 2 pairs; triple bond = 3 pairs.
- ★ Do not show carbon forming four ionic bonds in ordinary carbon compounds.

2. Properties of Covalent Compounds

Property	Board reason
Usually low melting/boiling points	Weak intermolecular forces compared with ionic compounds.
Poor conductors of electricity	Generally no free ions/electrons available to carry current.
Often gases/liquids or soft solids	Intermolecular attractions are relatively weak.
Exceptions exist	Graphite conducts electricity because of delocalised electrons.

[2M] Why are most carbon compounds poor conductors of electricity?

- They are covalent and generally do not produce free ions or mobile electrons; hence they cannot carry electric current effectively.

3. Versatile Nature of Carbon — two key reasons

3.1 Catenation

[1M] What is catenation?

→ The property of carbon to form strong covalent bonds with other carbon atoms, producing long chains, branched chains and rings, is called catenation.

3.2 Tetravalency

- Carbon has valency 4 and can bond with C, H, O, N, S, halogens, etc.
- Small atomic size gives strong C—C and C—other atom bonds.

Chains, branches and rings

- **Straight chain:** $\text{CH}_3\text{—CH}_2\text{—CH}_2\text{—CH}_3$ (butane)
- **Branched chain:** $\text{CH}_3\text{—CH(CH}_3\text{)—CH}_3$ (2-methylpropane / isobutane)
- **Cyclic compounds:** carbon atoms may join to form rings (e.g., cyclohexane concept)

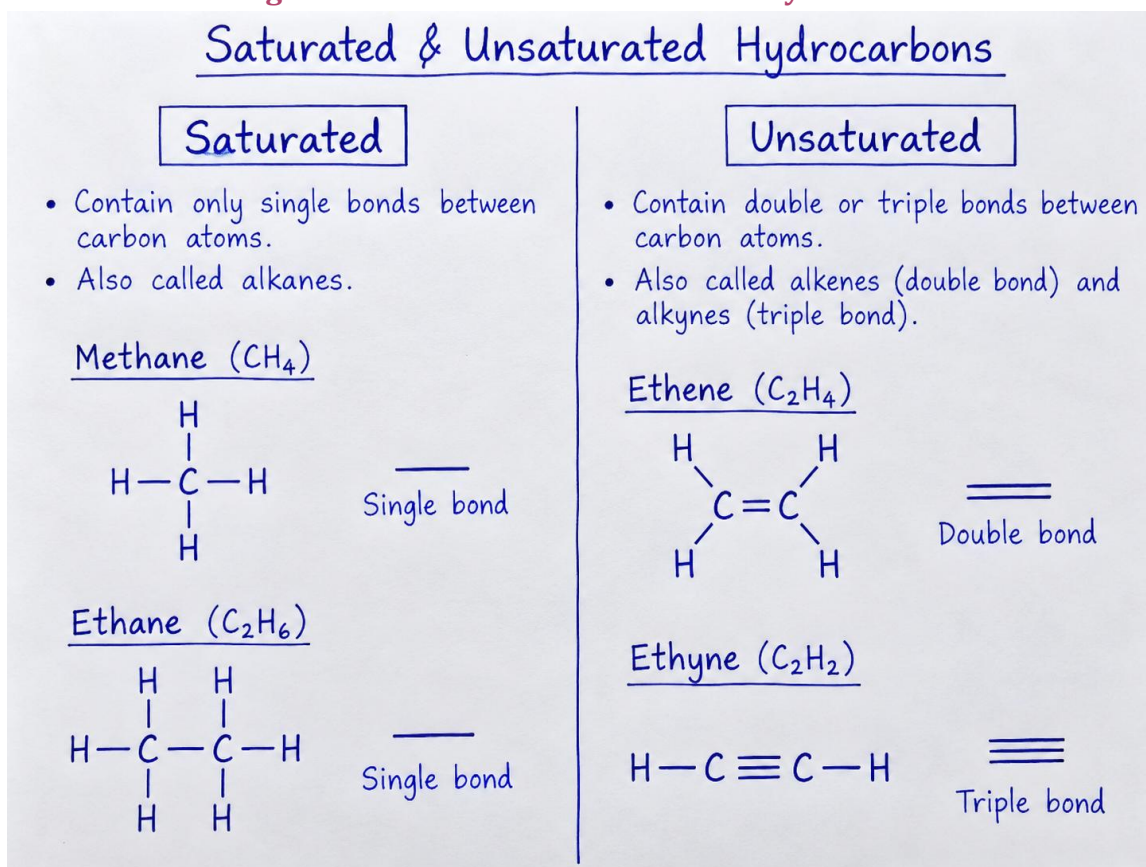
[2M] What are structural isomers?

→ Compounds having the same molecular formula but different structural arrangements of atoms are called structural isomers.

- Example: C_4H_{10} has n-butane and isobutane (2-methylpropane).

4. Saturated & Unsaturated Hydrocarbons

Handwritten Revision Diagram — Saturated & Unsaturated Hydrocarbons



Remember: alkanes have only single bonds, alkenes have double bonds and alkynes have triple bonds.

[1M] What is a hydrocarbon?

→ A compound made up only of carbon and hydrogen is called a hydrocarbon.

Type	Bonding	General idea	Example
Saturated	Only C—C single bonds	Alkanes	Ethane, C_2H_6
Unsaturated	At least one $\text{C}=\text{C}$ or	Alkenes / alkynes	Ethene C_2H_4 / ethyne C_2H_2

	$C \equiv C$		
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QUICK IDENTIFICATION

- ★ Alkane general formula: C_nH_{2n+2}
- ★ Alkene (one double bond): C_nH_{2n}
- ★ Alkyne (one triple bond): C_nH_{2n-2}
- ★ First possible alkene/alkyne has 2 carbon atoms: ethene / ethyne.

Bromine-water test for unsaturation

[2M] How can saturated and unsaturated compounds be distinguished experimentally?

→ Unsaturated compounds decolourise bromine water because bromine adds across the multiple bond; saturated compounds do not decolourise it under ordinary conditions.

5. Homologous Series — board favourite

[2M] Define a homologous series.

- A homologous series is a family of organic compounds having the same functional group and general formula, similar chemical properties, with successive members differing by $-\text{CH}_2-$ (14 u).
- Physical properties show a gradual change with increasing molecular mass.
- Chemical properties remain similar because the functional group is the same.

Examples

- Alkanes: CH_4 , C_2H_6 , C_3H_8 , C_4H_{10} ...
- Alcohols: CH_3OH , $\text{C}_2\text{H}_5\text{OH}$, $\text{C}_3\text{H}_7\text{OH}$...

[2M] Why do successive members differ in mass by 14 u?

→ They differ by one $-\text{CH}_2-$ unit; mass = $12 + 2(1) = 14$ u.

6. Naming Carbon Compounds — exam-ready method

6.1 Carbon-chain prefixes

C atoms	Prefix	C atoms	Prefix
1	meth-	6	hex-
2	eth-	7	hept-
3	prop-	8	oct-
4	but-	9	non-
5	pent-	10	dec-

6.2 Bond suffix

- Single bond → -ane | double bond → -ene | triple bond → -yne

6.3 Functional groups in syllabus

Functional group	Representation	Naming clue	Example
Halogen	$-\text{Cl}$ / $-\text{Br}$ / $-\text{I}$	chloro-, bromo-, iodo-	chloroethane
Alcohol	$-\text{OH}$	-ol	ethanol
Aldehyde	$-\text{CHO}$	-al	ethanal
Ketone	$>\text{C}=\text{O}$	-one	propanone

TOPPER NAMING RULE

- ★ Choose longest carbon chain → count C atoms → identify single/double/triple bond → identify functional group → name accordingly.
- ★ At Class 10 level, practise structures of methane, ethane, ethene, ethyne, chloroethane, ethanol, ethanal and propanone.

7. Important Structures You Must Recognise/Draw

Compound	Formula / condensed structure	Type
Methane	CH ₄	alkane
Ethane	CH ₃ —CH ₃	alkane
Ethene	CH ₂ =CH ₂	alkene
Ethyne	HC≡CH	alkyne
Chloroethane	CH ₃ —CH ₂ —Cl	halo compound
Ethanol	CH ₃ —CH ₂ —OH	alcohol
Ethanal	CH ₃ —CHO	aldehyde
Propanone	CH ₃ —CO—CH ₃	ketone
Ethanoic acid	CH ₃ —COOH	carboxylic acid

8. Chemical Properties of Carbon Compounds

8.1 Combustion

[2M] What happens when carbon compounds burn in oxygen?

→ Most carbon compounds burn in oxygen to produce carbon dioxide, water, heat and light. Combustion is an oxidation reaction.



- Saturated hydrocarbons generally burn with a clean blue flame when oxygen is sufficient.
- Unsaturated compounds may burn with a yellow, sooty flame because of incomplete combustion.

[2M] Why is a gas-stove flame sometimes yellow and sooty?

→ Air holes may be blocked, so oxygen supply is insufficient. Incomplete combustion forms unburnt carbon/soot, producing a yellow flame.

8.2 Oxidation

[1M] What is an oxidising agent?

→ A substance that supplies oxygen or removes hydrogen and causes oxidation is called an oxidising agent.



8.3 Addition reaction

[2M] What is an addition reaction?

→ A reaction in which atoms/groups add across a carbon-carbon double or triple bond to form a saturated product is called an addition reaction.



- Hydrogenation of vegetable oils converts unsaturated oils into saturated fats/vanaspati ghee.

8.4 Substitution reaction

[2M] What is a substitution reaction?

→ A reaction in which one atom or group in a saturated hydrocarbon is replaced by another atom/group is called a substitution reaction.



BOARD COMPARISON

- ★ Addition → characteristic of unsaturated compounds.
- ★ Substitution → characteristic of saturated hydrocarbons such as methane.
- ★ Combustion is always oxidation, but not every oxidation reaction is combustion.

9. Ethanol (C₂H₅OH) — properties + reactions + uses

[1M] Write two physical properties of ethanol.

→ Ethanol is a colourless liquid with a characteristic smell; it is soluble in water and is volatile.

9.1 Reaction with sodium



[2M] What is observed when sodium is added to ethanol?

→ Hydrogen gas is evolved and sodium ethoxide is formed. The gas can be confirmed by the pop test.

9.2 Dehydration of ethanol



[2M] What is dehydration?

→ Removal of a water molecule from ethanol to form ethene is called dehydration; concentrated sulphuric acid acts as a dehydrating agent.

9.3 Uses of ethanol

- Solvent in medicines, perfumes and tinctures.
- Fuel / blended with petrol in suitable applications.
- Raw material in chemical industry.

HEALTH NOTE

- ★ Ethanol affects the central nervous system; excessive consumption is harmful.
- ★ Denatured alcohol is ethanol made unfit for drinking by adding poisonous/unpleasant substances.

10. Ethanoic Acid / Acetic Acid (CH₃COOH)

[1M] Write key physical properties of ethanoic acid.

→ It is a colourless liquid with sour taste and pungent/vinegar-like smell; it mixes with water.

- 5–8% solution of ethanoic acid in water is called vinegar.
- Pure ethanoic acid freezes around 16.6°C and is called glacial acetic acid.

10.1 Reaction with sodium



10.2 Reaction with base



10.3 Reaction with carbonate / hydrogencarbonate



[2M] How can ethanol and ethanoic acid be distinguished experimentally?

→ Add sodium hydrogencarbonate/carbonate. Ethanoic acid gives brisk effervescence of CO₂, whereas ethanol does not.

- CO₂ turns lime water milky.

11. Esterification & Saponification — do not skip

[2M] What is esterification?

→ The reaction of an alcohol with a carboxylic acid in the presence of concentrated H₂SO₄ to form a sweet-smelling ester and water is called esterification.



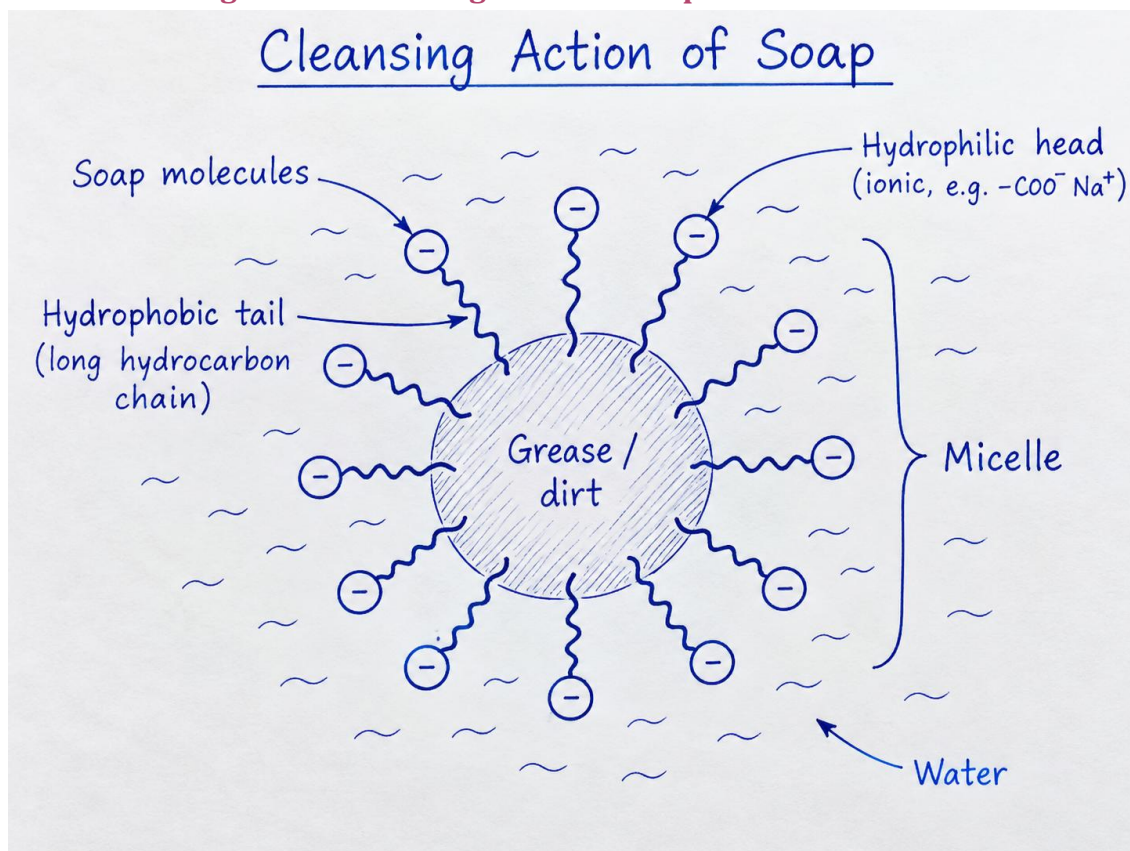
[2M] What is saponification?

→ The alkaline hydrolysis of an ester to form an alcohol and the salt of a carboxylic acid is called saponification; this principle is used in soap making.



12. Soaps & Detergents — board favourite

Handwritten Board Diagram — Cleansing Action of Soap



This helps in competency and explanation questions: soap forms micelles with hydrophobic tails inward and hydrophilic heads outward.

[1M] What is a soap?

→ A soap is the sodium or potassium salt of a long-chain carboxylic acid (fatty acid).

[1M] What is a detergent?

→ A detergent is a cleansing agent generally containing long hydrocarbon chains with ionic groups such as sulphate/sulphonate; it works effectively even in hard water.

12.1 Structure of a soap molecule

- Hydrophobic hydrocarbon tail: dissolves in oil/grease; avoids water.
- Hydrophilic ionic head: attracted to water.

12.2 Micelle formation / cleansing action

[3M] Explain how soap cleans oily dirt.

→ Soap molecules arrange around grease so that hydrophobic tails are embedded in oil while ionic heads remain in water. On agitation, the grease breaks into tiny droplets surrounded by soap molecules called micelles; these remain suspended and are washed away.

MICELLE DIAGRAM LABELS

- ★ Centre: grease/oil droplet.
- ★ Tails point inward into oil; ionic heads point outward toward water.
- ★ Micelles form above a certain concentration and trap oily dirt.

12.3 Soap in hard water

[2M] Why does soap not work well in hard water?

→ Ca^{2+} and Mg^{2+} ions react with soap to form insoluble calcium/magnesium salts called scum. This consumes soap and reduces lather/cleansing efficiency.

- Detergents do not form insoluble scum so readily and work better in hard water.

Soap	Detergent
Forms scum in hard water	Works well in hard water
Usually biodegradable more readily	Some detergents may be less biodegradable
Na^+/K^+ salts of fatty acids	Synthetic cleansing agents with sulphate/sulphonate-type ionic head

13. Important Allotropes of Carbon — useful board angle

[1M] What are allotropes?

→ Different structural forms of the same element having different physical properties but similar chemical nature are called allotropes.

Allotrope	Structure / key property
Diamond	Each C bonded to 4 C atoms; rigid 3-D network; very hard; poor conductor.
Graphite	Each C bonded to 3 C atoms in layers; one electron delocalised; conducts electricity; soft/slippery.
Fullerenes	Carbon atoms arranged in cage-like structures, e.g. C_{60} .

[2M] Why does graphite conduct electricity but diamond does not?

→ In graphite, each carbon forms three bonds and has a delocalised electron that can move through the structure. In diamond, all four valence electrons are locked in covalent bonds.

14. Must-Memorise Reactions — rapid board bank

Equation / reaction	Exam tag
$\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$	combustion
$\text{CH}_3\text{CH}_2\text{OH} + 2[\text{O}] \rightarrow \text{CH}_3\text{COOH} + \text{H}_2\text{O}$	oxidation of ethanol
$\text{CH}_2=\text{CH}_2 + \text{H}_2 \xrightarrow{\text{Ni}} \text{CH}_3-\text{CH}_3$	addition / hydrogenation
$\text{CH}_4 + \text{Cl}_2 \xrightarrow{\text{sunlight}} \text{CH}_3\text{Cl} + \text{HCl}$	substitution
$2\text{C}_2\text{H}_5\text{OH} + 2\text{Na} \rightarrow 2\text{C}_2\text{H}_5\text{ONa} + \text{H}_2$	ethanol + sodium
$\text{C}_2\text{H}_5\text{OH} \xrightarrow{\text{conc. H}_2\text{SO}_4, 443\text{K}} \text{CH}_2=\text{CH}_2 + \text{H}_2\text{O}$	dehydration
$\text{CH}_3\text{COOH} + \text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$	neutralisation
$\text{CH}_3\text{COOH} + \text{NaHCO}_3 \rightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O} + \text{CO}_2$	CO_2 test / acid
$\text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{OH} \xrightarrow{\text{conc. H}_2\text{SO}_4} \text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O}$	esterification
$\text{Ester} + \text{NaOH} \rightarrow \text{alcohol} + \text{carboxylate salt}$	saponification

15. Mark-wise Answer Bank — write like this

[1M] Why does carbon form a large number of compounds?

→ Mainly because of tetravalency and catenation.

[2M] Differentiate saturated and unsaturated hydrocarbons.

→ Saturated hydrocarbons contain only C—C single bonds; unsaturated hydrocarbons contain at least one C=C or C≡C bond.

- Saturated examples: ethane; unsaturated examples: ethene/ethyne.

[2M] Why are vegetable oils hydrogenated?

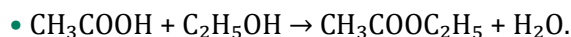
→ Hydrogen is added across double bonds in unsaturated vegetable oils using Ni catalyst, converting them into more saturated solid/semi-solid fats such as vanaspati.

[2M] Distinguish addition and substitution reactions.

→ Addition occurs at multiple bonds and converts an unsaturated compound to a more saturated product; substitution replaces an atom/group in a saturated compound without breaking the carbon skeleton.

[3M] Explain esterification with equation and observation.

→ Ethanoic acid reacts with ethanol in presence of concentrated H_2SO_4 to produce ethyl ethanoate and water.



• The ester has a pleasant/fruity smell.

[3M] Explain cleansing action of soap.

→ Soap has a water-loving ionic head and oil-loving hydrocarbon tail. Tails embed in grease while heads remain in water. Agitation forms micelles that suspend oily dirt so it can be rinsed away.

16. Competency / Case-based Thinking

IF CBSE GIVES A STRUCTURE

- ★ Count carbon atoms → locate multiple bond → identify functional group → classify → name.
- ★ Check carbon valency = 4 at every carbon atom.
- ★ Decide saturated/unsaturated from $\text{C}-\text{C}$, $\text{C}=\text{C}$ or $\text{C}\equiv\text{C}$.

IF CBSE GIVES AN UNKNOWN LIQUID

- ★ NaHCO_3 effervescence → likely carboxylic acid (CO_2 formed).
- ★ Oxidising agent converting alcohol to acid → oxidation.
- ★ Sweet-smelling product from alcohol + acid → esterification.

IF CBSE GIVES SOAP/HARD-WATER DATA

- ★ More scum / less lather → more $\text{Ca}^{2+}/\text{Mg}^{2+}$ hardness or poorer soap efficiency.
- ★ Detergent should perform better than soap in hard water.
- ★ Control variables in experiment: same water volume, cleansing-agent amount, shaking time and temperature.

17. Common Mistakes a Topper Avoids

- X Writing carbon valency as 2 instead of 4.
- X Calling every carbon compound a hydrocarbon — hydrocarbons contain only C and H.
- X Confusing ethene C_2H_4 with ethane C_2H_6 or ethyne C_2H_2 .
- X Forgetting that homologous members differ by CH_2 / 14 u.
- X Calling hydrogenation a substitution reaction — it is addition.
- X Writing concentrated H_2SO_4 as a catalyst in dehydration without noting its dehydrating role.
- X Confusing ethanol ($\text{C}_2\text{H}_5\text{OH}$) with ethanoic acid (CH_3COOH).
- X Writing esterification product without water or without the acid catalyst.
- X Reversing micelle orientation — tails point into grease, heads toward water.
- X Saying soap cleans better in hard water — detergents generally do.

18. 5-Minute Final Revision Grid

Memory cue	One-line recall
Carbon bonding	tetravalent + covalent
Versatility	catenation + tetravalency
Saturated	single bonds / alkane
Unsaturated	double/triple / alkene/alkyne
Homologues	same FG + general formula; differ by CH_2
Alcohol	$-\text{OH}$ / -ol
Aldehyde	$-\text{CHO}$ / -al
Ketone	$>\text{C}=\text{O}$ / -one

Addition	unsaturated + H ₂ → saturated
Substitution	H replaced in saturated hydrocarbon
Ethanol → ethene	dehydration, conc H ₂ SO ₄
Ethanoic acid + NaHCO ₃	CO ₂ effervescence
Esterification	acid + alcohol → ester + water
Soap	tail in grease; head in water
Hard water	soap scum; detergent better

19. Self-check — can you answer without looking?

- ☐ Explain why carbon forms covalent bonds and why it is tetravalent.
- ☐ Draw/recognise electron-dot structures of CH₄, C₂H₆, C₂H₄ and C₂H₂.
- ☐ Differentiate saturated and unsaturated compounds and state bromine-water test.
- ☐ Define homologous series; explain 14 u difference.
- ☐ Name simple compounds containing halo, alcohol, aldehyde and ketone groups.
- ☐ Write equations for combustion, oxidation, addition and substitution.
- ☐ Write ethanol + Na and dehydration equations.
- ☐ Write ethanoic acid reactions with NaOH and NaHCO₃.
- ☐ Explain esterification and saponification.
- ☐ Explain micelle formation and why soap fails in hard water.
- ☐ Differentiate diamond and graphite.
- ☐ Write all 10 reaction-bank equations from memory.

TOPPER RULE FOR THIS CHAPTER

- ★ For structure questions: count bonds carefully — carbon must complete valency 4.
- ★ For reactions: write reagent/condition over arrow and identify reaction type.
- ★ For 3-mark answers: definition/principle + equation + observation/reason.

CHAPTER 5 — LIFE PROCESSES

Topper-style revision • answer-ready board language • 2026–27 scope + sample-paper angles audited

BOARD SCOPE — what you must be able to do

- ★ Explain nutrition, respiration, transport and excretion in plants and animals with correct structures and functions.
- ★ Use equations, observations and experimental evidence for photosynthesis and respiration.
- ★ Explain human digestion organ-wise, including enzymes, bile, absorption and diet-related intestine length.
- ★ Compare aerobic/anaerobic respiration, breathing in aquatic/terrestrial organisms and gas exchange in alveoli.
- ★ Explain heart chambers, double circulation, blood vessels, blood/lymph, plant transport, nephron and urine formation.
- ★ Solve diagram-, experiment- and case-based questions by linking structure → function → consequence.

1. Life Processes — the maintenance functions

[1M] What are life processes?

- The basic processes that maintain the living state of an organism are called life processes.
- Core Class 10 processes: nutrition, respiration, transportation and excretion.
- Growth or visible movement alone cannot be used as the only criterion for life; continuous molecular activity is essential.

[2M] Why is diffusion alone insufficient in multicellular organisms?

- In large multicellular organisms, most cells are not directly in contact with the environment and distances are large. Diffusion is too slow to supply oxygen/food and remove wastes efficiently.
- Hence specialised surfaces and transport systems are needed.

2. Nutrition — obtaining food and raw materials

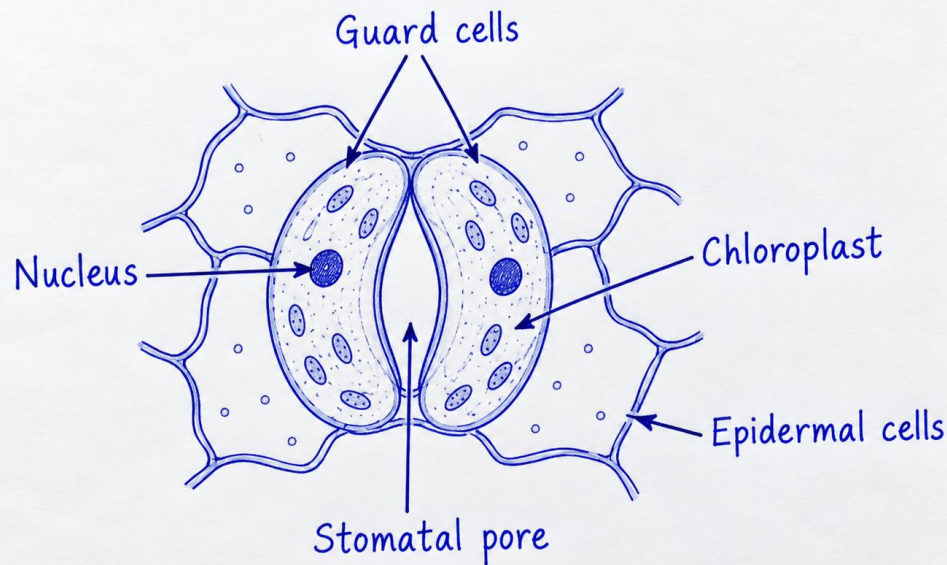
[1M] What is nutrition?

- Nutrition is the process by which an organism obtains food and utilises it for energy, growth, repair and maintenance.

2.1 Autotrophic nutrition — photosynthesis

Handwritten Board Diagram — Stomata

Stomata



A common board diagram: guard cells, stomatal pore, nucleus, chloroplast and epidermal cells should be labeled clearly.

[2M] What is autotrophic nutrition?

→ A mode of nutrition in which organisms make organic food from simple inorganic substances such as CO_2 and water using an external energy source is called autotrophic nutrition.



PHOTOSYNTHESIS — 3 events to write

- ★ Absorption of light energy by chlorophyll.
- ★ Conversion of light energy to chemical energy and splitting of water into hydrogen and oxygen.
- ★ Reduction of carbon dioxide to carbohydrates.

- CO_2 enters leaves mainly through stomata; water is absorbed by roots and transported through xylem.
- Guard cells regulate opening and closing of stomatal pores; stomata also permit water loss by transpiration.

[2M] How do desert plants reduce water loss while photosynthesising?

→ Many desert plants take up CO_2 at night and prepare an intermediate which is acted upon by sunlight during the day, allowing stomata to remain closed for much of the hot daytime.

- This is an adaptation to minimise water loss.

Photosynthesis experiments — current CBSE favourite

[2M] Why must a plant be destarched before testing a leaf for photosynthesis?

→ It is kept in darkness beforehand so stored starch is used up. Therefore any starch detected after the experiment can be attributed to photosynthesis during the test.

- Iodine test: blue-black colour indicates starch.
- Sunlight experiment: only the exposed part of a destarched leaf turns blue-black.
- CO_2 necessity experiment: KOH absorbs CO_2 ; the leaf without available CO_2 does not form starch.

3. Heterotrophic Nutrition & Human Digestion

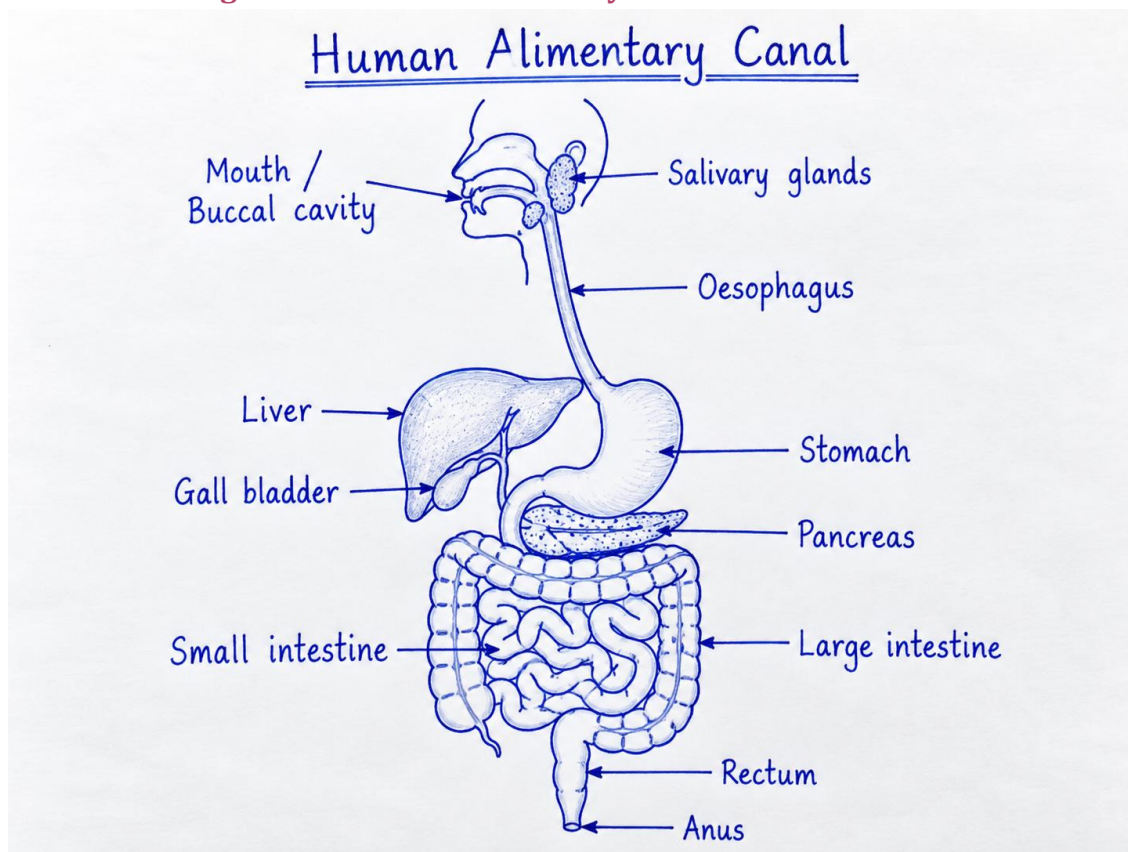
3.1 Main heterotrophic modes

Mode	Meaning / example
Holozoic	Food is ingested and digested inside the body; humans, Amoeba.

Saprophytic	Feeds on dead/decaying matter using extracellular digestion; fungi.
Parasitic	Obtains food from a living host; Cuscuta, tapeworm.

3.2 Human alimentary canal — organ → secretion → function

Handwritten Board Diagram — Human Alimentary Canal



Students should practice the organ sequence from mouth to anus and place accessory glands correctly.

Part	Key secretion/action	Must-write function
Mouth	Saliva; salivary amylase	Chewing; starch → simpler sugars.
Oesophagus	Peristalsis	Pushes food to stomach.
Stomach	HCl + pepsin + mucus	Acidic medium; protein digestion; mucus protects wall.
Liver/gall bladder	Bile	Neutralises acidic food; emulsifies fats.
Pancreas	Pancreatic juice	Digests carbohydrates, proteins and fats.
Small intestine	Intestinal enzymes + villi	Completes digestion; absorbs digested nutrients.
Large intestine	Water absorption	Absorbs water; waste forms faeces.

[2M] What is the role of bile in digestion?

→ Bile from the liver makes the food entering the small intestine alkaline and breaks large fat globules into smaller droplets (emulsification), increasing the surface area for lipase action.

[2M] If the bile duct is blocked, how is digestion affected?

→ Fat digestion is impaired because bile cannot enter the small intestine to emulsify fats. Neutralisation of the acidic chyme is also reduced, making conditions less suitable for intestinal/pancreatic enzymes.

[2M] Why is the small intestine of herbivores longer than that of carnivores?

→ Cellulose-rich plant food is harder and slower to digest, so herbivores need a longer small intestine for sufficient digestion and absorption. Meat is easier to digest, so carnivores have a shorter intestine.

DIAGRAM MUST-PRACTICE — HUMAN DIGESTIVE SYSTEM

★ Label: mouth/buccal cavity, oesophagus, stomach, liver, gall bladder, pancreas, small intestine, large intestine,

rectum, anus.

★ For diagram questions, attach one function to each labelled organ instead of writing only names.

4. Respiration — releasing usable energy

[1M] What is respiration?

→ Respiration is the cellular process in which food molecules are broken down to release energy for life activities.

→ **Glucose → Pyruvate (3-carbon molecule) + energy [in cytoplasm]**

Pathway	Condition	Products	Energy
Aerobic	O ₂ present; mitochondria	CO ₂ + H ₂ O	More energy
Anaerobic in yeast	No O ₂	Ethanol + CO ₂	Less energy
Muscle cells	O ₂ shortage	Lactic acid	Less energy

→ **C₆H₁₂O₆ + 6O₂ → 6CO₂ + 6H₂O + energy** *aerobic respiration*

[2M] Why do muscles sometimes develop cramps during vigorous exercise?

→ When oxygen supply is insufficient, muscle cells partly break down glucose anaerobically to lactic acid. Accumulation of lactic acid is associated with muscle cramps.

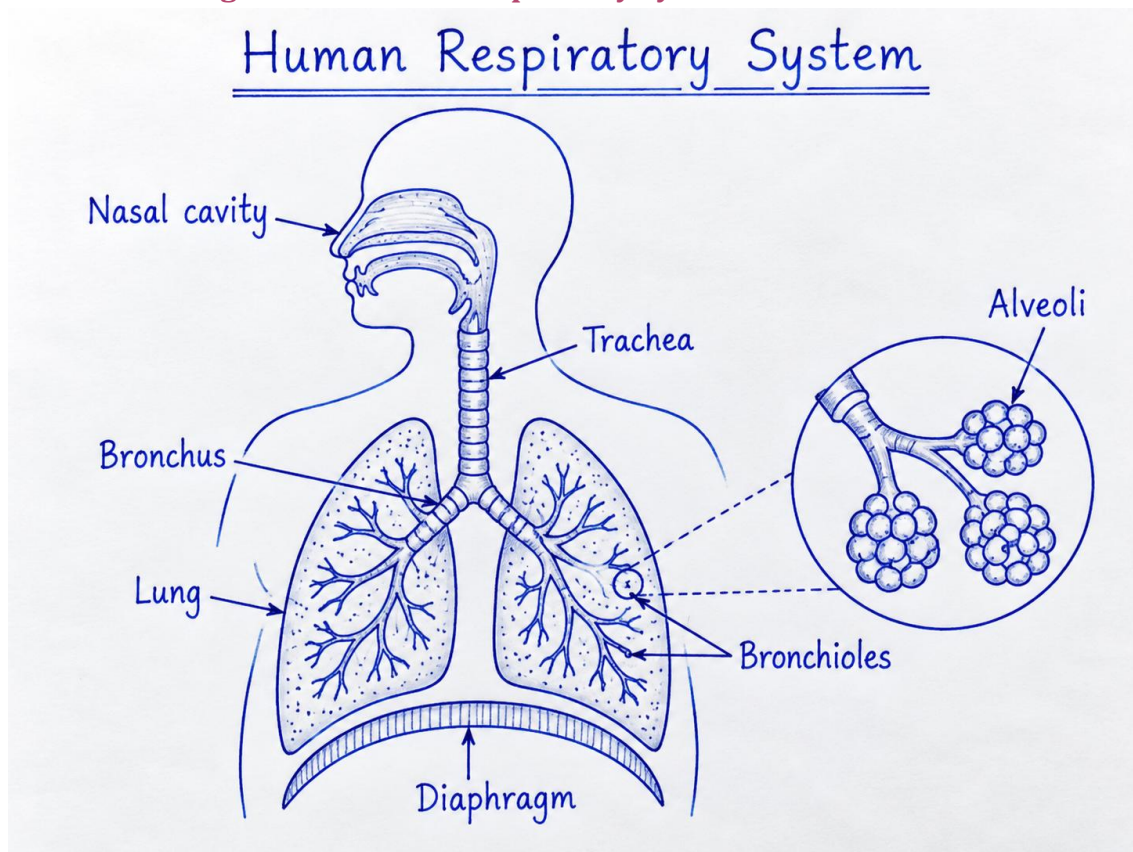
[2M] Differentiate breathing and respiration.

→ Breathing is the physical movement of air into and out of respiratory organs; respiration is the chemical breakdown of food inside cells to release energy.

- Breathing occurs at organ level; cellular respiration occurs inside cells.

4.1 Human breathing & gas exchange

Handwritten Board Diagram — Human Respiratory System



Focus on clean labels: nasal cavity, trachea, bronchi, bronchioles, lungs, alveoli and diaphragm.

- Air passage: nostrils → nasal passage → trachea → bronchi → bronchioles → alveoli.
- Nasal hairs/mucus filter dust; cartilage rings prevent the trachea from collapsing.

[3M] Describe inhalation.

→ During inhalation, ribs move up and outward and the diaphragm contracts and flattens. Chest-cavity volume increases, pressure inside falls and air is drawn into the lungs.

[2M] How are alveoli adapted for efficient gas exchange?

→ Alveoli are extremely numerous, providing a very large surface area. Their walls are thin and moist and are surrounded by a rich network of blood capillaries, giving a short diffusion distance and continuous concentration gradient.

- O_2 diffuses from alveoli into blood; CO_2 diffuses from blood into alveoli.
- Haemoglobin in red blood cells binds oxygen and transports it efficiently to tissues.
- CO_2 is more soluble in water than O_2 , so it is transported mainly in dissolved form in blood.

[2M] Why do aquatic organisms breathe faster than terrestrial organisms?

→ Water contains much less dissolved oxygen than air. Aquatic organisms must pass more water over respiratory surfaces to obtain sufficient oxygen, so their breathing rate is generally higher.

[2M] Why cannot diffusion alone deliver oxygen from lungs to toes?

→ Diffusion is effective only over very short distances. In a large body it would be far too slow; haemoglobin and the circulatory system rapidly transport oxygen to all tissues.

DIAGRAM MUST-PRACTICE — RESPIRATORY SYSTEM / ALVEOLUS

★ Label: nasal passage, trachea, bronchi, lungs, diaphragm; for an alveolus show thin wall and surrounding capillaries.

★ For inhalation diagrams, identify diaphragm flattening + ribs moving up/out + chest volume increase.

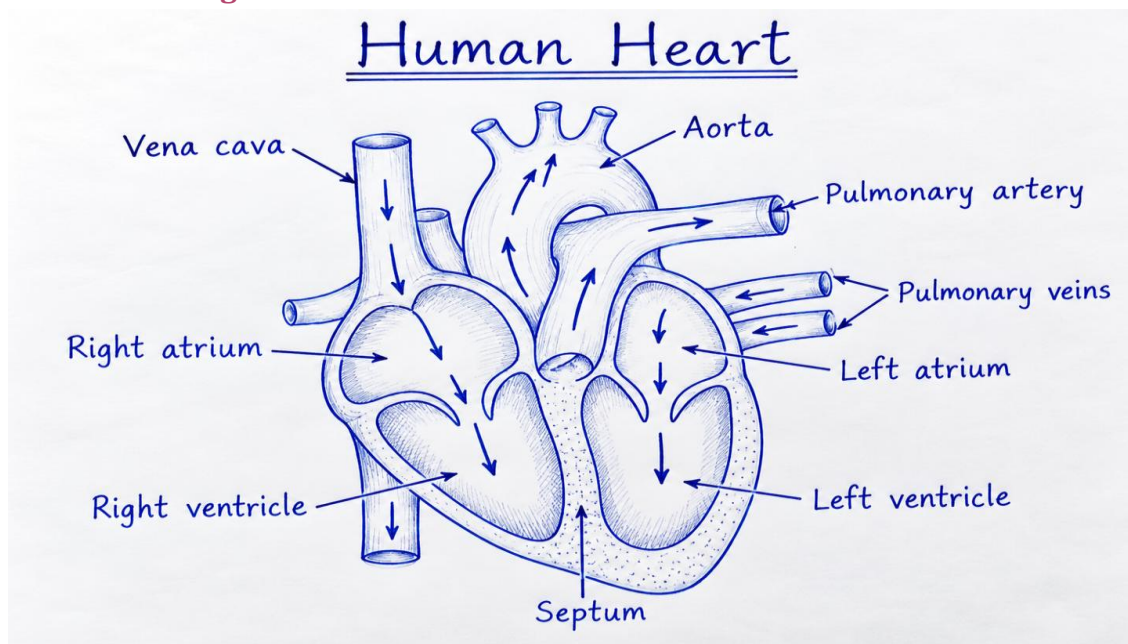
5. Transportation in Human Beings

5.1 Blood — fluid connective tissue

Component	Function
Plasma	Transports food, CO_2 , hormones, salts, wastes and blood cells.
RBCs	Haemoglobin carries oxygen.
WBCs	Defence against infection.
Platelets	Help blood clot and prevent excessive blood loss.

5.2 Heart and double circulation

Handwritten Board Diagram — Human Heart



In board exam, label the chambers and major vessels neatly and show direction of flow with arrows.

- Right atrium receives deoxygenated blood from body → right ventricle → pulmonary artery → lungs.
- Oxygenated blood returns by pulmonary veins → left atrium → left ventricle → aorta → body.

[2M] What is double circulation?

→ In one complete cycle, blood passes through the heart twice: once through pulmonary circulation (heart–lungs–heart) and once through systemic circulation (heart–body–heart).

[3M] Why is complete separation of oxygenated and deoxygenated blood important in birds and mammals?

→ Birds and mammals have high energy requirements for maintaining a constant body temperature. A four-chambered heart prevents mixing, ensuring an efficient supply of highly oxygenated blood for high rates of respiration.

Heart comparison — sample-paper angle

Group	Heart	Circulation / significance
Fish	2 chambers	Single circulation: heart → gills → body → heart.
Amphibians / many reptiles	3 chambers	Some mixing of oxygenated and deoxygenated blood.
Birds & mammals	4 chambers	Complete separation; efficient double circulation.

5.3 Blood vessels

Vessel	Direction / structure
Artery	Carries blood away from heart; thick elastic walls; high pressure; usually no valves.
Vein	Carries blood towards heart; thinner walls; valves prevent backflow.
Capillary	One-cell-thick wall; exchange of materials with tissues.

[1M] What is blood pressure?

- The force exerted by blood against the walls of blood vessels is called blood pressure.
- Normal adult arterial pressure is about 120/80 mm Hg (systolic/diastolic); measured by a sphygmomanometer.

5.4 Lymph / tissue fluid

[2M] State two functions of lymph.

- Lymph transports absorbed fats from the intestine and returns excess tissue fluid from intercellular spaces to the blood.
- It is colourless and contains less protein than blood plasma.

DIAGRAM MUST-PRACTICE — HUMAN HEART

- ★ Label 4 chambers, aorta, vena cava, pulmonary artery, pulmonary vein and septum.
- ★ Remember exception: pulmonary artery carries deoxygenated blood; pulmonary vein carries oxygenated blood.

6. Transportation in Plants

6.1 Xylem — water & minerals

- Root hairs absorb water and dissolved mineral ions from soil.
- Xylem vessels and tracheids form a continuous pathway from roots to leaves.
- Root pressure contributes, especially at night; transpiration pull is the major upward force during the day.

[1M] What is transpiration?

→ The loss of water as vapour from the aerial parts of a plant, mainly through stomata, is called transpiration.

[2M] State two functions of transpiration.

→ It creates a suction (transpiration pull) that helps upward transport of water/minerals and also helps cool the plant by temperature regulation.

6.2 Phloem — translocation of food

[2M] What is translocation?

→ The transport of soluble products of photosynthesis, mainly sugars, from leaves to other parts of the plant through phloem is called translocation.

- Phloem also carries amino acids and can transport materials both upward and downward according to need.
- Loading sucrose into phloem requires energy (ATP); osmotic pressure helps drive movement.

Xylem	Phloem
Water + minerals	Food/soluble organic substances
Mainly upward movement	Movement can be upward or downward
Largely driven by physical forces/transpiration	Translocation requires energy for loading

7. Excretion in Human Beings

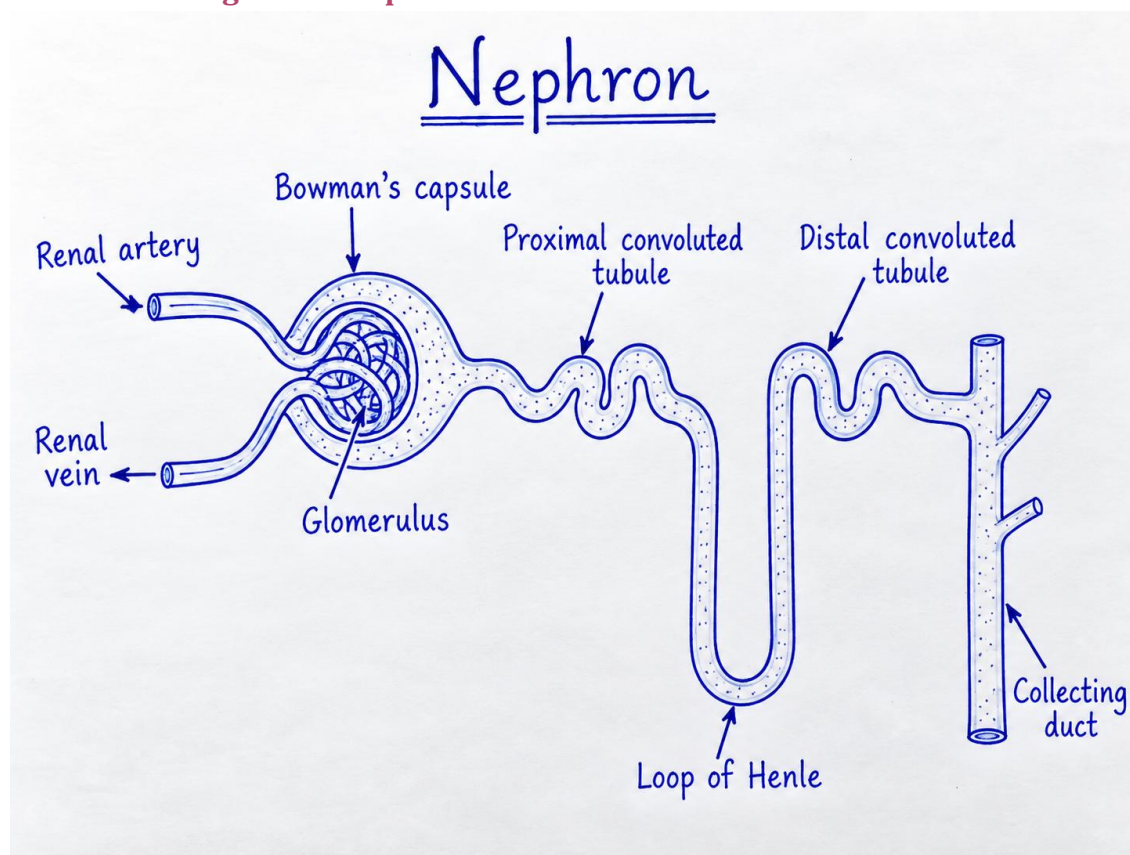
[1M] What is excretion?

→ Excretion is the biological process of removing harmful metabolic wastes from the body.

- Human urinary system: two kidneys, two ureters, urinary bladder and urethra.

7.1 Nephron — structural and functional unit of kidney

Handwritten Board Diagram — Nephron



Practice this exact label order for board diagrams: Bowman's capsule → glomerulus → PCT → loop of Henle → DCT → collecting duct.

[2M] What is a nephron?

→ A nephron is the microscopic filtration unit of the kidney, consisting of a capillary cluster (glomerulus) associated with Bowman's capsule and a long renal tubule.

URINE FORMATION — write these 3 ideas

- ★ Ultrafiltration: blood is filtered from the glomerular capillaries into Bowman's capsule.
- ★ Selective reabsorption: useful substances such as glucose, amino acids, salts and most water are taken back into blood.
- ★ Tubular secretion/remaining waste: excess ions and wastes stay/are added to the tubule; urine passes to collecting ducts → ureter.

[2M] Why is only about 1–2 L urine formed from a much larger daily filtrate?

→ Most water and useful solutes from the initial filtrate are selectively reabsorbed in kidney tubules. The amount of water reabsorbed depends on the body's water level and the amount of dissolved waste.

[1M] Which substances are normally reabsorbed in kidney tubules?

→ Glucose, amino acids, required salts and a major amount of water are selectively reabsorbed.

7.2 Artificial kidney / haemodialysis — NCERT box, competency-ready

[2M] What is haemodialysis?

→ Haemodialysis is the removal of nitrogenous wastes from the blood using an artificial kidney when natural kidneys fail.

- Blood flows through semi-permeable tubes in dialysing fluid; wastes diffuse out and purified blood is returned.
- Unlike a real nephron, dialysis does not perform selective reabsorption in the same way.

DIAGRAM MUST-PRACTICE — EXCRETORY SYSTEM + NEPHRON

- ★ Excretory system labels: kidneys, ureters, urinary bladder, urethra.
- ★ Nephron labels: glomerulus, Bowman's capsule, tubule, collecting duct and surrounding capillaries.

8. Excretion in Plants

- Gaseous wastes (O_2/CO_2) diffuse out through stomata/lenticels according to conditions.
- Excess water is removed by transpiration.
- Other wastes may be stored in vacuoles, old leaves that later fall, or as resins and gums in old xylem.
- Some waste substances are released into the surrounding soil.

9. Board-Exam Trigger Bank — observation → concept → reason

Trigger in question	What your answer must mention
Leaf exposed to sunlight gives starch	Destarch first; chlorophyll/light; iodine turns blue-black.
KOH in photosynthesis setup	KOH absorbs CO_2 → no starch where CO_2 unavailable.
Blocked bile duct	Poor fat emulsification + less neutralisation of acidic chyme.
Herbivore has long intestine	Cellulose takes longer to digest.
Aquatic animal breathes rapidly	Low dissolved O_2 in water.
Many alveoli	Huge surface area + thin moist wall + capillaries.
Four-chambered heart	No mixing → efficient O_2 supply → high metabolic rate.
Kidney filtrate vs urine volume	Selective reabsorption returns useful materials/water.
Wilting / stomata close	Reduced transpiration and gas exchange; guard-cell regulation.
Sugar moves from leaf to root/fruit	Phloem translocation; energy-dependent loading.

10. Mark-wise Answer Bank — write like this

[1M] Why are stomata important?

→ They allow gaseous exchange for photosynthesis/respiration and regulate water loss by transpiration.

[2M] Differentiate aerobic and anaerobic respiration.

→ Aerobic respiration uses oxygen and completely breaks down glucose to CO_2 and water with more energy; anaerobic respiration occurs without oxygen and releases less energy, producing ethanol + CO_2 in yeast or lactic acid in muscles.

[2M] Why is mucus secreted in the stomach?

→ It protects the stomach lining from the corrosive action of hydrochloric acid and digestive enzymes.

[2M] Why are capillaries one-cell thick?

→ Their thin walls provide a short diffusion distance for rapid exchange of oxygen, nutrients, carbon dioxide and wastes between blood and tissues.

[3M] Explain transportation of water in plants.

→ Root hairs absorb water; xylem forms a continuous conducting pathway through roots, stem and leaves. Root pressure contributes to upward movement, while evaporation from leaves produces transpiration pull, the major force during daytime.

[3M] Explain how food is transported in plants.

→ Sugars produced in leaves are actively loaded into phloem using ATP. Water enters, creating osmotic pressure that moves food to regions of lower pressure such as growing or storage tissues. Phloem transport can occur in either direction according to the plant's needs.

[3M] Explain urine formation in a nephron.

→ Blood is filtered at the glomerulus into Bowman's capsule. Useful substances and most water are selectively reabsorbed from the tubule into blood; additional ions/wastes may be secreted. The remaining fluid becomes urine and passes through collecting ducts and ureters.

[5M] How does inhalation occur and how are lungs adapted for gas exchange?

→ During inhalation the diaphragm contracts/flattens and ribs move up/out, increasing chest volume and lowering pressure so air enters. Airways conduct air to millions of alveoli. Alveoli provide a huge surface area, thin moist walls and a dense capillary network. O_2 diffuses into blood and binds haemoglobin; CO_2 diffuses into alveoli for exhalation.

11. Common Mistakes a Topper Avoids

- X Writing "bile digests fat" — bile emulsifies fat; lipase digests it.
- X Writing HCl as a digestive enzyme — HCl provides acidic medium and kills many germs; pepsin digests proteins.
- X Confusing breathing with cellular respiration.
- X Saying all arteries carry oxygenated blood — pulmonary artery is the key exception.
- X Saying all veins carry deoxygenated blood — pulmonary vein is the key exception.
- X Writing xylem carries food or phloem carries water as their main roles.
- X Forgetting that phloem translocation needs energy and may be bidirectional.
- X Saying urine is formed only by filtration — selective reabsorption and secretion are also important.
- X Confusing ureter (kidney→bladder) with urethra (bladder→outside).
- X Writing transpiration only as "water loss" without its transport/cooling significance.

12. 5-Minute Final Revision Grid

Memory cue	One-line recall
Photosynthesis	$CO_2 + H_2O \rightarrow \text{glucose} + O_2$ using light/chlorophyll
Stomata	Gas exchange + transpiration; guard cells regulate
Bile	Alkaline medium + emulsification of fats
Villi	Large surface area for absorption
Aerobic	O_2 ; mitochondria; $CO_2 + H_2O$; more energy
Anaerobic	Yeast→ethanol+ CO_2 ; muscle→lactic acid
Alveoli	Large area + thin moist wall + capillaries
Double circulation	Heart twice per body cycle
Xylem	Water/minerals; mainly upward
Phloem	Food; ATP-supported; both directions
Nephron	Filtration + reabsorption + secretion
Plant excretion	Diffusion, transpiration, vacuoles, leaves, gums/resins, soil

Self-check — can you answer without looking?

- 1. Define life processes and explain why diffusion is insufficient in multicellular organisms.
- 2. Write photosynthesis equation and explain the destarching/KOH experiments.
- 3. Explain digestion from mouth to small intestine and the roles of HCl, bile, enzymes and villi.
- 4. Explain aerobic vs anaerobic respiration and muscle cramps.
- 5. Describe inhalation, alveolar adaptations and O_2/CO_2 transport.
- 6. Draw/label the human heart and explain double circulation.
- 7. Compare artery, vein and capillary; state functions of blood components and lymph.
- 8. Explain xylem transport, transpiration pull and phloem translocation.
- 9. Draw/label nephron and explain urine formation.
- 10. List plant excretory strategies and explain dialysis.

13. 2026-27 Sample-Paper Drill — high-value question frames

[1M] During urine formation, which useful substances are reabsorbed?

→ Glucose, amino acids, required salts and most water are selectively reabsorbed from the filtrate into the blood.

[1M] Why is the assertion “aquatic organisms breathe slower than terrestrial organisms” incorrect?

→ Water has less dissolved oxygen than air, so aquatic organisms generally breathe faster to obtain enough oxygen.

[2M] A bile duct becomes narrowed. State two consequences for digestion.

→ Fat emulsification decreases and the acidic food from the stomach is neutralised less effectively, so digestion in the small intestine is affected.

[2M] A student sees a four-chambered heart. Which groups may it belong to and why is this design advantageous?

→ It may belong to a bird or mammal. Complete separation prevents mixing of oxygenated and deoxygenated blood and supports the high energy demand needed to maintain constant body temperature.

[3M] How would you identify inhalation in a diagram of the thorax?

→ Look for ribs moving upward/outward and the diaphragm becoming flatter. These changes increase thoracic volume, reduce internal pressure and draw air into the lungs.

[3M] Why is transport essential even though oxygen can diffuse across alveoli?

→ Diffusion transfers gases only across the very thin alveolar surface. Long-distance movement through the body would be far too slow by diffusion alone; haemoglobin and blood circulation rapidly carry oxygen to every tissue.

14. Last-Page Diagram & One-Mark Checklist

Diagram / term	Labels or fact you must recall
Stomata	Guard cells, stomatal pore; gas exchange + transpiration.
Digestive system	Mouth, oesophagus, stomach, liver, pancreas, small/large intestine.
Alveolus	Thin moist wall, capillary network, O_2 in / CO_2 out.
Heart	RA, RV, LA, LV, septum, aorta, vena cava, pulmonary artery/vein.
Plant transport	Xylem: water/minerals; phloem: food/translocation.
Nephron	Glomerulus, Bowman's capsule, tubule, collecting duct.

- Pepsin digests proteins in the stomach; salivary amylase acts on starch in the mouth.
- Bile is produced by liver and stored in gall bladder; it contains no digestive enzyme.
- Villi increase surface area for absorption in the small intestine.
- Pulmonary artery: deoxygenated blood; pulmonary vein: oxygenated blood.
- Platelets clot blood; haemoglobin in RBCs carries oxygen.

- Ureter carries urine to bladder; urethra carries urine outside the body.

TOPPER RULE FOR LIFE PROCESSES

★ For 2–5 mark biology answers: name the structure/process → state mechanism in sequence → add function/reason → use correct biological keyword.

★ For diagram/case questions: identify what the diagram shows first, then connect each visible change to the physiological consequence.

CHAPTER 6 — CONTROL AND COORDINATION

Topper-style revision • answer-ready board language • 2026–27 scope audited

BOARD SCOPE — what you must be able to do

- ★ Explain nervous control in animals: receptors, neurons, nerve impulse, synapse, reflex action and reflex arc.
- ★ Identify functions of forebrain, midbrain, hindbrain, cerebellum, medulla and spinal cord.
- ★ Distinguish voluntary, involuntary and reflex actions; explain muscle response to nerve impulses.
- ★ Explain plant coordination: immediate touch response, tropic movements and plant hormones.
- ★ Explain endocrine coordination in animals, major hormones, feedback control and common deficiency effects.
- ★ Solve observation/case-based questions by connecting stimulus → coordinator → response.

1. Why Control & Coordination?

[1M] Define control and coordination

Control and coordination is the process by which different organs and systems of an organism work together and respond appropriately to changes in the environment.

Underline: stimulus • response • coordination

Key words

TERM	BOARD MEANING
Stimulus	A detectable change in internal or external environment.
Response	Reaction of an organism/body part to a stimulus.
Receptor	Specialised cell/organ that detects a stimulus.
Effector	Muscle or gland that carries out the response.

FAST LOGIC

Stimulus → receptor → coordinating system → effector → response

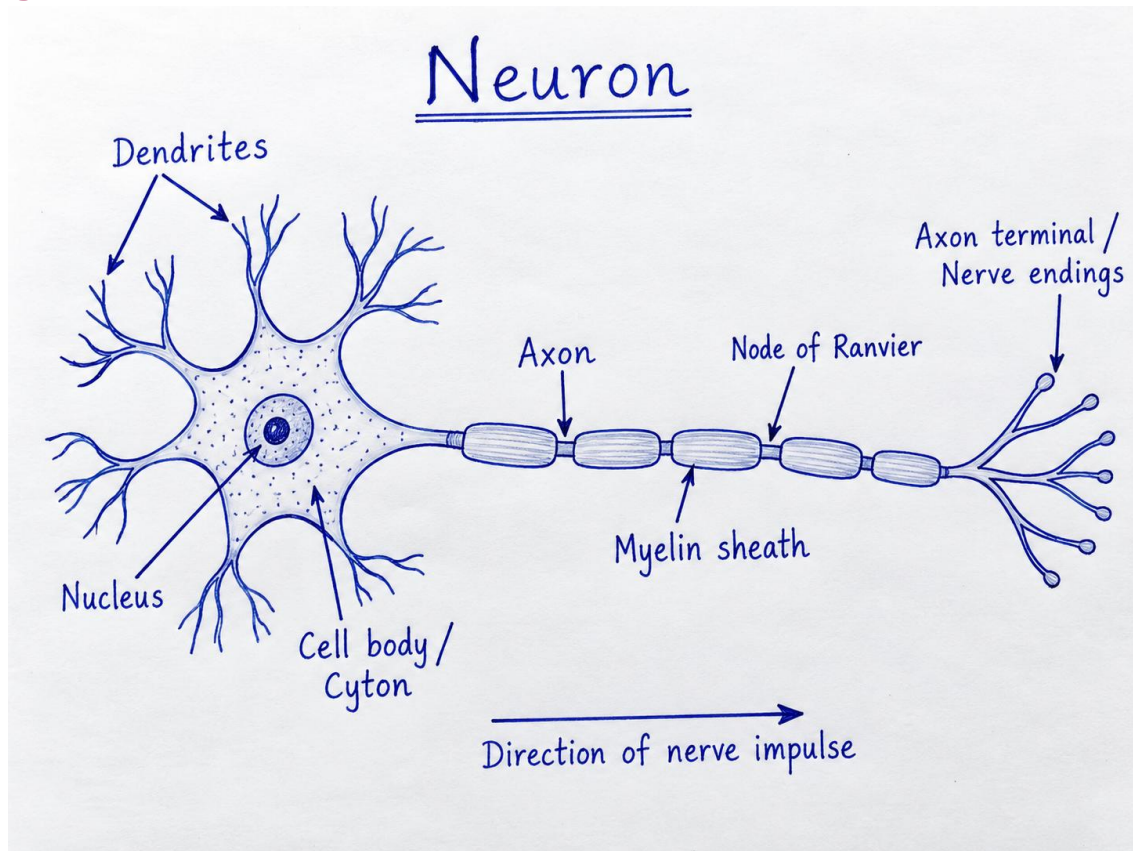
Animals mainly use nervous + hormonal coordination; plants mainly use chemical signals and changes in growth/turgor.

2. Nervous System — basic organisation

- Central nervous system (CNS): brain + spinal cord.
- Peripheral nervous system: nerves carrying information to/from CNS.
- Sense receptors: photoreceptors (eye), auditory receptors (ear), olfactory receptors (nose), gustatory receptors (tongue), touch/pressure receptors (skin).

3. Neuron — structural & functional unit

Exam Diagram — Labeled Neuron



Label-tip: Students should practice drawing 3–5 labels from memory for board diagrams.

[2M] Structure of a neuron

Dendrites receive information → cell body integrates it → axon carries the electrical impulse away → axon ending passes the signal to the next cell across a synapse.

Diagram labels: dendrite • cell body/cyton • nucleus • axon • nerve ending/synapse

How a nerve impulse travels

1. A stimulus causes a chemical/electrical change at the receptor or dendrite.
2. An electrical impulse travels along the neuron to the axon ending.
3. At the synapse, chemicals (neurotransmitters) are released.
4. These chemicals start a new electrical impulse in the next neuron/muscle cell.

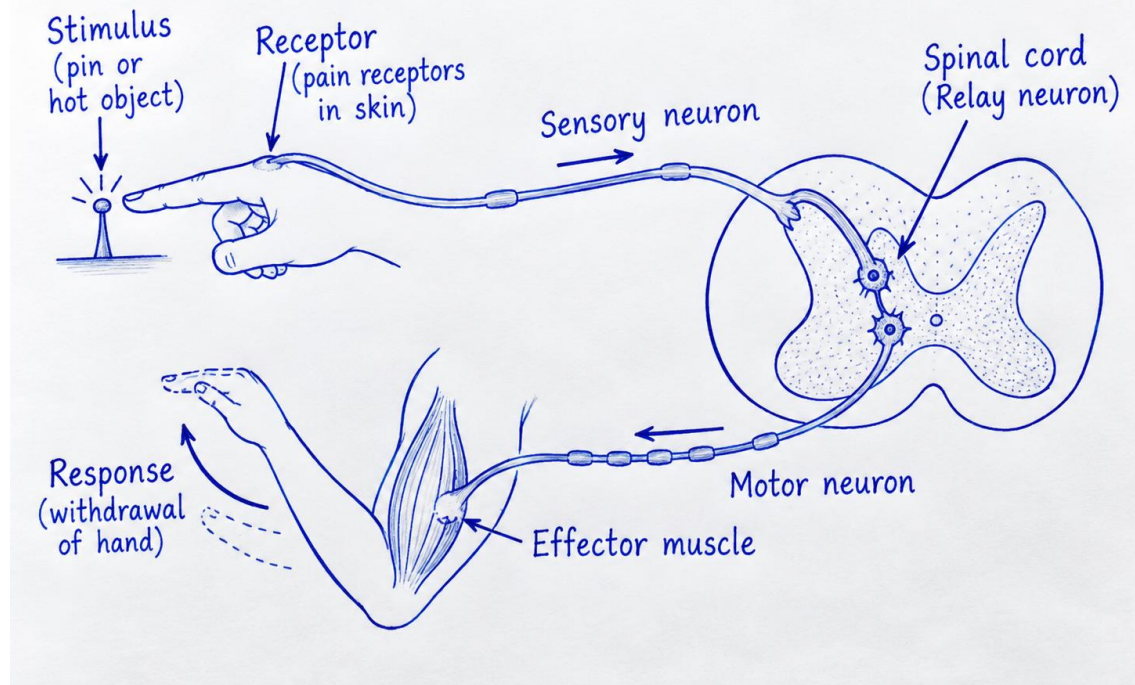
BOARD TRAP — electrical vs chemical

Along a neuron: mainly electrical signal. Across a synapse: chemical transmission. Do not write that the whole pathway is only electrical.

4. Reflex Action & Reflex Arc

Exam Diagram — Reflex Arc

Reflex Arc



Label-tip: Students should practice drawing 3–5 labels from memory for board diagrams.

[1M] Reflex action

A reflex action is a rapid, automatic and involuntary response to a stimulus, usually for protection, without waiting for conscious thought.

Receptor → sensory neuron → spinal cord/relay neuron → motor neuron → effector

[3M] Why is reflex action fast?

- ★ The spinal cord provides a short coordinating pathway, so the response can occur quickly.
- ★ The brain is informed of the event, but the immediate protective response does not wait for conscious decision.

Example: withdrawing the hand from a hot object.

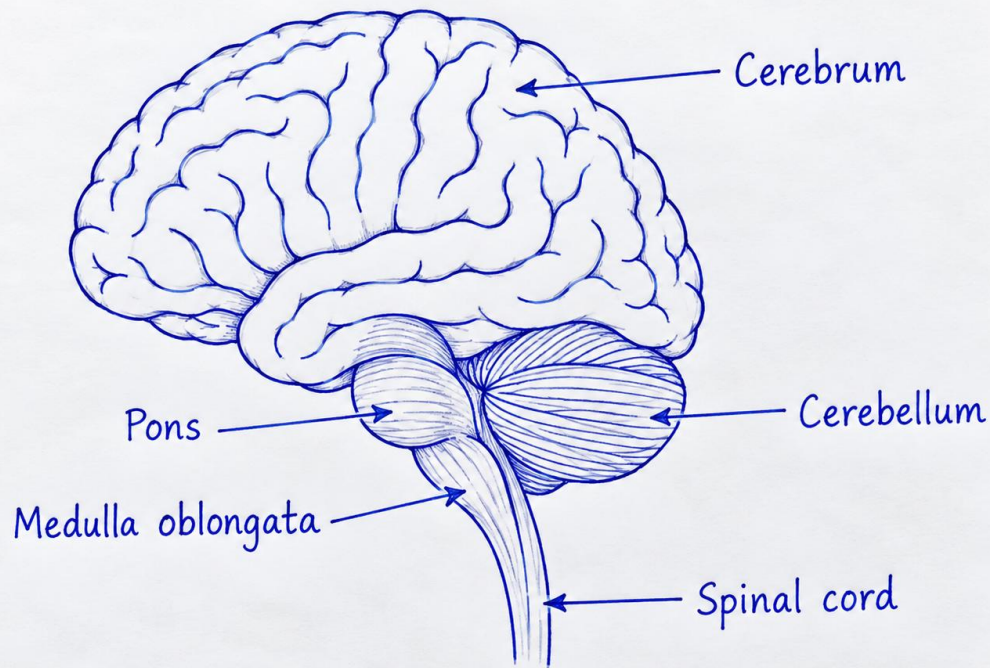
Reflex vs voluntary vs involuntary

TYPE	CONTROL	EXAMPLE
Voluntary	Conscious control; mainly forebrain	Writing, walking by choice
Involuntary	No conscious control; autonomic centres	Salivation, blood pressure, vomiting
Reflex	Rapid automatic protective response	Hand withdrawal, knee jerk

5. Human Brain — functions you must match

Exam Diagram — Human Brain

Human Brain



Label-tip: Students should practice drawing 3–5 labels from memory for board diagrams.

PART	MAIN BOARD FUNCTION
Forebrain / cerebrum	Thinking, memory, intelligence; receives/interprets sensory information; voluntary actions; centres associated with hunger/fullness.
Midbrain	Helps control some involuntary responses and relays information.
Hindbrain	Includes cerebellum, pons and medulla; important involuntary/coordination centres.
Cerebellum	Precision of voluntary actions, posture and balance.
Medulla	Involuntary functions such as blood pressure, salivation and vomiting.
Spinal cord	Carries impulses to/from brain; major centre for reflex actions.

2026-27 SAMPLE-PAPER ANGLE

Question: Which actions occur without thinking control and where are they controlled?

Answer frame: They are involuntary actions; control involves the midbrain and hindbrain. Examples: blood pressure, salivation, vomiting.

Protection of nervous tissue

- Brain is protected by the skull/cranium and cushioning fluid around it.
- Spinal cord is protected by the vertebral column.

6. How muscles produce movement

[3M] Nerve impulse → muscle contraction

When a nerve impulse reaches a muscle cell, special proteins in the muscle change their shape and arrangement.

This new arrangement makes the muscle fibre shorter, producing contraction and movement.

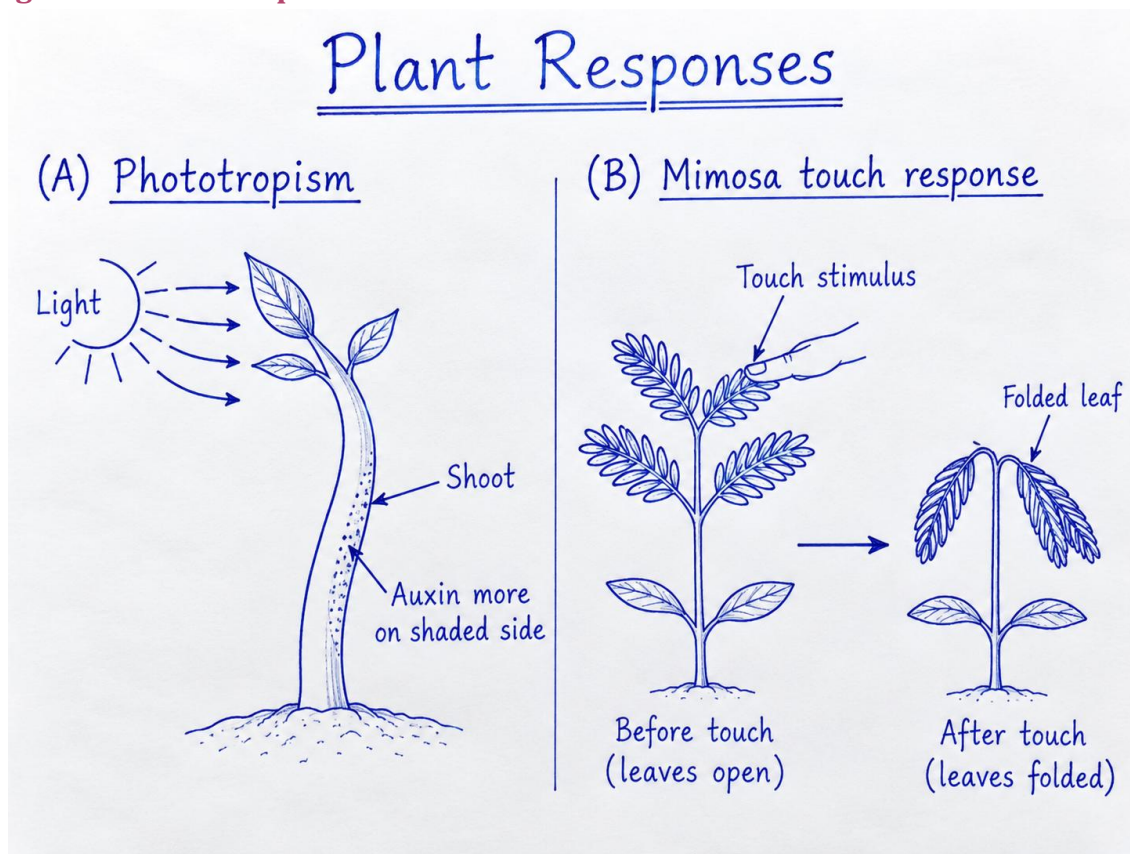
Exam phrase: nervous electrical impulse causes muscle proteins to rearrange → cell shortens.

Current board trigger

If the question asks “How does an electrical signal finally cause movement?”, do not stop at “motor neuron carries impulse”. Explain the protein rearrangement and shortening of muscle cells.

7. Coordination in Plants — no nerves, no muscles

Exam Diagram — Plant Responses



Label-tip: Students should practice drawing 3–5 labels from memory for board diagrams.

Two broad kinds of plant movement

1. Growth-independent / immediate movement — e.g. touch response in Mimosa (touch-me-not).
2. Growth-dependent directional movement — tropisms such as photo-, geo-, hydro-, chemo- and thigmotropism.

7.1 Touch response in Mimosa

[2M] How does Mimosa move without muscles?

Touch information is communicated from cell to cell by electrical-chemical means.

Certain cells change the amount of water in them; swelling/shrinking changes their shape, causing the leaf to fold.

Important: this is not growth movement.

Animal vs plant touch response

Animal: specialised neurons carry electrochemical impulses and muscles respond.

Plant: no nervous tissue; cells communicate signals and alter water content/turgor to change shape.

7.2 Tropic movements

TROPISM	STIMULUS	EXAMPLE
Phototropism	Light	Shoot bends towards light; roots generally away.
Geotropism / gravitropism	Gravity	Roots positive; shoots negative.
Hydrotropism	Water	Roots grow towards moisture.
Chemotropism	Chemical	Pollen tube grows towards ovule.
Thigmotropism	Touch/contact	Tendrils coil around a support.

Positive vs negative tropism

Growth towards stimulus = positive; growth away from stimulus = negative. Direction is judged relative to the stimulus, not “up/down”.

8. Plant Hormones

HORMONE	CORE ROLE
Auxin	Promotes cell elongation; important in shoot growth and phototropism.
Gibberellin	Promotes stem growth/elongation.
Cytokinin	Promotes cell division; abundant where rapid division occurs.
Abscissic acid (ABA)	Growth-inhibiting hormone; promotes dormancy/wilting-related responses and can help close stomata under water stress.

Phototropism — most asked mechanism

[3M] Why does a shoot bend towards light?

1. Light from one side causes auxin to accumulate more on the shaded side of the shoot.
2. Auxin promotes greater cell elongation on the shaded side.
3. Unequal growth bends the shoot towards light → positive phototropism.

Root caution

Auxin effects depend on organ and concentration. For board answers, use the standard shoot-phototropism explanation unless the question specifically asks about roots.

9. Chemical Coordination in Animals

[1M] Hormone

A hormone is a chemical messenger secreted in small amounts by an endocrine gland and carried by blood to a target organ/tissue where it produces a specific response.

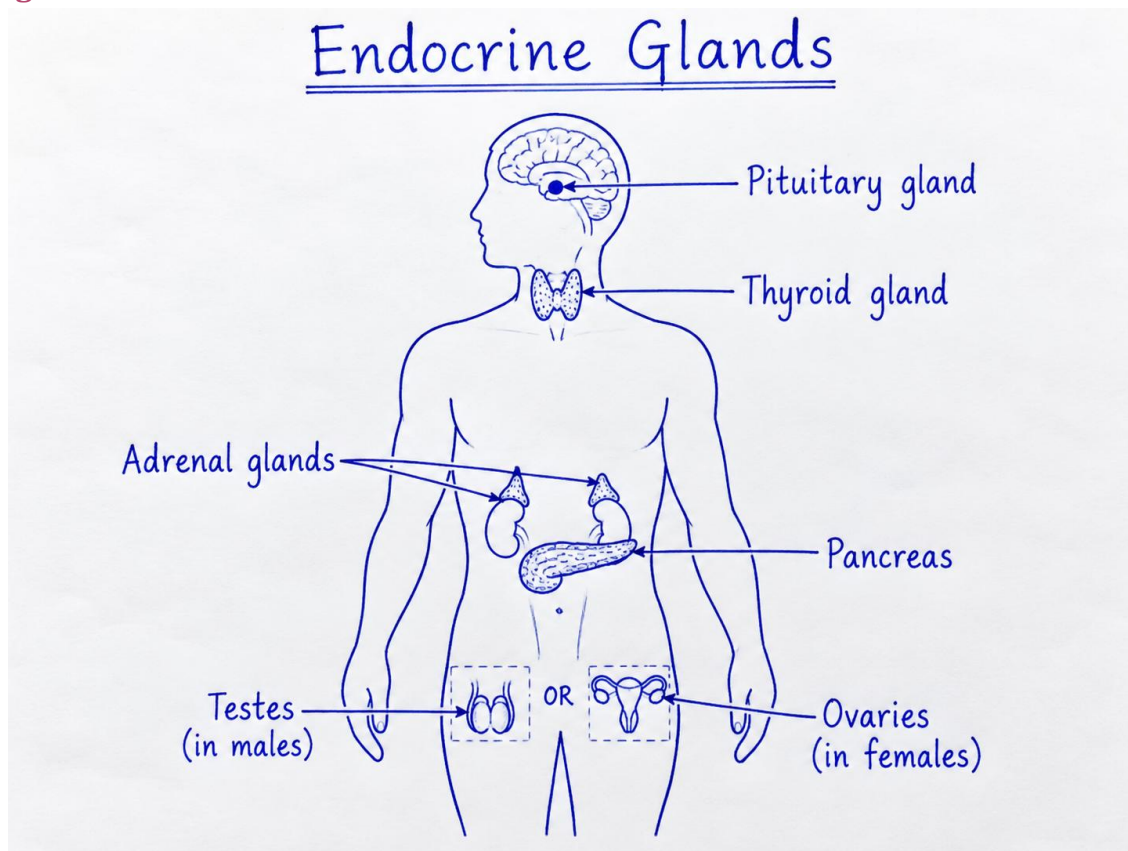
Nervous vs hormonal coordination

Nervous: fast, electrical/chemical, short-lived and targeted through neurons.

Hormonal: comparatively slower, chemical, carried in blood, often longer-lasting and may affect multiple target tissues.

10. Major Endocrine Glands & Hormones

Exam Diagram — Endocrine Glands



Label-tip: Students should practice drawing 3–5 labels from memory for board diagrams.

GLAND	HORMONE	BOARD FUNCTION / CLUE
Pituitary	Growth hormone	Controls body growth; pituitary is often called master gland because it influences other endocrine glands.
Thyroid	Thyroxine	Regulates metabolism; iodine is required for its synthesis. Iodine deficiency may cause goitre.
Adrenal	Adrenaline	Fight-or-flight: increases heart rate/breathing, diverts blood to muscles; prepares body for emergency.
Pancreas	Insulin	Helps regulate blood glucose; inadequate insulin can lead to diabetes.
Testes	Testosterone	Male secondary sexual characters and reproductive changes at puberty.
Ovaries	Oestrogen	Female secondary sexual characters and reproductive development.

[2M] Why is iodine important?

Iodine is needed to make thyroxine. Iodine deficiency reduces normal thyroxine production and can cause enlargement of the thyroid gland (goitre).

[2M] Adrenaline during emergency

Adrenaline increases heart rate and breathing rate, raises blood supply to skeletal muscles and prepares the body for rapid action.

11. Feedback Mechanism

[2M] Why is feedback needed?

Hormones must be maintained within suitable limits. The body senses the level/effect of a hormone and adjusts secretion accordingly.

Example: when blood glucose rises, insulin secretion increases; when glucose falls, insulin demand decreases.

12. Diagram / Labelling Checklist

Neuron — label these

Dendrite • cell body/cyton • nucleus • axon • nerve ending • synapse

Reflex arc — label these

Receptor in skin • sensory neuron • spinal cord/relay neuron • motor neuron • muscle/effector

Brain — label these

Forebrain/cerebrum • midbrain • cerebellum • medulla/hindbrain • spinal cord

Endocrine diagram — recognise positions

Pituitary (brain) • thyroid (neck) • adrenal (above kidneys) • pancreas (abdomen) • testes/ovaries

13. Mark-wise Answer Bank

[1M] Why are reflex actions useful?

They provide a rapid automatic response to potentially harmful stimuli, reducing reaction time.

[2M] Differentiate reflex and involuntary action

Reflex: sudden, rapid automatic response through a reflex arc, often protective.

Involuntary: routine body functions not under conscious control, e.g. salivation/blood pressure, controlled by brain centres/autonomic pathways.

[3M] Explain transmission across a synapse

Electrical impulse reaches axon ending → chemicals are released into synaptic gap → chemicals act on next cell membrane → a new electrical impulse is generated.

[3M] Explain phototropism in shoots

Auxin shifts/acts more on shaded side → shaded cells elongate more → unequal growth bends shoot towards light.

14. Competency / Case-based Thinking

Case 1 — hot object

Identify: stimulus = heat; receptor = skin; coordinator = spinal cord; effector = arm muscle; response = withdrawal. Then name it reflex action.

Case 2 — one-sided light on seedling

Predict shoot bends toward light. Reason: unequal auxin-mediated elongation on shaded side.

Case 3 — frightened student

Adrenaline released → heart/breathing rate rise → increased blood supply to muscles → body prepared for emergency.

Case 4 — neck swelling in iodine-poor diet

Link iodine deficiency → reduced thyroxine synthesis → enlarged thyroid/goitre.

15. Common Mistakes a Topper Avoids

- X Saying reflex action is “controlled by brain only” — spinal cord commonly coordinates the immediate reflex.
- X Saying all involuntary actions are reflexes — they are not the same category.
- X Writing signal is electrical across the synapse — transmission across synapse is chemical.
- X Saying plants have nerves/muscles — they use chemical signals, growth and turgor changes.
- X Confusing Mimosa folding with tropism — it is a rapid growth-independent response.
- X Writing auxin “moves plant toward light” — unequal cell elongation causes bending.
- X Confusing insulin with digestive enzyme — insulin is a hormone controlling blood glucose.
- X Forgetting target/effect when naming a hormone.

16. 5-Minute Final Revision Grid

TRIGGER	MUST WRITE
Neuron	dendrite → cell body → axon → synapse
Reflex	receptor → sensory → spinal cord → motor → effector
Forebrain	thinking + sensory interpretation + voluntary action
Cerebellum	balance + precision
Medulla	involuntary actions
Mimosa	electrochemical signal + water/turgor change
Phototropism	auxin + unequal elongation
Adrenaline	fight-or-flight
Thyroxine	iodine + metabolism
Insulin	blood glucose control

17. 2026–27 Board Drill

Q1. Give two involuntary actions and name the brain regions involved.

Blood pressure / salivation / vomiting; involuntary control involves midbrain and hindbrain.

Q2. Compare touch information in animals and plants.

Animals transmit information through electrochemical nerve impulses; plants transmit cell-to-cell signals and alter water content/turgor to produce movement.

Q3. How does a nerve impulse make a muscle move?

It causes special muscle proteins to change shape/arrangement, shortening the cell and producing contraction.

CHAPTER 7 — HOW DO ORGANISMS REPRODUCE?

Topper-style revision • 2026–27 CBSE scope audited • answer-ready + board-drawable diagrams

BOARD SCOPE — nothing important left out

- ★ Why reproduction is necessary for continuity of species; DNA copying and the role of variation.
- ★ Asexual reproduction: fission, fragmentation, regeneration, budding, vegetative propagation and spore formation.
- ★ Sexual reproduction in flowering plants: flower parts, pollination, fertilisation, seed/fruit formation and germination.
- ★ Human reproduction: puberty, male and female reproductive systems, fertilisation, implantation, placenta, menstruation and childbirth.
- ★ Reproductive health: contraception/family planning, safe sex, STIs including HIV/AIDS, and women's health during childbearing.
- ★ Competency patterns: emasculation + bagging, vasectomy/tubectomy logic, chromosome-number questions and post-implantation changes.

1. Why Do Organisms Reproduce?

[1M] Define reproduction.

→ Reproduction is the biological process by which living organisms produce new individuals of their own kind.

Underline: new individuals • continuity of species

- Reproduction is not essential for survival of an individual, but it is essential for continuation of a species.
- During reproduction, DNA is copied. DNA copying is highly accurate but not perfectly exact, so small variations can arise.
- Variations may help some individuals survive environmental changes; therefore they are important for long-term survival of a species.

DNA COPY → VARIATION → SURVIVAL

- ★ Exact copies preserve the basic body design of a species.
- ★ Small copying errors create variation.
- ★ Variation becomes especially useful when conditions change.

2. Asexual vs Sexual Reproduction

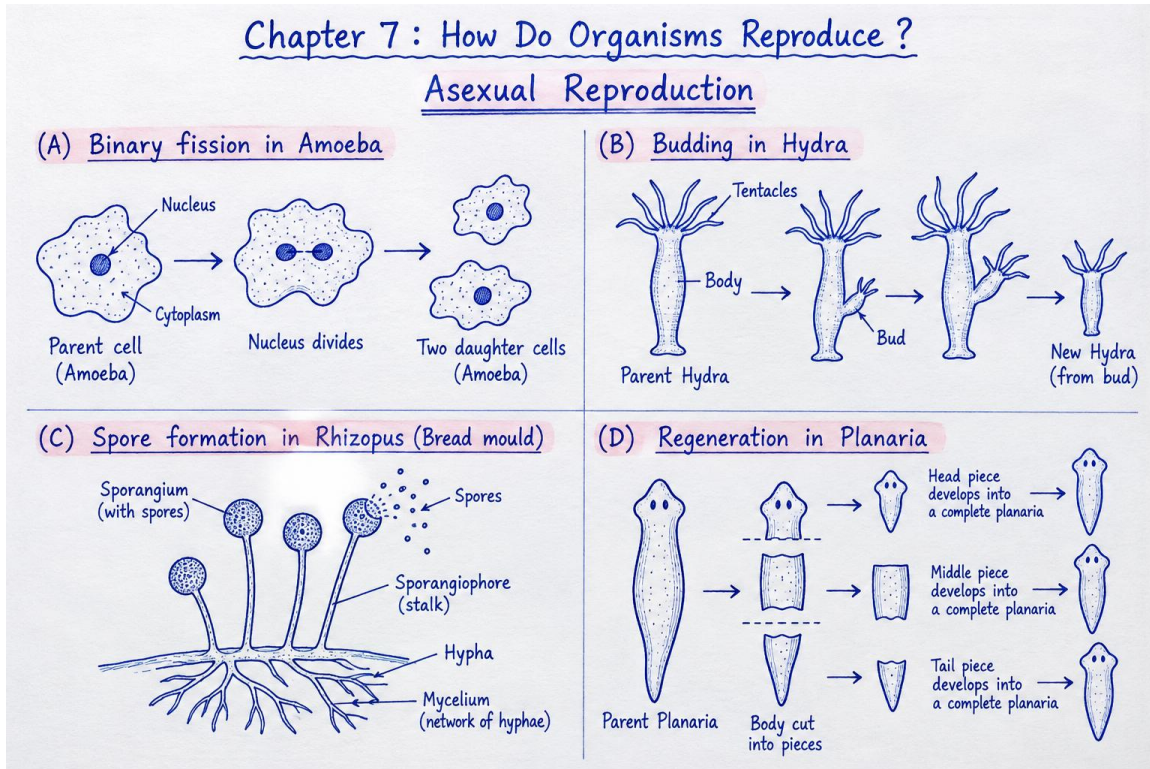
ASEXUAL	SEXUAL
One parent usually involved	Two parents / two gametes involved
No fusion of gametes	Male and female gametes fuse
Offspring are very similar to parent; little variation	Offspring show greater variation
Fast, simple; common in unicellular/simple organisms	More complex; common in higher plants and animals

3. Asexual Reproduction — learn method + example

3.1 Fission

- Binary fission: one parent cell divides into two daughter cells. Example: Amoeba.
- Leishmania divides in a definite plane because of its elongated body and flagellum.
- Multiple fission: one parent gives rise to many daughter cells. Example: Plasmodium.

Handwritten Board Diagram — Asexual Reproduction



Practice these simple line diagrams; label only what is asked and keep arrows neat.

3.2 Fragmentation — Spirogyra

- The filament breaks into two or more fragments; each fragment grows into a new individual.
- Works well in organisms with a simple body organisation.

3.3 Regeneration — Planaria

- If the body is cut into pieces, specialised cells proliferate and differentiate so each piece can form a complete organism.

Regeneration ≠ ordinary reproduction in all multicellular organisms

★ Complex multicellular organisms cannot generally reproduce simply by being cut into pieces because their body cells are highly specialised and organised.

3.4 Budding — Hydra / Yeast

- A small outgrowth called a bud develops because of repeated cell division, grows and separates as a new organism.

3.5 Spore formation — Rhizopus

- Spores are produced inside sporangia. They have thick protective walls and can survive unfavourable conditions.

- When conditions become suitable, spores germinate and form new individuals.

4. Vegetative Propagation in Plants

[2M] What is vegetative propagation?

→ It is a type of asexual reproduction in which new plants arise from vegetative parts such as root, stem or leaf.

EXAMPLE	VEGETATIVE PART / METHOD
Potato	Stem tuber / eyes
Ginger	Underground stem / rhizome
Onion	Bulb
Bryophyllum	Leaf buds
Sugarcane / rose / grapes	Stem cuttings

Advantages — common 2M reasoning

- Produces plants genetically similar to the parent, so desirable characters are preserved.
- Plants can be produced faster; many may flower and fruit earlier than seed-grown plants.
- Useful for plants that produce few/non-viable seeds or are conveniently propagated vegetatively.

Artificial methods / competency angle

- Cutting, layering and grafting are common horticultural methods.
- Tissue culture (micropropagation) can rapidly produce many disease-free plants from small pieces of plant tissue under sterile conditions.

[2M] Why does vegetative propagation produce little variation?

- Only one parent contributes genetic material and there is no fusion/recombination of gametes.
- New plants therefore remain genetically very similar to the parent.

5. Why Sexual Reproduction Produces More Variation

- Male and female gametes are formed with half the chromosome number (haploid).
- Fusion of two gametes restores the normal chromosome number in the zygote (diploid).
- DNA from two parents is combined, creating new gene combinations and greater variation.

Chromosome shortcut

★ If a gamete has n chromosomes, the other gamete of the same species also has n ; ordinary body cells have $2n$.

★ 2026–27 sample-paper pattern: male gamete 17 → female gamete 17 → adult plant cell 34.

6. Sexual Reproduction in Flowering Plants

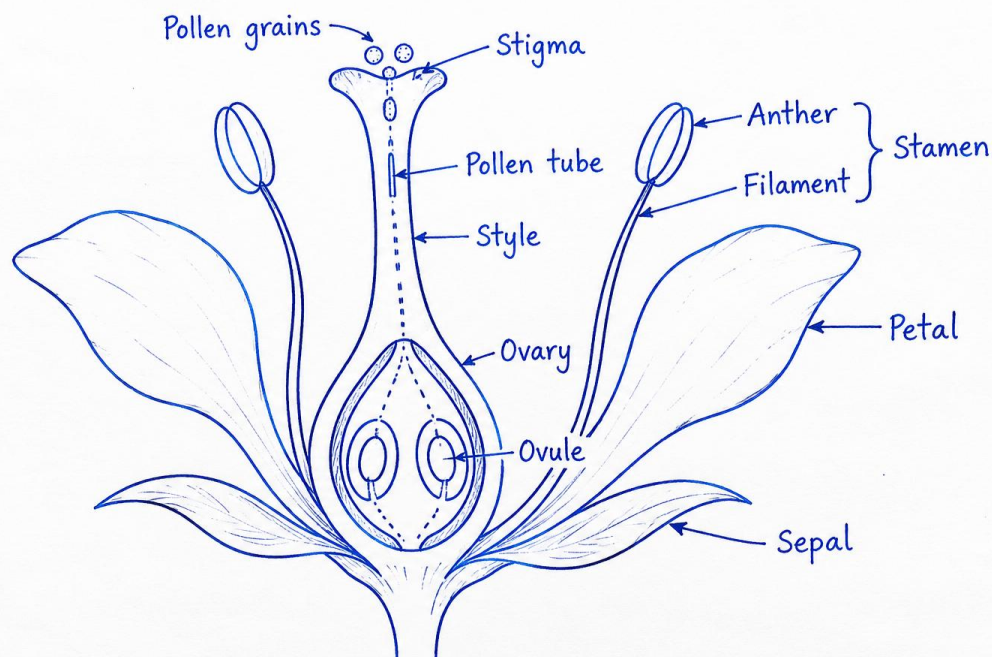
6.1 Reproductive parts of a flower

PART	FUNCTION
Stamen (male)	Anther + filament; anther produces pollen grains containing male gametes.
Pistil / carpel (female)	Stigma + style + ovary; ovary contains ovules with female gamete.
Petal / sepal	Help attract/protect; not the main reproductive organs.

- Bisexual flower: both stamen and pistil present. Unisexual flower: only one reproductive sex present.

Handwritten Board Diagram — L.S. of a Flower

L.S. of a Flower



Must-practice labels: stigma, style, ovary, ovule, anther, filament, petal and sepal.

6.2 Pollination

[1M] Define pollination.

→ Transfer of pollen grains from anther to stigma of a flower is called pollination.

TYPE	MEANING
Self-pollination	Pollen reaches stigma of the same flower or another flower on the same plant.
Cross-pollination	Pollen is transferred to a flower on another plant of the same species; agents include wind, water and animals/insects.

6.3 Fertilisation — stepwise board flow

- Pollen grain germinates on stigma and forms a pollen tube through the style.
- Male gamete travels through the pollen tube to the ovule.
- Male gamete fuses with the female gamete → zygote. This fusion is fertilisation.
- Zygote develops into embryo; ovule becomes seed; ovary develops into fruit.
- Seed germinates under suitable conditions and forms a new plant.

2026-27 SAMPLE-PAPER ANGLE — emasculation + bagging

- ★ If anthers are removed before maturity (emasculation), self-pollination is prevented.
- ★ If the flower is then bagged so outside pollen cannot reach the stigma, fertilisation will not occur; therefore seeds and fruits will not form.
- ★ In artificial hybridisation, desired pollen can later be introduced before re-bagging.

[3M] Differentiate pollination and fertilisation.

→ Pollination is transfer of pollen from anther to stigma.

- Fertilisation is fusion of male and female gametes, usually inside the ovule.

- Pollination occurs before fertilisation and enables the male gamete to reach the female gamete.

7. Reproduction in Human Beings — Puberty

- Sexual maturation occurs during adolescence. Reproductive organs mature and secondary sexual characters appear.

COMMON PUBERTY CHANGES

Boys

Facial/body hair, deeper voice, shoulders broaden; testes begin producing sperm; testosterone is important.

Girls

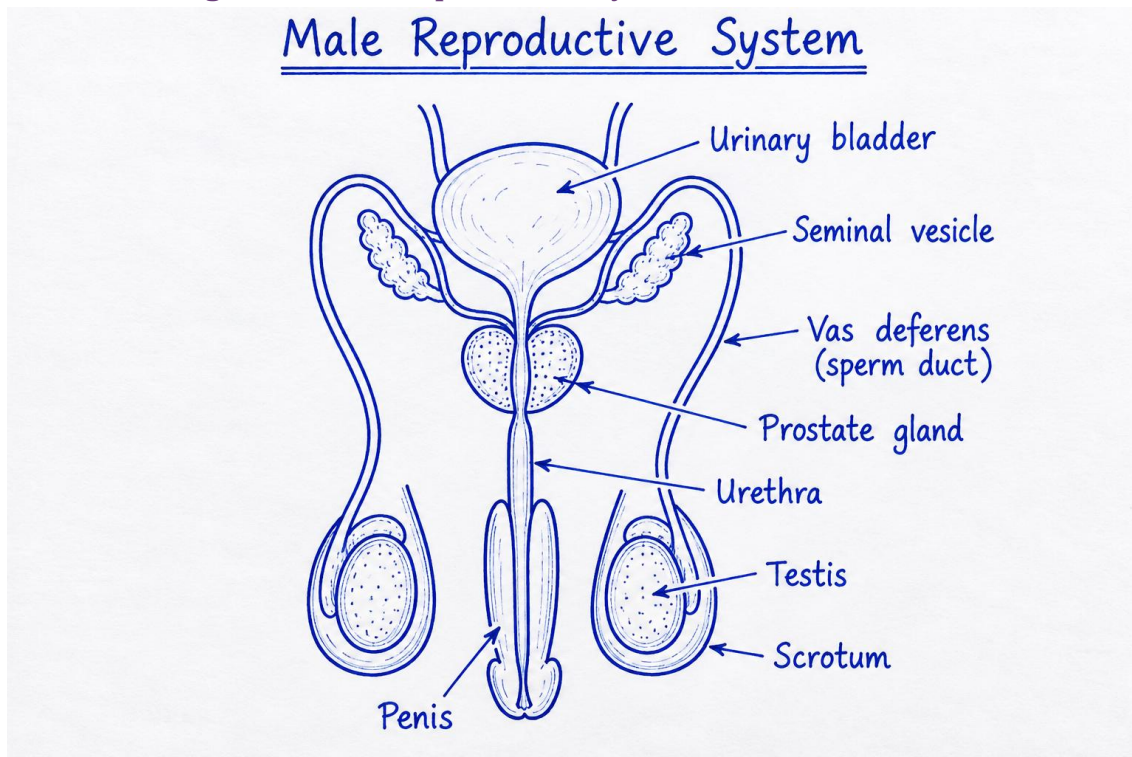
Breasts develop, body hair appears; ovaries begin releasing ova; menstruation begins; oestrogen/progesterone are important.

Exam caution

- ★ Puberty means attaining sexual maturity; it is not merely increase in height or body size.
- ★ Gametes are haploid; normal body cells are diploid.

8. Male Reproductive System

Handwritten Board Diagram — Male Reproductive System



Board labels: testis, scrotum, vas deferens, seminal vesicle, prostate gland, urethra and penis.

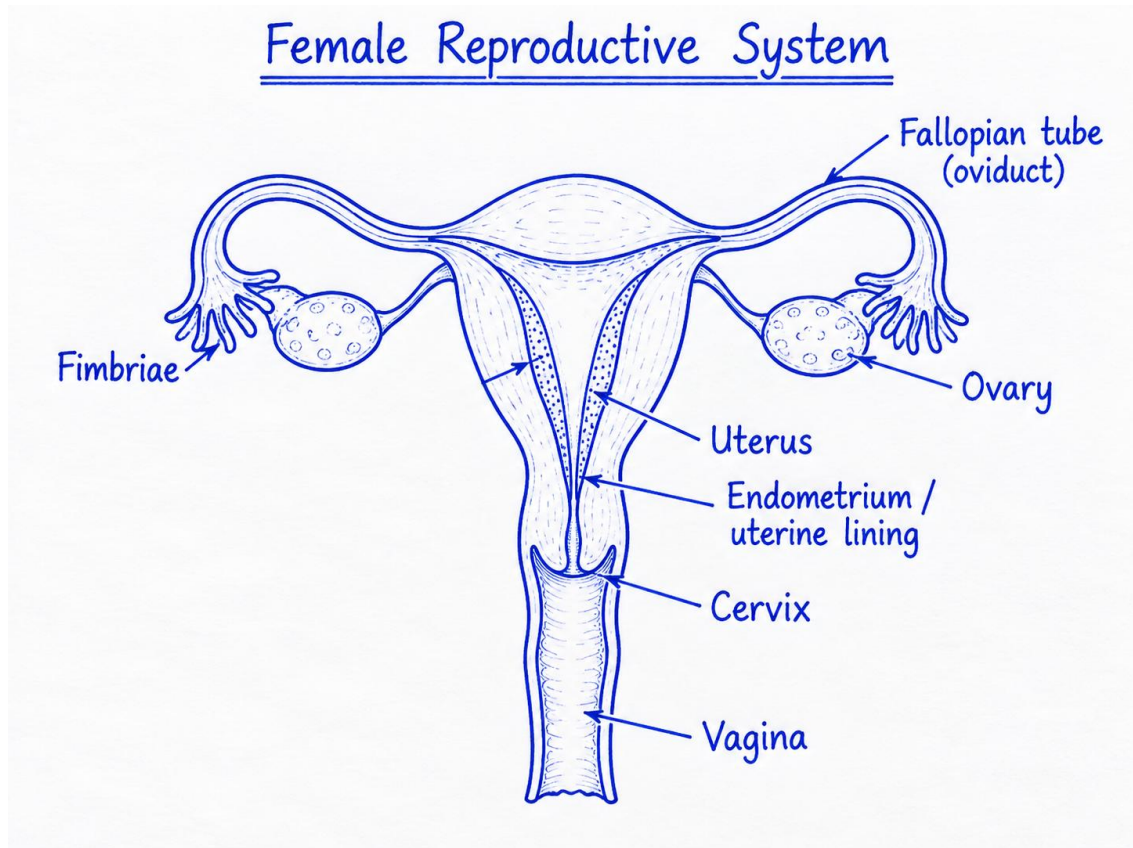
PART	FUNCTION
Testes	Produce sperms and testosterone.
Scrotum	Keeps testes at a temperature lower than body temperature, suitable for sperm formation.
Vas deferens	Carries sperms from testes.
Seminal vesicles + prostate	Add secretions that nourish sperms and form semen.

Urethra / penis

Passage through which semen is released.

9. Female Reproductive System

Handwritten Board Diagram — Female Reproductive System



Board labels: ovaries, oviducts/fallopian tubes, uterus, cervix and vagina.

PART	FUNCTION
Ovaries	Produce ova and female sex hormones.
Oviduct / fallopian tube	Receives ovum; usual site of fertilisation.
Uterus	Embryo implants and develops here.
Cervix	Narrow lower part of uterus.
Vagina	Passage opening to the outside; receives sperms and acts as birth canal.

9.1 Fertilisation → implantation → development

- Usually one mature ovum is released periodically from an ovary (ovulation).
- If sperm meets the ovum in the oviduct, fertilisation forms a zygote.
- Zygote divides repeatedly to form an embryo while moving toward the uterus.
- Embryo gets embedded in the thick uterine lining — implantation.

[2M] What changes occur in the uterus after implantation?

- The uterine lining remains thick and richly supplied with blood vessels to nourish the embryo.
- A placental connection develops between the uterus and embryo.

Underline: thick vascular uterine lining • placenta

10. Placenta, Menstruation & Childbirth

10.1 Placenta — exchange organ

- Placenta provides a large surface for exchange of nutrients and oxygen from mother to embryo/foetus.
- Waste substances from the developing child pass back into the mother's blood for removal.
- Maternal and foetal blood do not normally mix directly; materials are exchanged across placental surfaces.

10.2 If fertilisation does not occur — menstruation

- The thickened uterine lining is no longer required.
- It breaks down and is discharged along with blood and the unfertilised ovum — menstruation.
- The menstrual cycle is roughly about 28 days in many individuals, but normal cycle length can vary.

10.3 Pregnancy and childbirth

- After implantation, embryo develops into a foetus; human pregnancy lasts about nine months.
- At the end of gestation, rhythmic contractions of uterine muscles help in childbirth.

Sequence you should be able to write without looking

★ Ovulation → sperm meets ovum in oviduct → fertilisation → zygote → repeated divisions → embryo → implantation → placenta develops → foetal growth → childbirth.

11. Reproductive Health & Family Planning

Why family planning is needed

- Allows spacing/limiting of pregnancies and reduces health risks related to unwanted or very frequent pregnancies.
- Supports informed reproductive choices and maternal/child health.

Methods of contraception — know mechanism

TYPE	EXAMPLE	HOW IT WORKS
Barrier	Condom	Prevents sperm from reaching ovum; also reduces risk of many sexually transmitted infections.
Hormonal / chemical	Oral contraceptive pills	Alter hormonal control of ovulation/fertilisation; must be used appropriately under medical guidance.
IUCD	Copper-T	Placed in uterus; reduces likelihood of fertilisation/implantation.
Surgical	Vasectomy	Vas deferens is cut/blocked; sperm cannot reach semen pathway.
Surgical	Tubectomy	Oviducts are cut/blocked; sperm and ovum cannot meet.

2026–27 SAMPLE-PAPER TRAP — vasectomy vs tubectomy

- ★ Neither operation stops production of gametes or sex hormones.
- ★ Both prevent the successful meeting/fusion of male and female gametes by blocking their transport routes.

11.1 Safe sex, STIs & HIV/AIDS

- Sexually transmitted infections can spread through sexual contact; examples include gonorrhoea and syphilis (bacterial) and HIV/AIDS or genital warts (viral).
- Condom use lowers the risk of many STIs as well as unwanted pregnancy, though no method is 100% effective in all circumstances.
- HIV attacks the immune system; AIDS is the advanced disease condition caused by HIV infection.
- Safe practices, accurate health information, testing where appropriate, and medical care are important.

11.2 Childbearing & women's health

- Pregnancy places nutritional and physiological demands on the mother; adequate nutrition, rest and antenatal medical care are important.
- Avoiding very early, unwanted or closely spaced pregnancies helps protect maternal and child health.

12. Mark-wise Answer Bank — write to the marks

[1M] Why is reproduction important?

→ It ensures continuity of a species by producing new individuals.

[2M] Why are variations useful?

→ DNA copying is not perfectly exact, so small variations arise.

- If environmental conditions change, some variations may help individuals survive and reproduce.

[2M] Why are spores suited for survival?

→ Spores have thick protective walls and can remain dormant during unfavourable conditions.

- They germinate when conditions become suitable.

[2M] Why are testes present in the scrotum?

→ Sperm formation requires a temperature slightly lower than normal body temperature.

- The scrotum helps maintain this lower temperature.

[2M] What is the role of placenta?

→ It permits transfer of oxygen and nutrients from the mother to the developing embryo/foetus.

- It also allows wastes from the foetus to pass into the mother's blood for removal.

[3M] Explain what happens if the ovum is not fertilised.

→ The ovum does not form a zygote.

- The thick uterine lining is not required and starts breaking down.
- The lining, blood and unfertilised ovum are discharged as menstruation.

[3M] Compare asexual and sexual reproduction.

→ Asexual reproduction usually involves one parent and no gamete fusion; sexual reproduction involves gamete fusion.

- Asexual offspring are very similar to the parent; sexual reproduction produces greater variation.
- Asexual reproduction is usually faster/simpler, while sexual reproduction involves specialised reproductive organs/processes.

13. Competency / Case-based Thinking

Use this 4-step method

- ★ Identify the reproductive process / organ / event described.
- ★ Track the route: pollen or gamete → transport structure → site of fertilisation → next stage.
- ★ Apply chromosome logic: gamete = n ; zygote/body cell = $2n$.
- ★ For contraception questions, ask: Is gamete production stopped, or only gamete transport/fusion prevented?

14. 2026–27 Board Drill — current sample-paper patterns

[1M] Anthers of a mustard flower are removed before maturity and the bud is bagged. What happens?

- Self-pollination is prevented by removing anthers, and outside pollen is excluded by bagging.
- Therefore fertilisation does not occur and seeds/fruits do not form.

[1M] What common event is prevented by vasectomy and tubectomy?

- The successful meeting and fusion of the male and female gametes is prevented.

[1M] If a plant gamete has 17 chromosomes, how many are in the other gamete and adult plant cell?

- Other gamete = 17; adult body cell = 34.

[2M] State two uterine changes after implantation.

- Uterine lining stays thick and richly supplied with blood vessels.
- Placenta develops between the uterus and embryo.

15. Common Mistakes a Topper Avoids

- X Writing reproduction is necessary for survival of an individual — it is needed for continuity of the species.
- X Calling pollination fertilisation — pollination is pollen transfer; fertilisation is gamete fusion.
- X Writing ovary becomes seed — ovule becomes seed; ovary becomes fruit.
- X Saying vasectomy stops sperm production — it blocks the vas deferens; testes can still produce sperm and testosterone.
- X Saying tubectomy stops ovulation — it blocks oviducts; ovaries continue producing ova/hormones.
- X Forgetting that fertilisation usually occurs in the oviduct, while implantation occurs in the uterus.
- X Writing all asexual offspring are absolutely identical — DNA copying can still introduce minor variation.
- X Confusing regeneration with fragmentation: regeneration rebuilds a whole body from pieces using specialised cells; fragmentation is simple breaking into fragments that grow.

16. 5-Minute Final Revision Grid

TRIGGER	MUST WRITE
Reproduction	Continuity of species; DNA copying; variation
Fission	Amoeba / Leishmania; multiple fission — Plasmodium
Fragmentation	Spirogyra
Regeneration	Planaria

Budding	Hydra / yeast
Spore formation	Rhizopus; protective spores
Vegetative propagation	Potato, Bryophyllum, sugarcane/rose; tissue culture
Flower	Stamen + pistil; pollination → fertilisation → seed/fruit
Male system	Testis → vas deferens → urethra; glands add seminal fluid
Female system	Ovary → oviduct (fertilisation) → uterus (implantation)
Contraception	Condom, pills, IUCD, vasectomy, tubectomy
Reproductive health	Safe sex, STI prevention, maternal health

TOPPER RULE

★ For a 2–3 mark reproduction answer, write the process in sequence and name the exact organ/site.

★ For diagrams: use pencil/blue-pen style, keep labels horizontal, avoid crossing label lines, and label only requested structures.

CHAPTER 8 — HEREDITY

BOARD SCOPE — exact 2026–27 summative boundary

- ★ Heredity: how traits are transmitted from parents to offspring.
- ★ Mendel's contribution and laws of inheritance of traits — monohybrid and dihybrid reasoning are high priority.
- ★ Sex determination in humans — brief introduction using XX/XY chromosomes.
- ★ Evolution topics (acquired/inherited traits comparison, speciation, fossils, evolutionary relationships, human evolution, etc.) are formative-only and are NOT part of summative board assessment for 2026–27.

1. Heredity — the basic idea

[1M] Define heredity.

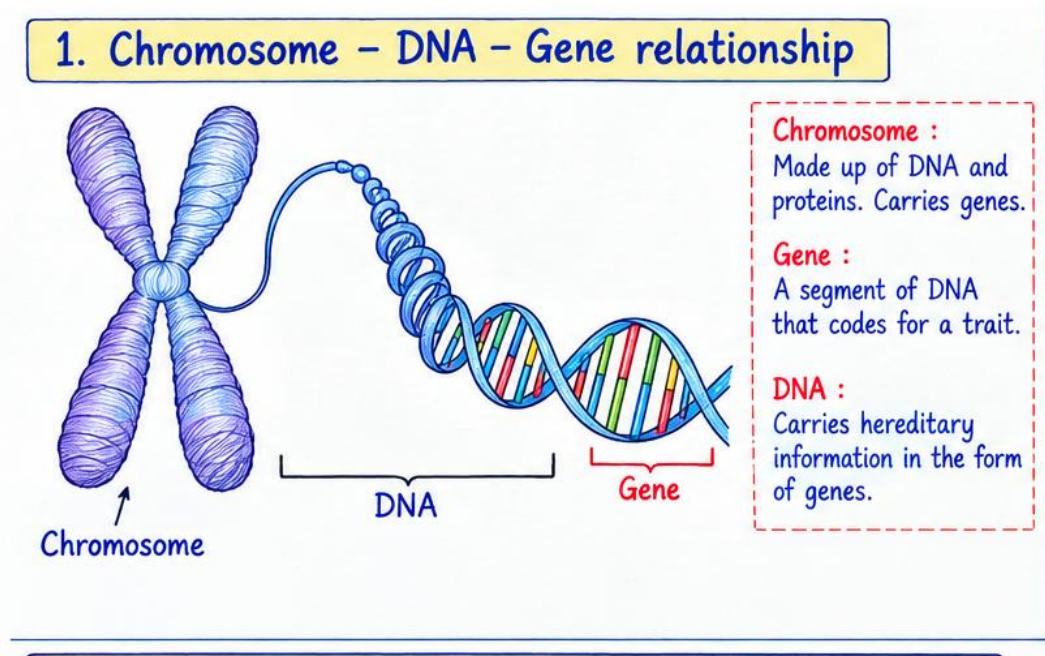
→ Heredity is the transmission of traits from parents to their offspring through genetic material.

Underline: traits • parents → offspring • genetic material

- A trait is a distinguishable feature of an organism, such as plant height, flower colour or seed shape.
- Offspring resemble parents because they inherit genes, but they are not always identical because new combinations of genes and DNA changes can create variation.

1.1 DNA → gene → chromosome → trait

Handwritten Concept Diagram — Chromosome, DNA & Gene



Core link: chromosomes contain DNA; a gene is a segment of DNA that carries information affecting a trait.

How does a gene influence a trait?

- ★ Genes contain instructions for making proteins or controlling cellular processes.
- ★ Differences in gene/allele information can change the amount or form of a protein and therefore alter the visible trait.
- ★ A trait may be influenced by one or more genes, but Class 10 Mendelian questions mainly use simple single-gene examples.

2. Essential Genetics Vocabulary — do not confuse these

TERM	BOARD MEANING
Gene	A segment of DNA carrying information for a character/trait.
Alleles	Alternative forms of the same gene, e.g. T and t.
Genotype	Genetic combination, e.g. TT, Tt or tt.
Phenotype	Observable expression, e.g. tall or dwarf.
Homozygous / pure	Both alleles same: TT or tt.
Heterozygous / hybrid	Alleles different: Tt.
Dominant allele	Expresses even when only one copy is present, e.g. T in Tt.
Recessive allele	Expresses only when dominant allele is absent, e.g. tt.
P generation	Original parental generation used in a cross.
F ₁ / F ₂	First and second filial generations.
Gamete	Haploid reproductive cell carrying one allele for each gene considered.

Exam traps

- ★ Dominant does NOT mean stronger, better, more common or more useful.
- ★ Recessive does NOT mean weak. It is simply masked in a heterozygote.
- ★ Phenotype and genotype are not the same: TT and Tt have different genotypes but both can be tall.

3. Mendel's Contribution — why pea plants?

- Gregor Mendel studied inheritance by performing controlled crosses on garden pea plants.
- Pea plants have clear contrasting traits, short generation time, many offspring and can self-pollinate as well as be cross-pollinated deliberately.
- He used true-breeding (pure) parents so the starting genotypes were known.

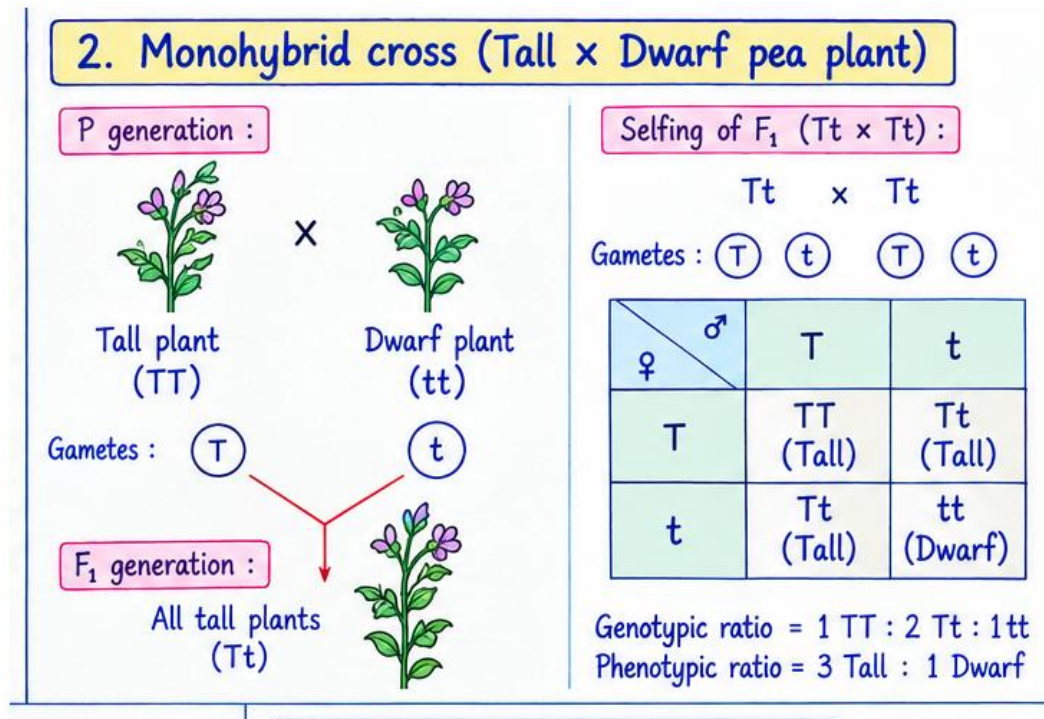
Seven contrasting characters often linked with Mendel

CHARACTER	DOMINANT EXAMPLE	RECESSIVE EXAMPLE
Seed shape	Round	Wrinkled
Seed colour	Yellow	Green
Flower colour	Violet	White

Pod shape	Inflated	Constricted
Pod colour	Green	Yellow
Flower position	Axial	Terminal
Stem height	Tall	Dwarf

4. Monohybrid Cross — one contrasting character

Handwritten Board Diagram — Monohybrid Cross



Board target: P → gametes → F₁ → selfing → Punnett square → F₂ ratios.

[2M] What happens when pure tall (TT) is crossed with pure dwarf (tt)?

- All F₁ offspring receive T from tall parent and t from dwarf parent, so all are Tt.
- All F₁ plants are tall because T is dominant over t.

Underline: TT × tt → all Tt • 100% tall

[2M] What does F₂ show after selfing Tt × Tt?

- Genotypes: TT, Tt, Tt, tt → 1 : 2 : 1.
- Phenotypes: 3 tall : 1 dwarf → 75% tall and 25% dwarf.

Underline: genotype 1:2:1 • phenotype 3:1

Why there is NO “medium-height” plant in Mendel’s simple cross

- ★ Inheritance is particulate, not blending. The dominant allele T masks t in Tt.
- ★ Therefore Tt is tall, not intermediate between tall and dwarf.

5. Mendel's Laws — state + connect to the cross

5.1 Law of Dominance

- When two unlike alleles occur together in a heterozygote, one may express itself (dominant) while the other is masked (recessive).
- Example: Tt is tall because T is dominant over t.

5.2 Law of Segregation / Purity of Gametes

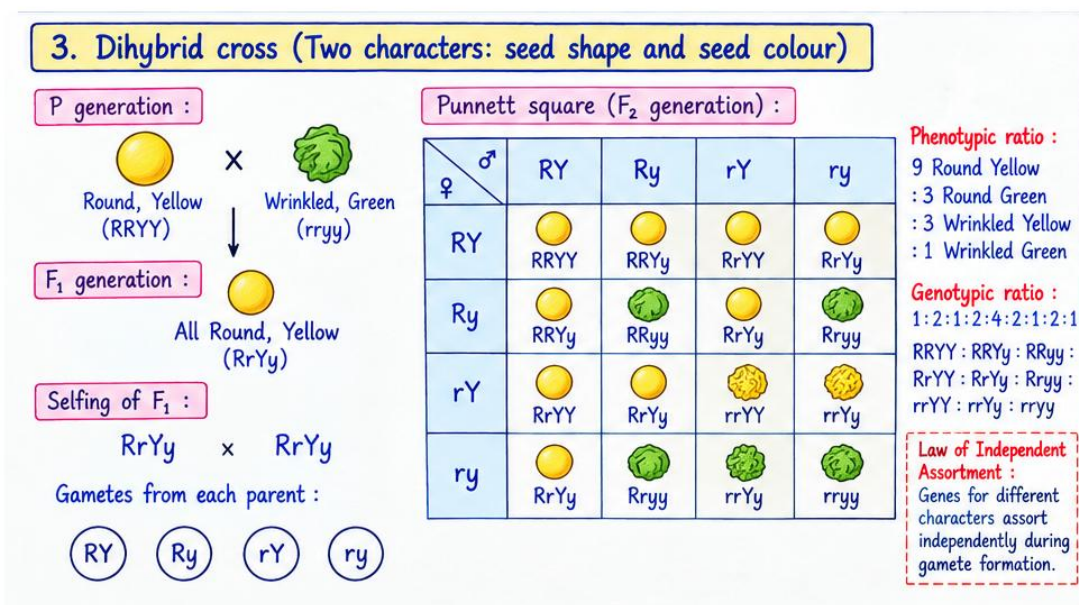
- The two alleles of a gene separate during gamete formation, so each gamete receives only one allele.
- A Tt plant therefore makes two kinds of gametes: T and t.

5.3 Law of Independent Assortment

- For two different genes considered together, allele pairs assort independently into gametes in Mendel's dihybrid model.
- An RrYy plant can form four gamete types: RY, Ry, rY and ry.

6. Dihybrid Cross — two characters together

Handwritten Board Diagram — Dihybrid Cross



Classic F₂ phenotypic ratio: 9 : 3 : 3 : 1.

The ratio you must know

- ★ RrYy × RrYy → F₂ phenotypes = 9 dominant-dominant : 3 dominant-recessive : 3 recessive-dominant : 1 recessive-recessive.
- ★ This ratio supports independent assortment of the two character pairs.

7. Probability Shortcuts — fastest way to solve board questions

OUTCOME	FRACTION	PERCENT
Monohybrid dominant phenotype in F ₂	3/4	75%

Monohybrid recessive phenotype in F_2	$1/4$	25%
Two dominant phenotypes in dihybrid F_2	$3/4 \times 3/4 = 9/16$	56.25%
Dominant first + recessive second	$3/4 \times 1/4 = 3/16$	18.75%
Recessive first + dominant second	$1/4 \times 3/4 = 3/16$	18.75%
Both recessive	$1/4 \times 1/4 = 1/16$	6.25%

Topper method

- ★ If two traits assort independently, solve each trait as a monohybrid probability and multiply the probabilities.
- ★ Convert the final fraction to percentage only at the end.

8. 2026–27 Sample-Paper Pattern — Mendelian percentage problem

Experiment 1: flower colour

- VV (violet) $\times$ vv (white) $\rightarrow F_1$ all $Vv \rightarrow 100\%$ violet.
- F_1 selfing $Vv \times Vv \rightarrow F_2 = 1 VV : 2 Vv : 1 vv \rightarrow 75\%$ violet, 25% white.

Experiment 2: colour + height

- $VVTT \times vvtt \rightarrow F_1$ all $VvTt \rightarrow$ all violet and tall.
- F_1 selfing $VvTt \times VvTt$ produces 9:3:3:1 phenotypic pattern in F_2 .

[1M] Percentage of tall plants with white flowers in F_2 ?

$\rightarrow$ Tall probability = $3/4$ and white probability = $1/4$. Therefore $3/4 \times 1/4 = 3/16 = 18.75\%$.

[2M] Percentage of short plants with white flowers in F_1 and F_2 ?

$\rightarrow F_1 = 0\%$, because all F_1 show dominant traits.

- $F_2 = 1/4 \times 1/4 = 1/16 = 6.25\%$.

[2M] Can medium-height plants with light-violet flowers appear in this simple Mendelian cross?

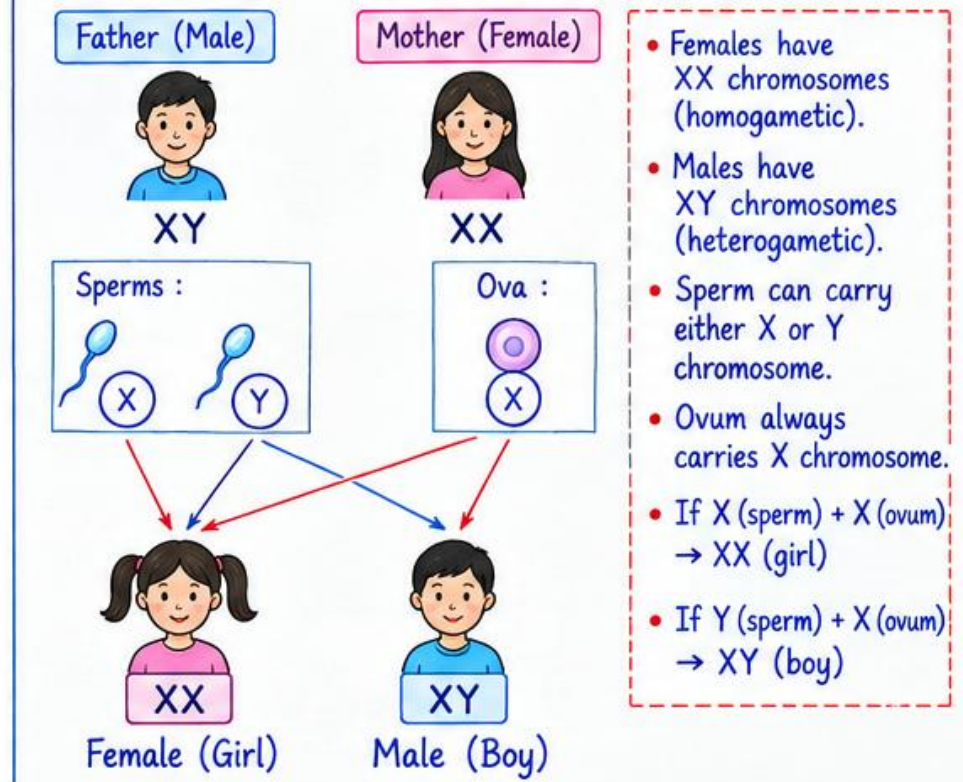
$\rightarrow$ No. In complete dominance, heterozygotes show the dominant phenotype; there is no blending/intermediate phenotype.

- Therefore such intermediate combinations are 0% in both F_1 and F_2 for these simple Mendelian traits.

9. Human Sex Determination — XX / XY system

Handwritten Board Diagram — Sex Determination in Humans

4. Human sex determination



Mother contributes X in every ovum; father contributes either X or Y through sperm.

- Humans normally have 23 pairs of chromosomes in body cells: 22 pairs of autosomes + 1 pair of sex chromosomes.
- Female: XX. Male: XY.
- Every ovum carries X. Roughly half the sperms carry X and half carry Y.
- X-sperm + X-ovum → XX; Y-sperm + X-ovum → XY.
- Thus each fertilisation has approximately 50% probability of XX and 50% probability of XY in the simple textbook model.

Very important social-science link

- ★ The mother does not determine whether the child is XX or XY because all ova carry X.
- ★ The type of sperm (X-bearing or Y-bearing) that fertilises the ovum determines the XX/XY combination.
- ★ Each pregnancy is an independent event; previous children do not change the 50:50 probability for the next fertilisation.

[2M] Explain sex determination in humans.

- Female parent (XX) produces ova carrying only X chromosome; male parent (XY) produces X-bearing and Y-bearing sperms.
- Fertilisation by X-bearing sperm gives XX, while Y-bearing sperm gives XY; hence the chance is about 50% each.

Underline: XX mother • XY father • X/Y sperm • 50%

10. Mark-wise Answer Bank — write to the marks

[1M] What is a dominant trait?

→ A trait expressed in a heterozygous condition is called dominant.

[1M] What is a recessive trait?

→ A trait expressed only when both alleles are recessive is called recessive.

[1M] Differentiate genotype and phenotype.

→ Genotype is the allele combination; phenotype is the observable expression of that genotype.

[2M] How did Mendel show that traits do not blend?

→ Crossing pure tall and dwarf plants gave all tall F_1 plants rather than medium-height plants.

- The dwarf trait reappeared in F_2 , showing that it had remained intact but was masked in F_1 .

[2M] How does the law of segregation explain F_2 ?

→ The two alleles in a heterozygote separate during gamete formation.

- Random fusion of T and t gametes produces TT, Tt, Tt and tt, giving 1:2:1 genotypic and 3:1 phenotypic ratios.

[3M] Explain a dihybrid cross briefly.

→ Pure parents differing in two traits are crossed, e.g. $RRYY \times rryy$, producing F_1 $RrYy$ with both dominant phenotypes.

- F_1 forms RY, Ry, rY and ry gametes and is self-crossed.
- F_2 shows 9:3:3:1 phenotypic ratio, supporting independent assortment.

[3M] How are traits inherited from parents?

→ Genes are segments of DNA located on chromosomes and carry information for traits.

- An offspring receives one allele of a gene from each parent through gametes.
- The allele combination and its dominance relationship determine the genotype and phenotype.

11. Competency / Case-based Thinking

Question decoding method

- ★ Step 1 — Identify whether the question is monohybrid or dihybrid.
- ★ Step 2 — Assign symbols clearly: capital letter for dominant, small letter for recessive.
- ★ Step 3 — Write parent genotypes and gametes before jumping to the ratio.
- ★ Step 4 — For two independent traits, use probability multiplication when faster than a 4×4 Punnett square.
- ★ Step 5 — State the biological reason: dominance, segregation or independent assortment.

12. Common Mistakes a Topper Avoids

- ✗ Writing dominant = stronger/better/common. Dominance only describes expression in a heterozygote.
- ✗ Confusing genotype (TT/Tt/tt) with phenotype (tall/dwarf).
- ✗ Writing F_2 monohybrid genotypic ratio as 3:1 — 3:1 is phenotypic; genotype is 1:2:1.
- ✗ Forgetting that F_1 from pure dominant $\times$ pure recessive is 100% heterozygous.
- ✗ Using 9:3:3:1 for a monohybrid cross — that ratio belongs to a standard dihybrid F_2 .
- ✗ Adding probabilities instead of multiplying independent probabilities in dihybrid outcomes.
- ✗ Saying a recessive allele disappears in F_1 — it is present but masked and can reappear in F_2 .
- ✗ Saying the mother determines the sex chromosome combination of the child — all ova carry X; sperms carry X or Y.

- X Studying evolution as a board-summative unit for 2026–27 — CBSE lists it as formative-only.

13. Final Board Drill — answer without notes

[1M] $TT \times tt$: genotype and phenotype of F_1 ?

→ All Tt ; all tall.

[1M] $Tt \times Tt$: probability of dwarf offspring?

→ $1/4 = 25\%$.

[1M] $VvTt \times VvTt$: probability of white and short offspring?

→ $1/4 \times 1/4 = 1/16 = 6.25\%$.

[1M] $VvTt \times VvTt$: probability of violet and short offspring?

→ $3/4 \times 1/4 = 3/16 = 18.75\%$.

[1M] Which Mendelian law is demonstrated by 9:3:3:1?

→ Law of independent assortment.

[1M] Which parent contributes either X or Y chromosome in humans?

→ The male parent/father through sperm.

14. 5-Minute Final Revision Grid

TOPIC	MUST REMEMBER
Heredity	Transmission of traits from parents to offspring
Gene	Segment of DNA controlling/influencing a trait
Dominant / recessive	Expressed in heterozygote / masked in heterozygote
Monohybrid F_2	Phenotype 3:1; genotype 1:2:1
Dihybrid F_2	Phenotype 9:3:3:1
Laws	Dominance, segregation, independent assortment
Sex determination	XX female, XY male; ovum X; sperm X or Y; ~50:50
2026–27 boundary	Evolution is formative-only, not summative-board content

TOPPER RULE

- ★ For every genetics calculation: write symbols → parents → gametes → cross/probability → ratio/percentage → biological reason.
- ★ Never give only the final percentage when the question asks “give a reason”.

CHAPTER 9 — LIGHT: REFLECTION AND REFRACTION

Chapter Coverage

- ★ Reflection by curved surfaces: spherical mirrors, image formation, centre of curvature, principal axis, principal focus, focal length, mirror formula and magnification.
- ★ Refraction: laws of refraction and refractive index.
- ★ Spherical lenses: image formation, lens formula, magnification and power of a lens.
- ★ Applications of spherical mirrors and lenses. Derivations of mirror and lens formulae are NOT required.
- ★ Prescribed practical links: focal length of concave mirror/convex lens using a distant object, and ray tracing through a rectangular glass slab.

1. Reflection of Light — foundation

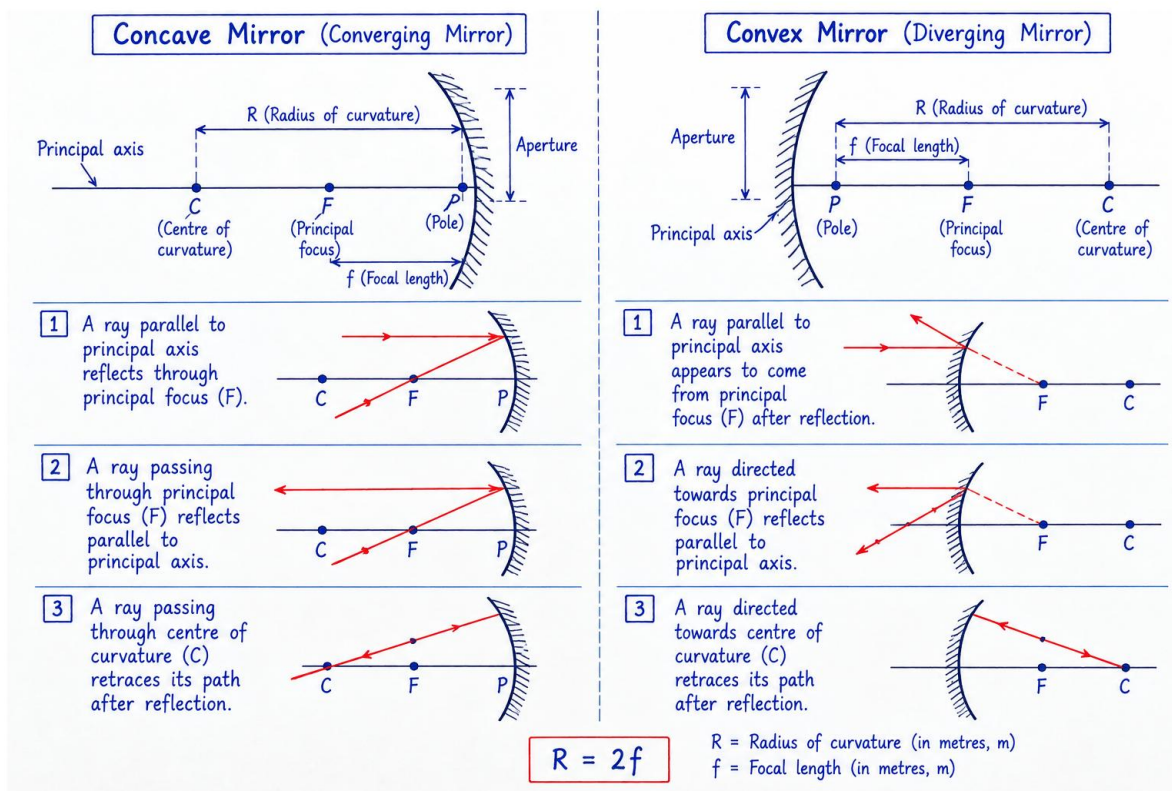
- Reflection is the bouncing back of light into the same medium after striking a surface.
- Law 1: angle of incidence = angle of reflection, $i = r$.
- Law 2: incident ray, reflected ray and normal at the point of incidence lie in the same plane.

1.1 Real image vs virtual image

IMAGE	HOW FORMED	SCREEN?	USUAL ORIENTATION
Real image	Rays actually meet	Can be obtained on a screen	Usually inverted for a single mirror/lens in this chapter
Virtual image	Rays only appear to meet	Cannot be obtained on a screen	Erect in the standard cases studied here

2. Spherical Mirrors — Terms and Ray Rules

Diagram — Concave & Convex Mirrors + Principal Rays



Memorise P, F, C, principal axis, aperture and $R = 2f$; then practise the three principal rays.

2.1 Mirror Terms — exact 1-mark definitions

TERM	DEFINITION
Pole (P)	Geometric centre of the reflecting surface of a spherical mirror.
Centre of curvature (C)	Centre of the sphere of which the reflecting surface of the mirror forms a part.
Radius of curvature (R)	Distance PC between the pole and centre of curvature.
Principal axis	Straight line passing through the pole P and centre of curvature C.
Principal focus (F)	Point on the principal axis where rays parallel to it meet after reflection (concave) or appear to diverge from (convex).
Focal length (f)	Distance PF between pole and principal focus. For a spherical mirror, $R = 2f$.
Aperture	Effective diameter of the reflecting surface of the mirror.

3. Image Formation by Spherical Mirrors

3.1 Concave mirror — all six object positions

Diagram — Concave Mirror Image Formation

3. Image formation by Concave mirror (Ray diagrams)				
Position of object	Position of image	Nature of image	Size	Diagram
At infinity	At F	Real Inverted	Highly diminished	
Beyond C	Between C and F	Real Inverted	Diminished	
At C	At C	Real Inverted	Same size	
Between C and F	Beyond C	Real Inverted	Enlarged	
At F	At infinity	Real Inverted	Highly enlarged	
Between F and P	Behind the mirror	Virtual Erect	Enlarged	

The position of the object decides position, nature and size of the image.

OBJECT	IMAGE	NATURE	SIZE
At infinity	At F	Real, inverted	Highly diminished / point-sized
Beyond C	Between C and F	Real, inverted	Diminished
At C	At C	Real, inverted	Same size
Between C and F	Beyond C	Real, inverted	Enlarged
At F	At infinity	Real, inverted	Highly enlarged
Between F and P	Behind mirror	Virtual, erect	Enlarged

3.2 Convex mirror — one rule covers all positions

Convex mirror result

- ★ For any finite object position, image forms behind the mirror between P and F.
- ★ Image is always virtual, erect and diminished.
- ★ It gives a wide field of view, so it is used as a rear-view/side-view mirror in vehicles.
- ★ At infinity → image at F behind the mirror; virtual, erect and highly diminished / point-sized.

4. New Cartesian Sign Convention

For mirrors

- ★ Take pole P as origin and principal axis as reference.
- ★ Measure all distances from P. Distances measured in the direction of incident light are positive; opposite direction is negative.
- ★ Heights above principal axis are positive; below it are negative.
- ★ For a real object placed in front of a mirror: u is negative. Concave mirror f is negative; convex mirror f is positive.

4.1 Mirror formula & magnification

Formula bank

- ★ Mirror formula: $1/f = 1/v + 1/u$
- ★ Magnification: $m = h_i/h_o = -v/u$
- ★ For mirrors: $m > 0$ means erect/virtual; $m < 0$ means inverted/real.
- ★ $|m| > 1$ enlarged; $|m| = 1$ same size; $|m| < 1$ diminished.
- ★ For spherical mirrors: $R = 2f$.

[3M] Example: concave mirror with $u = -30$ cm and $f = -15$ cm. Find v and nature.

→ $1/f = 1/v + 1/u \rightarrow -1/15 = 1/v - 1/30$.

- $1/v = -1/15 + 1/30 = -1/30 \rightarrow v = -30$ cm.
- Negative v means image forms in front of mirror: real and inverted; object is at C, so image is same size.

[2M] Why are convex mirrors preferred in vehicles?

- They form virtual, erect and diminished images for all object positions.
- Their wide field of view lets the driver see a larger area behind the vehicle.

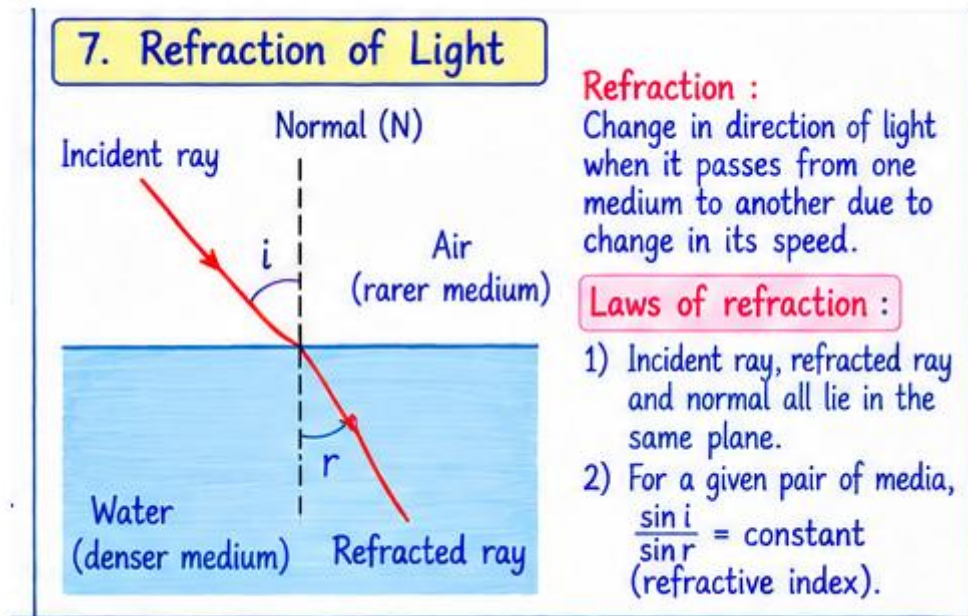
4.2 Common applications

DEVICE	USE	WHY
Concave mirror	Shaving/makeup mirror; dentist mirror	Object within F gives enlarged, erect virtual image.
Concave mirror	Headlight/torch reflector	Source near F sends approximately parallel beam.
Concave mirror	Solar furnace	Parallel sunlight converges near F.
Convex mirror	Vehicle rear/side-view mirror	Wide field; erect and diminished image.

5. Refraction of Light — why a ray bends

- Refraction is the change in direction of light when it passes obliquely from one transparent medium to another because its speed changes.
- Rarer → denser: ray bends towards the normal. Denser → rarer: ray bends away from the normal.
- If light enters normally ($i = 0^\circ$), its speed changes but direction does not change.

Diagram — Laws of Refraction



The normal is drawn perpendicular to the interface at the point of incidence.

5.1 Laws of refraction / Snell's Law

- Incident ray, refracted ray and normal at the point of incidence lie in the same plane.
- For a fixed pair of media and a fixed colour of light: $\sin i / \sin r = \text{constant}$.

Relative refractive index

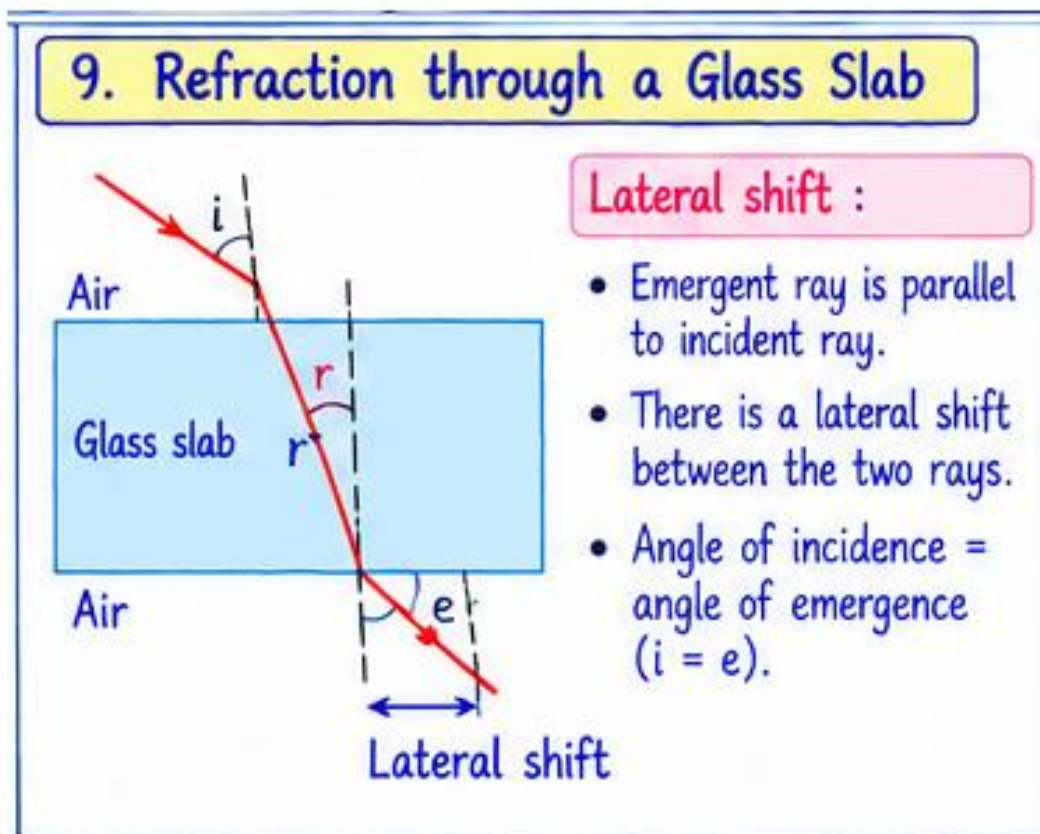
- ★ For light going from medium 1 to medium 2: $n_{21} = \sin i / \sin r = v_1 / v_2$.
- ★ Absolute refractive index of a medium: $n = c/v$, where c is speed of light in vacuum and v is speed in the medium.
- ★ Higher refractive index means lower speed of light and greater optical density. Optical density is not the same as mass density.

[1M] Why does a coin under water appear raised?

- Light travels from water to air and bends away from the normal. The backward extensions of refracted rays meet at a point above the real coin, so apparent depth is less than real depth.

6. Refraction through a Rectangular Glass Slab

Diagram — Rectangular Glass Slab



Labels: incident ray, normal, i , r , e , emergent ray and lateral displacement.

- At the first surface (air $\rightarrow$ glass), ray bends towards normal; at the second surface (glass $\rightarrow$ air), it bends away from normal.
- Because the two faces are parallel, the emergent ray is parallel to the incident ray and angle of emergence e = angle of incidence i .
- The sideways separation between incident-ray direction and emergent ray is called lateral displacement.

6.1 Prescribed practical — tracing a ray through glass slab

Procedure logic

- ★ Trace slab outline; draw an incident ray and fix two pins on it.
- ★ Looking through the opposite face, align two more pins with the images of first two pins.
- ★ Join the pin points to obtain refracted and emergent rays; draw normals at both surfaces.
- ★ Measure i , r and e . Verify approximately $i = e$ and emergent ray $\parallel$ incident ray.

[2M] State Snell's law and define relative refractive index.

$\rightarrow$ For a fixed pair of media, $\sin i / \sin r$ is constant.

- The refractive index of medium 2 with respect to medium 1 is $n_{21} = \sin i / \sin r = v_1 / v_2$.

QUESTION PATTERN

- ★ Assertion: apparent depth of a coin in water is less than real depth — TRUE.
- ★ Reason: when light goes from water to air, it bends away from normal — TRUE and explains the apparent-depth effect.

7. Spherical Lenses — terminology + principal rays

Diagram — Convex & Concave Lens + Image Rules

7. Lenses

Convex lens (Converging)

Thicker at centre
Converges parallel rays to a point.

Concave lens (Diverging)

Thinner at centre
Diverges parallel rays as if from a point.

Basic terms :

O : Optical centre (rays pass undeviated)
F : Principal focus
R : Radius of curvature
f : Focal length
2F : Twice the focal length

Lens formula : $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

Magnification : $m = \frac{v}{u} = \frac{h'}{h}$

Power of a lens : $P = \frac{1}{f}$ (f in metres)
Unit : dioptre (D) 1 D = 1 m⁻¹

8. Image formation by Convex lens

Position of object	Position of image	Nature of image	Size	Diagram
At infinity	At F ₂	Real Inverted	Highly diminished	
Beyond 2F ₁	Between F ₂ and 2F ₂	Real Inverted	Diminished	
At 2F ₁	At 2F ₂	Real Inverted	Same size	
Between F ₁ and 2F ₁	Beyond 2F ₂	Real Inverted	Enlarged	
At F ₁	At infinity	Real Inverted	Highly enlarged	
Between O and F ₁	On the same side of object	Virtual Erect	Enlarged	

9. Image formation by Concave lens

For any position of object, image is always :

- Between optical centre (O) and focus (F₁)
- Virtual and erect
- Diminished in size

Convex lens is converging; concave lens is diverging. Note F₁/F₂, 2F₁/2F₂ and optical centre O.

7.1 Essential lens terms

TERM	MEANING
Optical centre O	A ray through O passes approximately undeviated for a thin lens.
Principal axis	Straight line through optical centre and centres of curvature.
Principal focus F	Point where parallel rays converge, or appear to diverge from.
Focal length f	Distance between optical centre O and principal focus F.
Convex lens	Thicker at centre; converging; f positive.
Concave lens	Thinner at centre; diverging; f negative.

7.2 Three standard rays for lenses

- Parallel to principal axis → passes through F₂ after a convex lens; appears to diverge from F₁ after a concave lens.
- Through/towards principal focus → emerges parallel to principal axis.
- Through optical centre O → passes approximately undeviated.

8. Image Formation by Lenses

8.1 Convex lens — all object positions

OBJECT	IMAGE	NATURE	SIZE
At infinity	At F ₂	Real, inverted	Highly diminished
Beyond 2F ₁	Between F ₂ and 2F ₂	Real, inverted	Diminished
At 2F ₁	At 2F ₂	Real, inverted	Same size

Between F_1 and $2F_1$	Beyond $2F_2$	Real, inverted	Enlarged
At F_1	At infinity	Real, inverted	Highly enlarged
Between O and F_1	Same side as object	Virtual, erect	Enlarged

8.2 Concave lens — one rule covers all finite object positions

Concave lens result

- ★ Image always forms between O and F_1 on the same side as the object.
- ★ Image is always virtual, erect and diminished.
- ★ At infinity → image at F_1 on the same side as the object; virtual, erect and highly diminished / point-sized.

8.3 Applications of Spherical Lenses

LENS	COMMON USE	WHY IT MATTERS
Convex lens	Magnifying glass	Object within focal length gives a virtual, erect and enlarged image.
Convex lens	Camera / projector	Converges light to form a real image; exact arrangement depends on the optical device.
Concave lens	Spectacles for myopia	Diverges incoming rays; detailed eye-defect correction is covered in Chapter 10.

9. Lens Formula, Magnification & Power

Formula bank

- ★ Lens formula: $1/f = 1/v - 1/u$
- ★ Magnification: $m = h_i/h_o = v/u$
- ★ Power: $P = 1/f$ when f is in metres. Unit = diopetre (D).
- ★ If f is in centimetres, $P(D) = 100/f(\text{cm})$.
- ★ Convex lens: f and P positive. Concave lens: f and P negative.
- ★ 1 diopetre (1 D) is the power of a lens whose focal length is 1 metre; $1 \text{ D} = 1 \text{ m}^{-1}$.

10. Lens Sign Convention — do not lose numerical marks

For lenses

- ★ Take optical centre O as origin. Measure distances from O along the principal axis.
- ★ Direction of incident light is positive; opposite direction is negative.
- ★ For a real object on the left: u is negative.
- ★ Convex lens f is positive; concave lens f is negative.
- ★ Real image on the opposite side: v positive. Virtual image on object side: v negative.

[3M] Example: a concave lens forms an image 1/3 the object size. Object is 24 cm in front. Find v and f.

- For concave lens image is virtual and erect, so $m = +1/3$. With $u = -24$ cm, $m = v/u \Rightarrow v = (1/3)(-24) = -8$ cm.
- Lens formula: $1/f = 1/v - 1/u = -1/8 - (-1/24) = -1/12$.
- Therefore $f = -12$ cm. Negative focal length confirms a concave lens.

[3M] Practice example: convex lens, $u = -24$ cm, $f = +12$ cm. Find v and nature.

- $1/f = 1/v - 1/u \Rightarrow 1/12 = 1/v + 1/24$.
- $1/v = 1/24 \Rightarrow v = +24$ cm.
- Positive v means a real image on the opposite side; since object is at $2F_1$, image is real, inverted and same size.

[1M] A lens has power -2 D. Identify lens and focal length.

- Negative power means concave lens. $f = 1/P = -1/2$ m = -0.50 m = -50 cm.

11. Practical Skills

11.1 Focal length of convex lens using a distant object

Correct practical procedure

- ★ Face the convex lens toward a distant object (or the Sun only with strict safety precautions).
- ★ Place a screen/paper on the other side and move it until the smallest, sharpest, brightest image/spot is obtained.
- ★ For a distant object, incident rays are nearly parallel and the image forms near principal focus F_2 .
- ★ Distance between optical centre of lens and sharp image $\approx$ focal length.

[1M] Which procedure gives the most accurate focal length of a convex lens using sunlight?

- Adjust the lens-screen distance until the smallest, brightest sharp spot is formed, then measure that distance.

11.2 Focal length of concave mirror using distant object

- Face concave mirror toward a distant object; adjust screen in front of mirror to obtain sharp image. Mirror-screen distance $\approx$ focal length.
- Never look directly at the Sun through a lens or into a concentrated reflected/refracted beam.

12. Common Question Patterns

QUESTION CUE	WHAT THEY MAY ASK	KEY RESPONSE
Focal-length practical	Why smallest sharp image?	Distant rays are nearly parallel; they converge at focus.
Apparent depth	Coin appears raised	Refraction from denser water to rarer air bends rays away from normal.
Snell's law	State law + formula	$\sin i / \sin r = \text{constant}$ for a fixed pair of media.
Relative refractive index	Compare speeds	$n_{21} = v_1/v_2$.
Concave-lens numerical	Use magnification first	$m = v/u$, then lens formula.
Ray diagram	Infer nature/size/position	Use any two principal rays correctly.

13. Mark-wise Answer Practice

[1M] Define principal focus of a concave mirror.

→ The point on the principal axis where rays parallel to the principal axis meet after reflection from a concave mirror.

[1M] Define power of a lens.

→ Power is the reciprocal of focal length in metres: $P = 1/f$. SI unit: dioptre (D).

[1M] Define one dioptre.

→ One dioptre is the power of a lens of focal length 1 metre. Thus, $1 \text{ D} = 1 \text{ m}^{-1}$.

[2M] Differentiate convex and concave lenses.

→ Convex lens is thicker at centre, converges parallel rays and has positive focal length.

- Concave lens is thinner at centre, diverges parallel rays and has negative focal length.

[2M] Why does light bend on entering another medium?

→ Its speed changes when it enters a medium of different optical density.

- At oblique incidence this speed change changes the direction of propagation, producing refraction.

[3M] Explain image formation by a convex mirror.

→ A ray parallel to principal axis reflects as if it came from F; another suitable ray is reflected according to the standard ray rule.

- Backward extensions of reflected rays meet behind mirror between P and F.

- Hence image is always virtual, erect and diminished.

[3M] Explain refraction through a rectangular glass slab.

→ At first face, light bends toward normal on entering glass; at second face, it bends away from normal on entering air.

- Parallel faces make emergent ray parallel to incident ray, so $i \approx e$.

- The emergent ray is laterally displaced from the original direction.

14. Application & Case-Based Questions

For any ray-diagram question

- ★ First identify device: concave/convex mirror or convex/concave lens.

- ★ Mark P/O, F, C or 2F points before drawing rays.

- ★ Use two standard rays only; extend reflected/refracted rays backward with dotted lines for virtual images.

- ★ Then state three things explicitly: position, nature (real/virtual; erect/inverted) and size.

For any numerical

- ★ Write sign convention first: u , v and f signs.

- ★ Write formula before substitution.

- ★ Keep units consistent. Convert cm to m only when calculating power.

- ★ Interpret sign of final answer — do not stop at a number.

15. Common Mistakes

- ✗ Writing mirror magnification as $+v/u$. Correct for mirrors: $m = -v/u$.

- ✗ Using mirror formula for lens or lens formula for mirror.

- X Forgetting that u is negative for a real object in the standard Cartesian setup.
- X Calling every positive magnification “real”. Positive m generally indicates erect image; check device and v sign.
- X Saying convex mirror can form a real image of a real object — it cannot in the Class 10 cases.
- X Saying concave lens can form a real image of a real object — it forms virtual, erect, diminished image.
- X Using centimetres directly in $P=1/f$. Focal length must be in metres for dioptries.
- X Confusing optical density with mass density.
- X Drawing the emergent ray from a rectangular glass slab non-parallel to incident ray.
- X Forgetting dotted backward extensions for virtual image construction.

16. Quick Revision

TOPIC	KEY POINT
Spherical mirror	$R = 2f$; $1/f = 1/v + 1/u$; $m = -v/u$
Concave mirror	Can form real/inverted or virtual/erect image depending on object position.
Convex mirror	Always virtual, erect, diminished for real object.
Snell's law	$\sin i / \sin r = \text{constant}$; $n_{21} = v_1/v_2$
Absolute RI	$n = c/v$
Glass slab	Emergent ray $\parallel$ incident ray; $i=e$; lateral displacement.
Lens formula	$1/f = 1/v - 1/u$; $m = v/u$
Convex lens	Usually real/inverted; virtual/erect only when object is within F_1 .
Concave lens	Always virtual, erect, diminished for real object.
Power	$P=1/f(m)$; convex +, concave -; unit D

- **Check: ray diagrams need labels/arrows; numericals need sign convention, formula, unit and interpretation.**

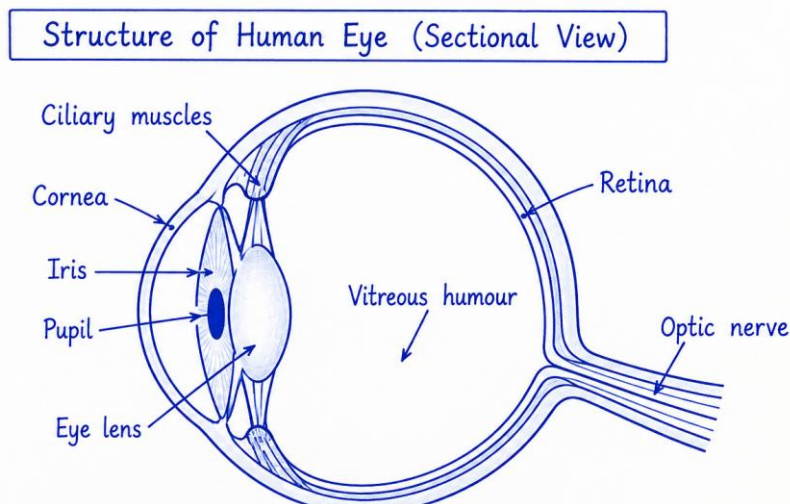
CHAPTER 10 — THE HUMAN EYE AND THE COLOURFUL WORLD

Chapter Coverage

- ★ Functioning of the lens in the human eye; power of accommodation; near point and far point.
- ★ Defects of vision and their correction: myopia, hypermetropia and presbyopia.
- ★ Refraction through a glass prism; dispersion of white light and daily-life applications.
- ★ Atmospheric refraction and scattering of light: twinkling of stars, advanced sunrise/delayed sunset, blue sky, Tyndall effect and red warning signals.
- ★ Not required for the 2026–27 board exam: colour of the Sun at sunrise and sunset.

1. Human Eye — structure + function

Diagram — Human Eye



Labels to include: cornea, iris, pupil, eye lens, ciliary muscles, retina and optic nerve. Keep labels horizontal and avoid crossing label lines.

PART	BOARD FUNCTION
Cornea	Transparent front surface; provides most of the refraction of incoming light.
Iris	Muscular coloured diaphragm that controls the size of the pupil.
Pupil	Opening in iris through which light enters; regulates amount of light entering by change in pupil size.
Eye lens	Transparent flexible convex lens; fine-focuses light on retina by changing curvature/focal length.
Ciliary muscles	Change curvature/thickness of eye lens for accommodation.
Retina	Light-sensitive screen containing photoreceptors; a real, inverted and diminished image is formed here.
Optic nerve	Carries visual signals from retina to brain.
Aqueous / vitreous humour	Transparent media that help maintain the eyeball and allow light to pass through.

1.1 How do we see an object?

- Light reflected from the object enters mainly through the cornea and pupil.
- Cornea and eye lens refract the light and focus it on the retina.
- The retinal image is real, inverted and diminished; photoreceptors convert light into electrical signals.
- The optic nerve carries signals to the brain, which interprets the visual information.

Remember: the pupil does NOT change the focal length. The iris controls pupil size; ciliary muscles change the curvature/focal length of the eye lens.

2. Power of Accommodation

Definition [1M]: Accommodation is the ability of the eye lens to change its focal length so that objects at different distances are focused sharply on the retina.

OBJECT	CILIARY MUSCLES	LENS	FOCAL LENGTH / POWER
Distant object	Relax	Becomes thinner / less curved	Focal length increases; converging power decreases
Near object	Contract	Becomes thicker / more curved	Focal length decreases; converging power increases

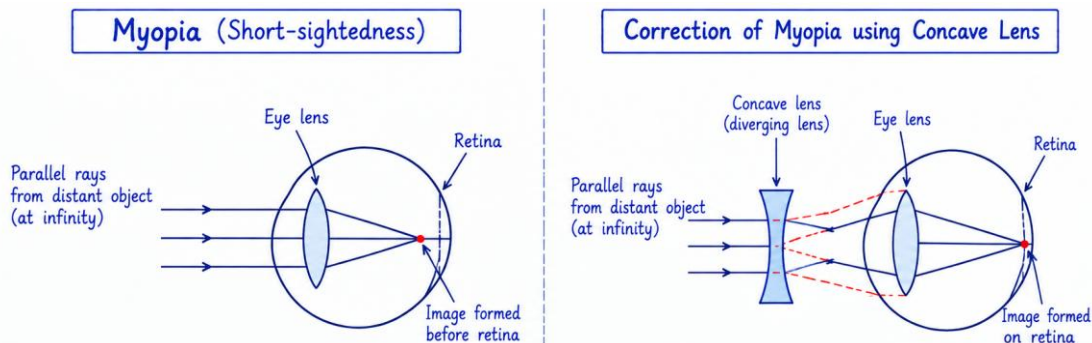
- Near point / least distance of distinct vision of a normal adult eye ≈ 25 cm.
- Far point of a normal eye = infinity.
- Far point [1M]: Farthest point seen clearly without strain; for a normal eye, it is at infinity.
- Normal range of distinct vision: about 25 cm to infinity.

3. Defects of Vision — identify cause $\rightarrow$ image position $\rightarrow$ correcting lens

DEFECT	WHAT IS BLURRED?	IMAGE FORMS	MAIN CAUSE	CORRECTION
Myopia (short-sightedness)	Distant objects	Before retina	Eyeball too long OR eye lens too powerful / focal length too small	Concave (diverging) lens; negative power
Hypermetropia (long-sightedness)	Near objects	Behind retina	Eyeball too short OR eye lens has insufficient converging power / focal length too large	Convex (converging) lens; positive power
Presbyopia	Usually near objects; may coexist with distance defect	Near point recedes	Ageing: ciliary muscles weaken and lens loses flexibility	Convex reading lens; bifocal/progressive lens if two corrections are needed

3.1 Myopia — short-sightedness

Diagram — Myopia & Correction

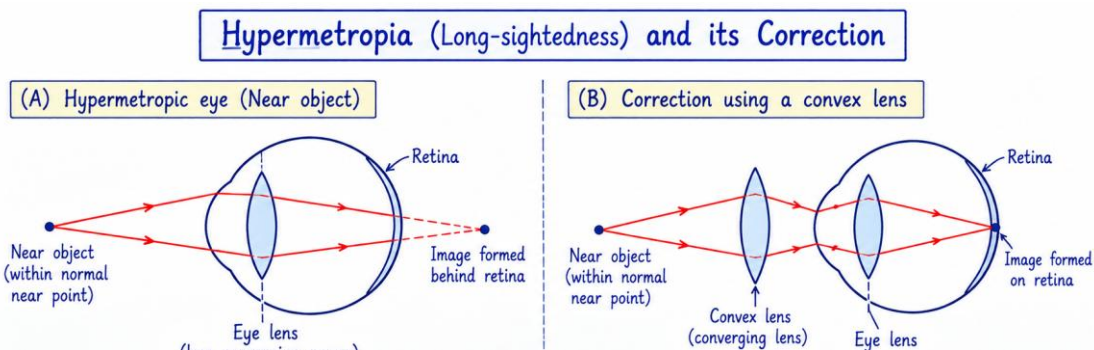


Exam diagram: show parallel rays from a distant object; without correction they focus before retina. A concave lens diverges them so the eye focuses them on retina.

- A myopic person has a finite far point; distant objects beyond it are not seen distinctly.
- Concave corrective lens has negative focal length and negative power.

3.2 Hypermetropia — long-sightedness

Diagram — Hypermetropia & Correction



Exam diagram: rays from a near object would meet behind retina. A suitable convex lens increases convergence so image forms on retina.

- Near point of a hypermetropic eye is farther than the normal 25 cm.
- Convex corrective lens has positive focal length and positive power.

3.3 Presbyopia — age-related loss of accommodation

- With age, ciliary muscles weaken and eye lens becomes less flexible; the power of accommodation decreases.
- The near point moves farther away, causing difficulty in reading nearby objects.
- A convex reading lens is used for near vision. If presbyopia coexists with myopia, bifocal/progressive lenses can provide separate distance and near corrections.

Cataract ≠ refractive defect: in cataract the eye lens becomes cloudy/opaque. It is treated medically, commonly by replacing the cloudy lens; it is not corrected merely by spectacles.

3.4 Lens Power Numericals

[3M] Myopia: far point = 80 cm. Find lens power for seeing distant objects.

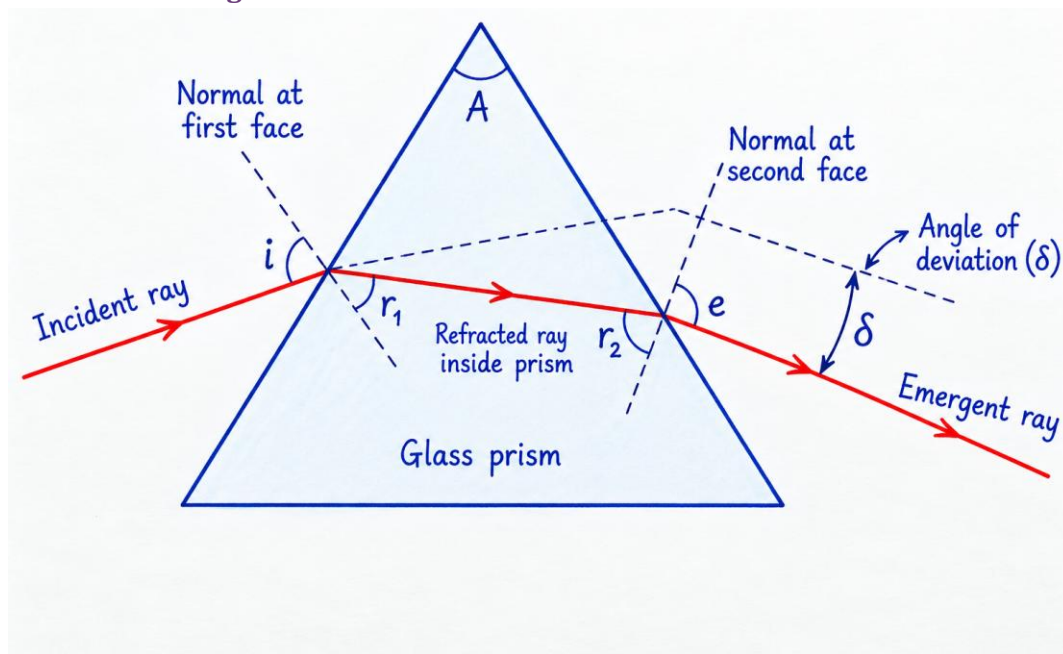
→ For a distant object, $u = -\infty$ and corrective lens must form a virtual image at the far point $v = -0.80$ m. Thus $f \approx -0.80$ m and $P = 1/f = -1.25$ D. Use a concave lens.

[3M] Hypermetropia: near point = 100 cm; person wants to read at 25 cm.

→ Corrective convex lens should form a virtual image at $v = -100$ cm for an object at $u = -25$ cm. $1/f = 1/v - 1/u = -1/100 + 1/25 = 3/100 \text{ cm}^{-1}$, so $f \approx +33.3 \text{ cm} = +0.333 \text{ m}$ and $P \approx +3.0$ D.

4. Refraction through a Glass Prism

Diagram — Refraction through a Prism



Labels: incident ray, emergent ray, normals, angle of prism A , i , r_1 , r_2 , e and angle of deviation δ .

- A glass prism has two inclined refracting surfaces; the emergent ray is deviated from the original direction.
- Angle between the direction of the incident ray (produced forward) and emergent ray is called angle of deviation (δ).

4.1 Prescribed practical — tracing a ray through a glass prism

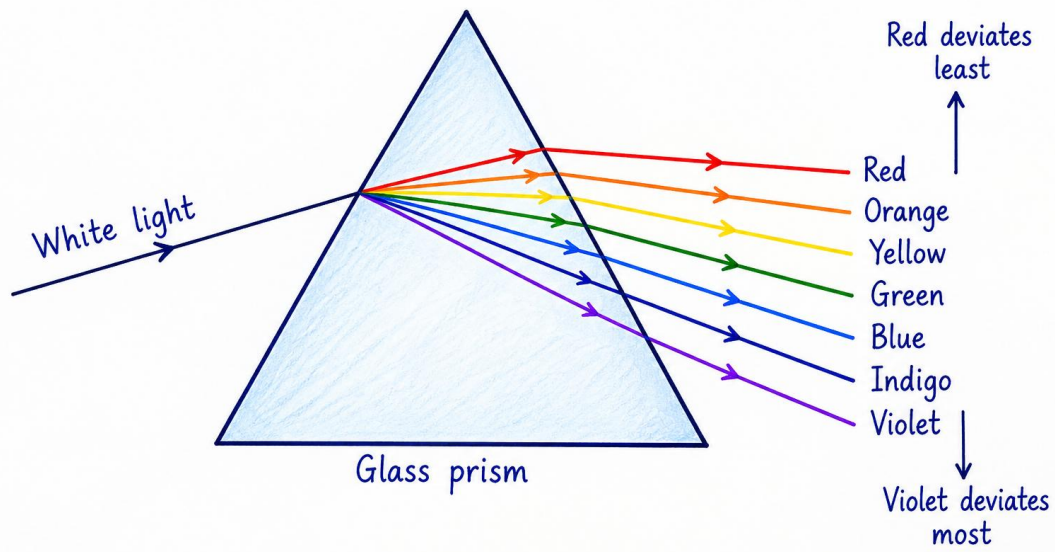
- Trace the triangular outline of the prism on paper and draw an incident ray on one face.
- Fix two pins vertically on the incident ray. Looking through the second face, align two more pins with the images of the first two pins.
- Remove prism/pins; join points to obtain refracted and emergent paths. Draw normals at both faces.
- Mark i , r and e and observe that the prism deviates the ray toward its base.

5. Dispersion of White Light

Dispersion [1M]: splitting of white light into its component colours due to wavelength-dependent refraction.

Diagram — Dispersion through a Glass Prism

Dispersion of White Light Through a Glass Prism



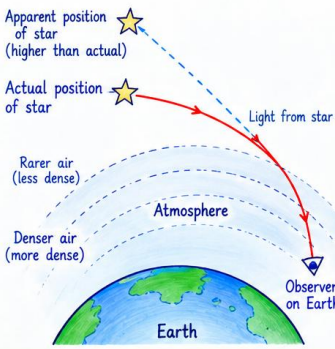
Remember: VIBGYOR. In the usual prism diagram, red deviates least and violet deviates most.

- Different colours have different refractive indices/speeds in glass, so they bend by different amounts.
- Order of spectrum: Violet – Indigo – Blue – Green – Yellow – Orange – Red (VIBGYOR).
- Violet has the shortest visible wavelength and usually deviates most; red has the longest visible wavelength and deviates least.
- A second identical prism placed in inverted orientation can recombine the spectrum to give white light.
- Rainbow: a natural spectrum produced by refraction, dispersion and internal reflection of sunlight in water droplets; red is outer/top and violet inner/bottom in the primary rainbow.

6. Atmospheric Refraction — daily-life applications

Handwritten Concept Sheet — Atmospheric Refraction, Scattering & Tyndall Effect

1. Atmospheric Refraction and Twinkling of Stars



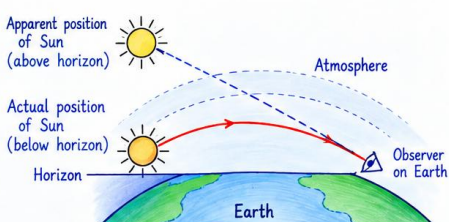
Why stars twinkle?

- Starlight passes through different layers of air with varying refractive index.
- Continuous change in direction (due to refraction) makes the apparent position and brightness of star fluctuate.
- Hence stars appear to twinkle.

Why planets do not twinkle appreciably?

- Planets are much closer to Earth.
- They appear as extended sources (not point-sized).
- Fluctuations from different parts average out, so no appreciable twinkling.

2. Advanced Sunrise and Delayed Sunset

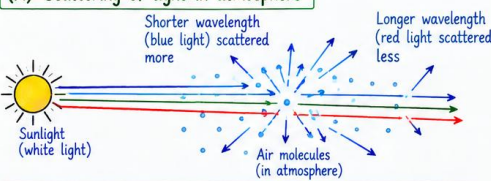


Due to atmospheric refraction, the Sun appears above the horizon even when it is actually below the horizon.

- Hence we see the Sun about **2 minutes before** the actual sunrise (**advanced sunrise**).
- Similarly, the Sun remains visible for about 2 minutes after the actual sunset (**delayed sunset**).

3. Scattering of Light and Tyndall Effect

(A) Scattering of light in atmosphere



Why does the sky appear blue?

- Blue light (shorter wavelength) is scattered much more by air molecules in all directions, so the sky appears blue.

Why are danger/stop signals red?

- Red light (longer wavelength) is scattered the least, so it can be seen from a long distance even in fog or smoke.

Why does the sky appear dark in space?

- In space, there are almost no air molecules.
- Hence there is no scattering of sunlight, so the sky appears dark (black).

(B) Tyndall Effect



Tyndall Effect:

- Scattering of light by colloidal particles makes the path of light **visible** when a beam of light is passed through a colloidal solution.

Use the first two panels to revise atmospheric refraction; the lower panel covers scattering/Tyndall effect. The syllabus excludes the colour of the Sun at sunrise/sunset.

6.1 Why do stars twinkle?

- Earth's atmosphere has layers of continuously changing density and refractive index.
- Starlight is refracted repeatedly; the star's apparent position and apparent brightness fluctuate.

[1M] Why does a star appear slightly higher than its actual position?

→ Starlight bends continuously toward the normal through denser atmospheric layers; its backward extension makes the star appear slightly higher than its actual position.

- Stars are effectively point sources, so these fluctuations are noticeable as twinkling.

Why do planets usually not twinkle appreciably?

→ Planets appear as extended sources. Light from different parts undergoes different fluctuations that average out, so appreciable twinkling is not seen.

6.2 Advanced sunrise and delayed sunset

- Near the horizon, atmospheric refraction makes the Sun appear slightly above its actual position.
- Therefore the Sun can be seen about 2 minutes before actual sunrise and about 2 minutes after actual sunset.

[2M] Why does the Sun appear slightly flattened near the horizon?

→ Atmospheric refraction raises the lower edge of the Sun slightly more than the upper edge, reducing its apparent vertical diameter; hence the disc looks flattened.

Not required for the 2026–27 board exam: colour of the Sun at sunrise and sunset.

7. Scattering of Light & Tyndall Effect

7.1 Scattering — wavelength matters

- Fine particles/molecules scatter light in different directions. Shorter wavelengths are scattered more strongly than longer wavelengths.
- Blue/violet light is scattered much more than red. Among visible colours, violet has the shortest wavelength and red the longest.

Particle size → scattering rule

- Very fine particles: scatter shorter wavelengths much more strongly.
- Larger particles: scatter a broader range of wavelengths.
- Very large particles/water droplets: scatter most visible colours nearly equally → white appearance.

[2M] Why does the sky appear blue?

→ Air molecules scatter shorter wavelengths (blue and violet) more strongly than longer wavelengths. Human eyes are more sensitive to blue than violet, so the sky appears blue.

[1M] Why is the sky dark to an astronaut in space?

→ There is essentially no atmosphere to scatter sunlight, so the sky appears dark/black.

[1M] Why are danger/stop signals red?

→ Red has the longest visible wavelength and is scattered least, so it remains visible over longer distances even through fog, smoke or dust.

Why do clouds/fog often appear white?

→ Relatively large droplets/particles scatter different visible wavelengths nearly equally, so the scattered light appears white.

7.2 Tyndall Effect — make the light path visible

Tyndall effect: scattering of light by colloidal/fine suspended particles so that the path of a light beam becomes visible.

- Examples: sunlight in a dusty room, beam of headlights in fog/smoke, light passing through a colloidal solution such as milk diluted in water.
- A true solution with particles too small to scatter visible light appreciably does not show the same visible beam path.

8. Common Question Patterns

QUESTION CUE	WHAT TO FIND	KEY RESPONSE
Accommodation	Define it	Ability of eye lens to change focal length to focus different distances on retina.
Near point	Value + meaning	≈25 cm; closest comfortable distinct-vision point for a normal adult eye.
Cannot see distant objects	Identify correction	Myopia → concave/diverging lens; draw correction diagram.
Hypermetropia	Cause + correction	Near image tends behind retina due short eyeball/weak convergence → convex lens.
Dispersion	Definition + least deviation	White light splits into colours; red deviates least.
Astronaut	Why sky dark?	No atmosphere → no appreciable scattering.

QUESTION CUE	WHAT TO FIND	KEY RESPONSE
Tyndall effect	Reason	Atmosphere/colloid has minute particles that scatter light and make beam path visible.
Signal colour	Why red?	Longest wavelength, least scattering, visible from far away.
Blue sky	Reason	Shorter wavelengths scattered strongly; eye more sensitive to blue than violet.

9. Mark-wise Answer Practice

[1M] What is power of accommodation?

→ Ability of the eye lens to change its focal length to focus objects at different distances sharply on the retina.

[1M] State near point and far point of a normal eye.

→ Near point $\approx$ 25 cm; far point = infinity.

[2M] Differentiate myopia and hypermetropia.

→ Myopia: distant objects blurred; image tends before retina; corrected by concave lens. Hypermetropia: near objects blurred; image tends behind retina; corrected by convex lens.

[2M] Why does a prism disperse white light?

→ Different colours have different refractive indices/speeds in glass, so they refract through different angles. Hence white light separates into a spectrum.

[3M] Explain twinkling of stars.

→ Atmospheric layers have continuously changing refractive index. Starlight undergoes repeated refraction, causing fluctuations in apparent position and intensity. Since a star is effectively a point source, the fluctuations are noticeable as twinkling.

[3M] Explain Tyndall effect and give two examples.

→ Fine colloidal/suspended particles scatter a beam of light, making its path visible. Examples: a sunbeam in a dusty room and headlights in fog/smoke.

[5M] Typical integrated question: eye defect OR colourful-world option

→ Eye option: define accommodation → state near point → identify myopia and concave correction → draw labelled correction diagram. Colourful-world option: define dispersion → red deviates least → no atmosphere means dark sky in space → draw labelled prism spectrum.

10. Common Mistakes

- X Saying pupil changes focal length. Ciliary muscles change lens curvature/focal length; iris controls pupil size.
- X Reversing defect corrections: myopia → concave; hypermetropia → convex.
- X Saying myopia image forms behind retina — it forms before retina for distant objects.
- X Saying hypermetropia image forms before retina — it tends to form behind retina for near objects.
- X Confusing presbyopia with hypermetropia: presbyopia is age-related loss of accommodation.
- X Reversing prism deviation: red least, violet most.
- X Saying stars twinkle because they emit changing light — twinkling is due to atmospheric refraction.
- X Saying the sky is blue because blue is the “only” colour scattered — shorter wavelengths are scattered more; our eyes favour blue over violet.
- X Treating the colour of the Sun at sunrise/sunset as a required board topic; it is excluded for 2026–27.

11. Quick Revision

TOPIC	KEY POINT
Eye image	Real, inverted, diminished on retina
Accommodation	Ciliary muscles change lens curvature/focal length
Normal near / far point	25 cm / infinity
Myopia	Before retina; concave lens; negative power
Hypermetropia	Behind retina for near object; convex lens; positive power
Presbyopia	Ageing → reduced accommodation; reading/bifocal/progressive correction
Prism	Deviation toward base; trace incident/refracted/emergent rays
Dispersion	VIBGYOR; red least deviation; violet most
Twinkling	Atmospheric refraction of point-source starlight
Advanced sunrise	Sun appears above actual position; ~2 min early/late
Blue sky	Shorter wavelengths scattered more
Red signals	Red scattered least
Space sky	Dark because no atmosphere for scattering
Tyndall effect	Colloidal particles scatter light and reveal beam path

Self-check before you leave the chapter

- ☐ Can I label the eye, explain accommodation, and state near/far points?
- ☐ Can I identify myopia, hypermetropia and presbyopia and draw correction diagrams?
- ☐ Can I trace prism refraction and draw VIBGYOR dispersion with red least/violet most?
- ☐ Can I explain star position/twinkling, sunrise delay, scattering, Tyndall effect and the syllabus exclusion?

CHAPTER 11 — ELECTRICITY

Chapter Coverage

★ Electric current & potential difference • ★ Ohm's law • ★ Resistance & resistivity • ★ Factors affecting resistance • ★ Series/parallel combinations & daily-life applications • ★ Heating effect & Joule's law • ★ Electric power and P-V-I-R relations • ★ Practicals: V-I graph and equivalent resistance

1. Electric Current, Charge & Potential Difference

1.1 Electric current — flow of charge

- Electric current is the rate of flow of electric charge through a conductor.

$$I = Q / t$$

I in ampere (A), Q in coulomb (C), t in second (s)

- 1 ampere = 1 coulomb of charge flowing per second: $1 \text{ A} = 1 \text{ C s}^{-1}$.
- Conventional current is taken from positive terminal to negative terminal in the external circuit; electrons drift in the opposite direction.
- Charge of one electron = $1.6 \times 10^{-19} \text{ C}$ in magnitude. Number of electrons transferred: $n = Q/e$.

1.2 Potential difference — energy supplied per unit charge

- Potential difference between two points is the work done (or energy transferred) per unit charge in moving a test charge between them.

$$V = W / Q$$

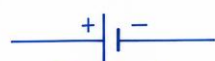
1 volt = 1 joule per coulomb = 1 J C^{-1}

- A cell/battery maintains potential difference and supplies energy to charges in a closed circuit.
- Electric circuit: a continuous conducting path through which electric current can flow.
- Closed circuit: the conducting path is complete, so current flows. Open circuit: the path is broken, so current does not flow.

1.3 Measuring instruments + circuit symbols

INSTRUMENT	CONNECTION	WHY / EXAM KEYWORD
Ammeter	Series	Measures current through the circuit; ideally very low resistance.
Voltmeter	Parallel across component	Measures potential difference; ideally very high resistance.
Rheostat	Series	Variable resistance used to change current in a circuit.

Diagram — Common Circuit Symbols

Common Circuit Symbols — Electricity

Electric cell



Battery (combination of cells)



Open key (switch)



Closed key (switch)



Resistor



Variable resistor / Rheostat



Ammeter



Voltmeter



Electric bulb / Lamp



Fuse

2. Ohm's Law — the V-I relationship

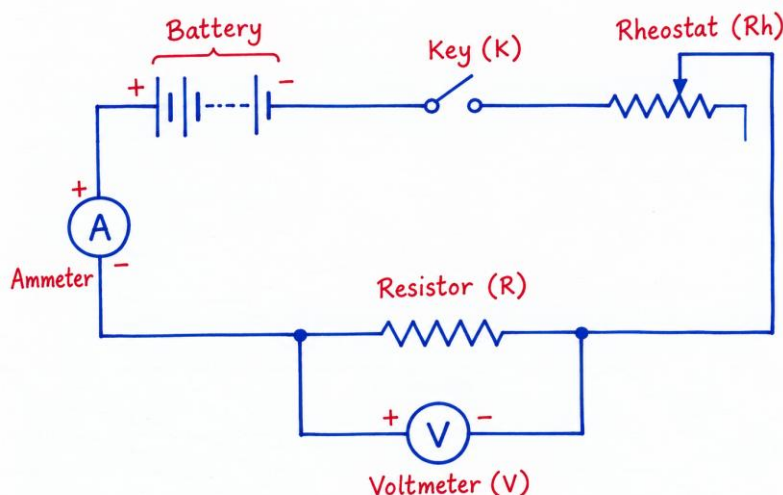
At constant temperature, the potential difference V across a conductor is directly proportional to the current I through it.

$$V \propto I \Rightarrow V = IR \Rightarrow R = V/I$$

R is resistance; SI unit = ohm (Ω). $1 \Omega = 1 \text{ V A}^{-1}$.

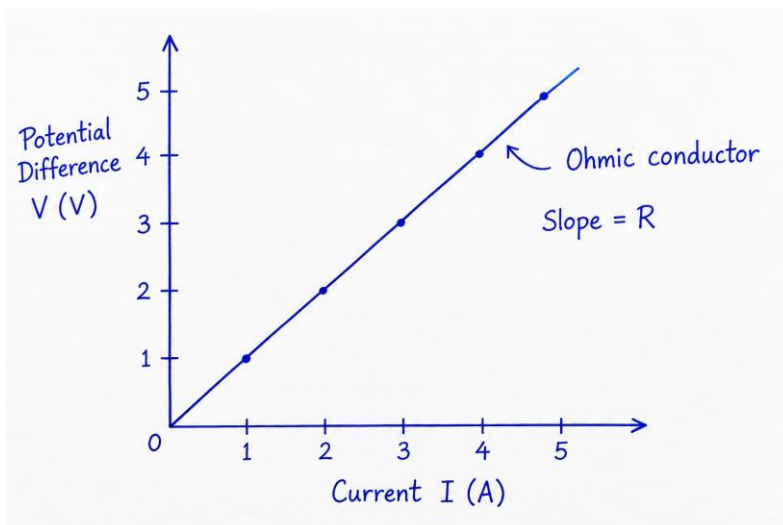
- An ohmic conductor gives a straight-line V-I graph through the origin, provided physical conditions (especially temperature) remain constant.
- If V is plotted on the y-axis and I on the x-axis, slope $\Delta V / \Delta I = R$.

Diagram — Ohm's Law Experimental Circuit



Remember: ammeter in series; voltmeter in parallel; rheostat in series to vary current.

Graph — V-I Characteristic of an Ohmic Conductor



Graph question: straight line through origin $\Rightarrow V/I$ is constant $\Rightarrow$ resistance is constant.

2.1 Prescribed practical — V versus I

1. Connect battery, key, rheostat, resistor and ammeter in series; connect voltmeter across the resistor.
2. Close the key briefly. Adjust rheostat and record corresponding V and I readings.
3. Take several observations, tabulate V and I, and calculate $R = V/I$ for each pair.
4. Plot V on y-axis against I on x-axis. A straight line through origin verifies Ohm's law; slope gives R.

Practical precautions: correct polarity of meters • tight connections • do not keep key closed unnecessarily (heating changes resistance) • choose suitable meter range.

3. Resistance & Resistivity

3.1 Resistance — opposition to current

- Resistance is the property of a conductor by virtue of which it opposes the flow of electric current.
- For a given potential difference, larger resistance gives smaller current: $I = V/R$.

3.2 Factors affecting resistance of a uniform conductor

FACTOR	RELATION	REASONING
Length l	$R \propto l$	Longer conductor $\rightarrow$ more collisions/path $\rightarrow$ larger R.
Cross-sectional area A	$R \propto 1/A$	Thicker conductor $\rightarrow$ more paths for charge $\rightarrow$ smaller R.
Material	Depends on ρ	Different materials have different resistivities.
Temperature	Usually changes R	For metallic conductors, resistance generally increases with temperature.

$$R = \rho l / A$$

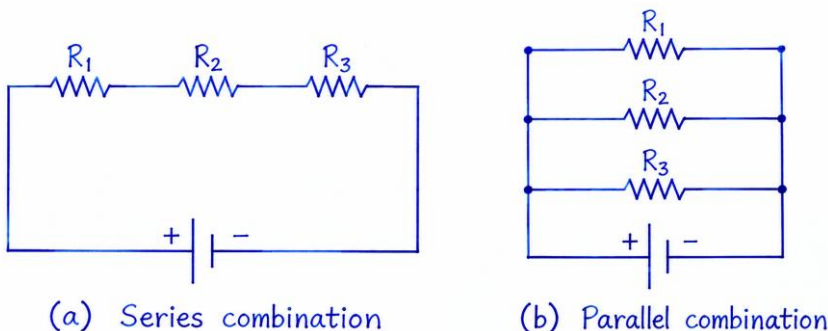
ρ (rho) = resistivity; SI unit = $\Omega \text{ m}$

- Resistivity is a characteristic/intrinsic property of a material at a given temperature; it does not depend on the length or cross-sectional area of the specimen.
- Low resistivity $\rightarrow$ good conductor (e.g., copper/aluminium). High resistivity $\rightarrow$ poor conductor/insulator. Alloys such as nichrome have relatively high resistivity and are useful as heating elements.
- Wire stretching: if a wire is stretched to n times its original length without change in volume, $A' = A/n$; hence $R' = \rho(nl)/(A/n) = n^2R$.

Quick relations: l doubled (same A) $\Rightarrow R$ doubles. A doubled (same l) $\Rightarrow R$ halves. Radius doubled $\Rightarrow$ area becomes $4A \Rightarrow R$ becomes $R/4$.

4. Combination of Resistors

Diagram — Series and Parallel Combinations



Learn current/voltage rules along with the diagram.

4.1 Resistors in series

- Same current flows through every resistor: $I = I_1 = I_2 = \dots$
- Total potential difference is the sum of individual drops: $V = V_1 + V_2 + \dots$

Derivation — equivalent resistance in series

- $V = V_1 + V_2 + V_3$; using $V = IR$ and the same current I : $IR_s = IR_1 + IR_2 + IR_3 \Rightarrow R_s = R_1 + R_2 + R_3$.

$$R_s = R_1 + R_2 + R_3 + \dots$$

Equivalent resistance in series is greater than any individual resistance.

- For n identical resistors each R in series: $R_s = nR$.

4.2 Resistors in parallel

- Same potential difference acts across every branch: $V = V_1 = V_2 = \dots$
- Total current splits among branches: $I = I_1 + I_2 + \dots$

Derivation — equivalent resistance in parallel

- $I = I_1 + I_2 + I_3$; using $I = V/R$ and the same potential difference V : $V/R_p = V/R_1 + V/R_2 + V/R_3 \Rightarrow 1/R_p = 1/R_1 + 1/R_2 + 1/R_3$.

$$1/R_p = 1/R_1 + 1/R_2 + 1/R_3 + \dots$$

For two resistors: $R_p = R_1 R_2 / (R_1 + R_2)$. For n identical resistors: $R_p = R/n$.

- Equivalent resistance in parallel is smaller than the smallest branch resistance.

4.3 Why household appliances are connected in parallel

- Each appliance gets the full supply potential difference.
- Each appliance can be switched on/off independently; failure of one branch does not stop the others.
- Different appliances can draw different currents according to their power/resistance ratings.

4.4 Prescribed practical — equivalent resistance

- Measure V and I for two resistors connected in series and calculate $R_s = V/I$; compare with $R_1 + R_2$.
- Repeat for parallel connection and compare with $1/R_p = 1/R_1 + 1/R_2$.

5. Heating Effect of Electric Current — Joule's Law

- When current passes through a resistor, electrical energy is converted into heat because moving charges lose energy in collisions with the conductor lattice.

$$H = I^2 R t$$

Joule's law of heating: heat $\propto I^2$, R and t

$$H = V I t = (V^2 / R) t$$

Use the form that matches the quantities held constant in the question.

CONDITION	IF R IS DOUBLED	RESULT FOR HEAT
I and t constant	$H = I^2 R t$	Heat doubles.
V and t constant	$H = V^2 t / R$	Heat becomes half.

5.1 Daily-life applications

- Electric iron, heater, toaster, kettle and geyser use a high-resistance heating element to produce heat.
- Why nichrome/alloys are used as heating elements: high resistivity, high melting point, and they do not oxidise/burn readily at high temperature.
- Bulb filament becomes white-hot and emits light; tungsten is used because of its very high melting point (modern lighting may use other technologies).
- Fuse wire works on heating effect: excess current heats and melts the fuse, breaking the circuit. Detailed domestic-circuit protection is revised in the next chapter.

6. Electric Power & Electrical Energy

- Electric power is the rate at which electrical energy is consumed or work is done.
- 1 watt (1 W): power when 1 joule of work/energy is transferred in 1 second. $1 \text{ W} = 1 \text{ J s}^{-1} = 1 \text{ V} \times 1 \text{ A}$.

$$P = W/t = VI = I^2 R = V^2 / R$$

SI unit: watt (W). 1 kW = 1000 W.

- An appliance rated 1000 W, 250 V draws $I = P/V = 4 \text{ A}$ — a useful exam pattern.
- Electrical energy consumed: $E = Pt$. In SI units, E is in joules when P is in watts and t in seconds.
- Commercial unit — 1 kilowatt-hour (1 kWh): electrical energy consumed by a 1 kW appliance operating for 1 hour. $1 \text{ kWh} = 1 \text{ unit} = 3.6 \times 10^6 \text{ J}$.

$$1 \text{ kWh} = 1 \text{ unit} = 3.6 \times 10^6 \text{ J}$$

Commercial electrical energy = power in kW $\times$ time in h

Electricity bill: Units consumed = Σ (power in kW $\times$ time in h). Cost = units $\times$ tariff per unit.

7. Common Question Patterns

QUESTION CUE	WHAT TO FIND	KEY RESPONSE
1000 W at 250 V	Find current	$I = P/V = 1000/250 = 4 \text{ A}$.
Joule's law	State + change R	$H = I^2 R t$; if I, t constant and R doubles, H doubles.
Resistivity	Define + intrinsic	$\rho = RA/l$; material property at fixed temperature, independent of specimen dimensions.

QUESTION CUE	WHAT TO FIND	KEY RESPONSE
15 V with 2Ω , 3Ω , 5Ω in series	Total R, I, voltage drop	$R=10\Omega$; $I=1.5\text{A}$; V across $3\Omega = 4.5\text{V}$.
Circuit diagram	Find total current / P.D.	Reduce series/parallel blocks stepwise, then apply $V=IR$ branch-wise.

8. Numerical Practice

GIVEN → REQUIRED → FORMULA → SUBSTITUTION WITH SI UNITS → FINAL ANSWER + UNIT

8.1 Core patterns

- [2M] Charge/current: 600 C passes in 5 min. $I = Q/t = 600/(5 \times 60) = 2\text{ A}$.
- [2M] Ohm's law: 12 V across 4Ω . $I = V/R = 3\text{ A}$.
- [3M] Resistivity: wire length 2 m, area $1 \times 10^{-6}\text{ m}^2$, $R = 3.4\Omega$. $\rho = RA/l = 1.7 \times 10^{-6}\Omega\text{ m}$.
- [3M] Two 6Ω resistors: series = 12Ω ; parallel = 3Ω . At same V, the parallel combination draws more current.
- [3M] Heater: $I = 5\text{ A}$, $R = 20\Omega$, $t = 2\text{ min}$. $H = I^2Rt = 5^2 \times 20 \times 120 = 60,000\text{ J}$.
- [3M] Energy bill: 1.5 kW heater for 2 h/day for 30 days → 90 kWh = 90 units.

8.2 Power-comparison traps

- At the same voltage: $P = V^2/R \Rightarrow$ smaller resistance consumes more power.
- At the same current: $P = I^2R \Rightarrow$ larger resistance consumes more power.
- Appliance rating "P at V" tells the intended operating voltage and power; current at rated voltage is $I = P/V$.

9. Application & Case-Based Questions

- Graph: steeper V–I line (V on y-axis) means larger resistance because slope = R.
- Wire comparison: keep material and temperature fixed; decide R using l/A before doing arithmetic.
- Bulbs/appliances in parallel: full voltage across each branch; total current equals sum of branch currents.
- Heating: first identify what is held constant. Doubling R has opposite effects in I^2Rt and V^2t/R situations.
- Mixed network: mark obvious series/parallel groups, find equivalent resistance, total current, then expand backward to branch values.

10. Mark-wise Answer Practice

[1M] Define electric current.

- Rate of flow of electric charge through a conductor: $I = Q/t$.

[1M] Define one ohm.

- A conductor has resistance 1Ω if a potential difference of 1 V produces a current of 1 A through it.

[1M] Define one watt.

- Power is 1 W when 1 J of electrical work is done in 1 s: $1\text{ W} = 1\text{ J s}^{-1}$.

[1M] What is one commercial unit of electrical energy?

- 1 commercial unit = 1 kWh, the energy used by a 1 kW appliance in 1 h = $3.6 \times 10^6\text{ J}$.

[2M] State Ohm's law and one condition.

- At constant temperature/physical conditions, $V \propto I$ for a conductor; hence $V = IR$.

[2M] Define resistivity and give SI unit.

- Resistivity ρ is the material constant in $R = \rho l/A$; equivalently $\rho = RA/l$. SI unit: $\Omega \text{ m}$.

[3M] Why are domestic appliances connected in parallel?

- Each gets full supply voltage; independent switching/failure; each draws current according to its own resistance/power.

[3M] State Joule's law and give two applications.

- $H = I^2 R t$. Applications include electric heaters/irons and fuse protection (any correct examples).

11. Common Mistakes

- ✗ Connecting ammeter in parallel or voltmeter in series.
- ✗ Forgetting "constant temperature" while stating Ohm's law.
- ✗ Using slope = I/V when the graph has V on y-axis and I on x-axis; here slope = $V/I = R$.
- ✗ Confusing resistance R (depends on dimensions) with resistivity ρ (material property at fixed temperature).
- ✗ Adding parallel resistances directly instead of reciprocals.
- ✗ Using minutes/hours without converting to seconds when calculating heat in joules.
- ✗ Writing kW as an energy unit; kW is power, kWh is electrical energy.
- ✗ Applying $H \propto R$ without checking whether current or voltage is held constant.

12. Quick Revision

TOPIC	KEY POINT
Current	$I = Q/t$; $1 \text{ A} = 1 \text{ C/s}$
Potential difference	$V = W/Q$; $1 \text{ V} = 1 \text{ J/C}$
Ohm's law	$V = IR$ at constant temperature
Resistance	$R = V/I$; unit Ω
Resistivity	$R = \rho l/A$; unit $\Omega \text{ m}$
Series	same I ; $R_s = \Sigma R$
Parallel	same V ; $1/R_p = \Sigma(1/R)$
Heating	$H = I^2 R t = V I t = V^2 t/R$
Power	$P = VI = I^2 R = V^2/R$
Energy	$E = Pt$; $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$

Self-check before you leave the chapter

- ☐ Can I draw the Ohm's-law circuit with correct meter connections?
- ☐ Can I read slope from a V-I graph?
- ☐ Can I use $R = \rho l/A$ for length/area changes?
- ☐ Can I reduce series, parallel and mixed resistor networks?

- ☐ Can I choose the correct Joule-heating formula for what is held constant?
- ☐ Can I convert $W \leftrightarrow kW$, $s \leftrightarrow h$ and $J \leftrightarrow kWh$ correctly?

Answer tip: In every electricity numerical, write the formula before substitution, convert units first, and finish with the correct unit. In circuit reasoning, state “same current in series” or “same voltage in parallel” before calculating.

- ☐ Can I derive R_s and R_p from Ohm's law instead of only memorising the formulas?
- ☐ Can I define 1 W and 1 kWh and solve the n-times stretched-wire resistance case?

13. Rapid Recall

1-mark definitions you should be able to write exactly

- Electric circuit: a continuous conducting path through which current can flow. Closed circuit: complete path, current flows. Open circuit: broken path, no current.
- 1 watt: 1 J of electrical work done per second; $1 W = 1 J s^{-1}$. 1 commercial unit: $1 kWh = 3.6 \times 10^6 J$.
- Heating element reason: nichrome/alloys have high resistivity, high melting point and resist oxidation at high temperature.

Derivation check

- Series: $V = V_1 + V_2 + V_3$ and $V = IR \Rightarrow R_s = R_1 + R_2 + R_3$. Parallel: $I = I_1 + I_2 + I_3$ and $I = V/R \Rightarrow 1/R_p = 1/R_1 + 1/R_2 + 1/R_3$.

Wire stretching

- If a wire is stretched to 3 times its length at constant volume, area becomes $A/3$ and resistance becomes $3^2 R = 9R$. In general, $l' = nl \Rightarrow R' = n^2 R$.

CHAPTER 12 — MAGNETIC EFFECTS OF ELECTRIC CURRENT

Chapter Coverage

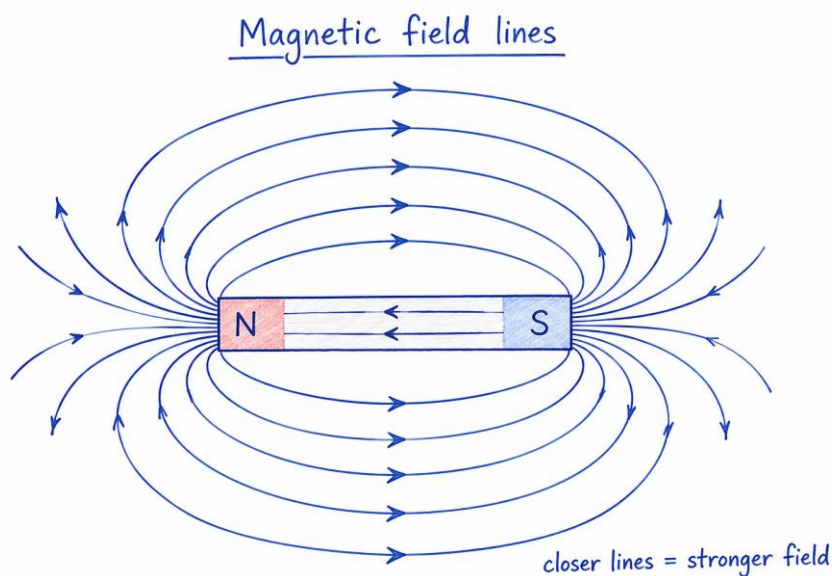
Core idea	What you should be able to do
Magnetic field & field lines	Draw/interpret field lines and identify stronger or weaker regions.
Current produces magnetic field	Use the right-hand thumb rule for a straight conductor; compare field strength with current and distance.
Circular loop & solenoid	Sketch field patterns, identify poles and explain why the field becomes stronger.
Force on a conductor	Use Fleming's left-hand rule and predict reversal of force when current or field is reversed.
DC and AC	Differentiate them; know AC frequency in India and the transmission advantage of AC.
Domestic electric circuits	Explain live, neutral and earth wires, parallel connection, fuse/MCB, overloading, short circuit and earthing.
2026–27 year-end boundary: Electric motor, electromagnetic induction and electric generator are listed for formative learning only. The chapter below focuses on the summative-board content.	

1. Magnetic Field and Magnetic Field Lines

- A magnetic field is the region around a magnet or a current-carrying conductor in which a magnetic force can be experienced. It is represented by B ; SI unit: tesla (T).
- A compass needle placed in a magnetic field aligns approximately along the field direction. Oersted's observation showed that electric current produces a magnetic effect.

1.1 Properties of magnetic field lines

- Outside a bar magnet, field lines emerge from the North pole and enter the South pole; inside the magnet they return from South to North, so magnetic field lines form closed curves.
- The tangent to a field line at any point gives the direction of the magnetic field at that point.
- Field lines never intersect; otherwise the field would have two directions at one point.
- Closer field lines indicate a stronger magnetic field. The lines are most crowded near the poles of a bar magnet.

Diagram — Magnetic Field Lines Around a Bar Magnet

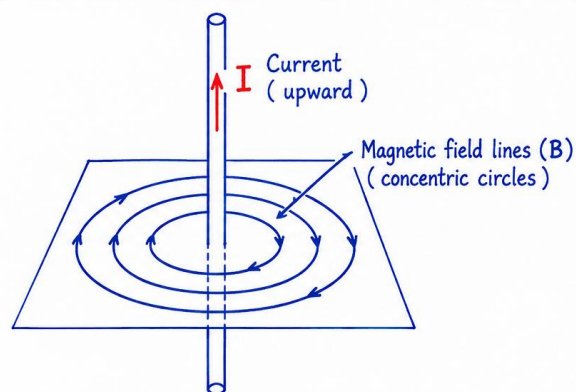
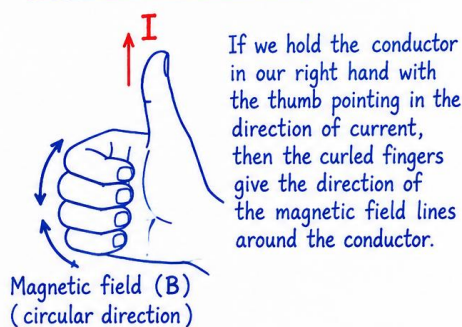
Draw smooth closed curves; arrows outside the magnet must point N → S.

2. Magnetic Field Due to a Straight Current-Carrying Conductor

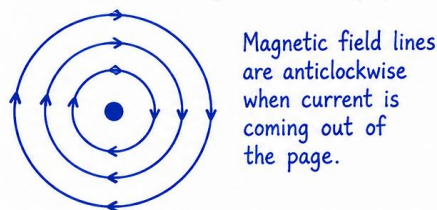
- The magnetic field lines around a straight current-carrying conductor are concentric circles centred on the conductor.
- Reversing the current reverses the direction of the magnetic field.
- For the same distance, increasing current makes the field stronger. For the same current, the field becomes weaker as distance from the conductor increases.

2.1 Right-Hand Thumb Rule

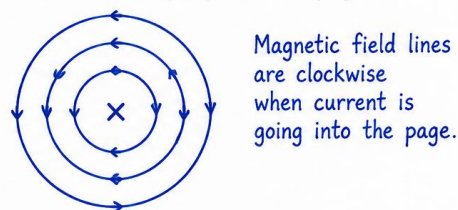
- Hold the straight conductor in your right hand with the thumb pointing in the direction of current. The curled fingers give the direction of the magnetic field lines around the conductor.
- Dot (•) represents current coming out of the page; cross (×) represents current going into the page.

Diagram — Straight Conductor and Right-Hand Thumb RuleMagnetic field around a straight current-carrying conductorRight-Hand Thumb Rule

(•) Current coming out of the page



(X) Current going into the page



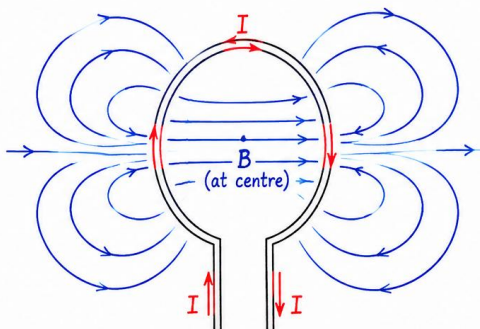
Typical 3-mark pattern: state the rule, draw the field direction, then explain that field strength decreases with distance.

3. Magnetic Field Due to a Circular Loop and a Solenoid**3.1 Current-carrying circular loop**

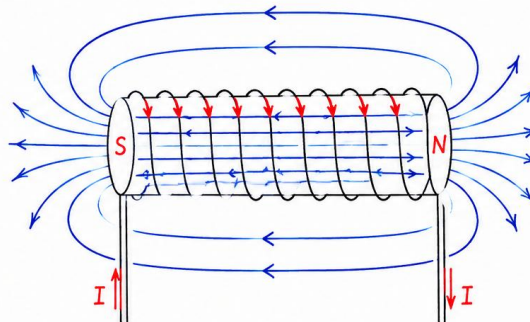
- Each small part of the loop produces a magnetic field; at the centre, these contributions reinforce one another and the field lines are nearly parallel.
- The magnetic field at the centre becomes stronger when current is increased or when the number of turns is increased. For similar coils, a smaller radius gives a stronger central field.
- Viewed from one face of a coil: anticlockwise current makes that face behave as North; clockwise current makes it behave as South.

3.2 Solenoid

- A solenoid is a long coil containing many closely wound turns of insulated wire.
- Its magnetic field pattern resembles that of a bar magnet: one end acts as North and the other as South.
- Inside a long current-carrying solenoid, field lines are nearly parallel and equally spaced, indicating a strong, almost uniform magnetic field.
- A soft-iron core placed inside a solenoid greatly strengthens the field and forms an electromagnet. Reversing current reverses the poles.

Diagram — Circular Loop and SolenoidCurrent-carrying circular loop and solenoidCurrent-carrying circular loop

Magnetic field lines at the centre are nearly parallel and the field is strong. The strength increases with larger current and with more turns (if a coil is used).

Current-carrying solenoid

Inside a current-carrying solenoid, the magnetic field lines are nearly parallel and equally spaced, so the field is strong and almost uniform. The solenoid behaves like a bar magnet with South pole (S) at one end and North pole (N) at the other.

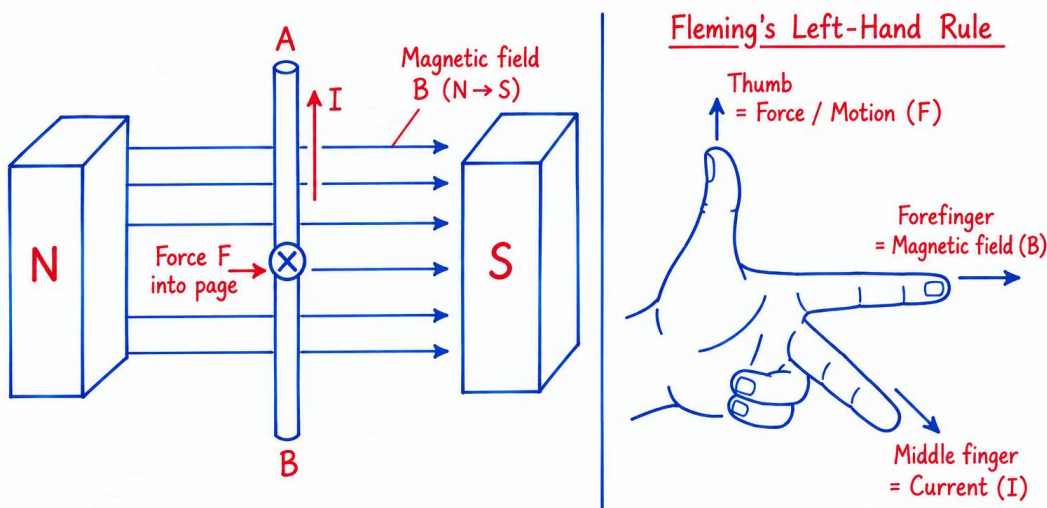
For a solenoid, draw nearly parallel field lines inside and closed curves outside.

4. Force on a Current-Carrying Conductor in a Magnetic Field

- A current-carrying conductor placed in an external magnetic field experiences a force. The force is greatest when the conductor is perpendicular to the magnetic field.
- If either the current direction or the magnetic-field direction is reversed, the force reverses. If both are reversed together, the force remains in the same direction.

4.1 Fleming's Left-Hand Rule

- Stretch the thumb, forefinger and middle finger of the left hand mutually perpendicular to each other: forefinger → magnetic field, middle finger → current, thumb → force/motion.

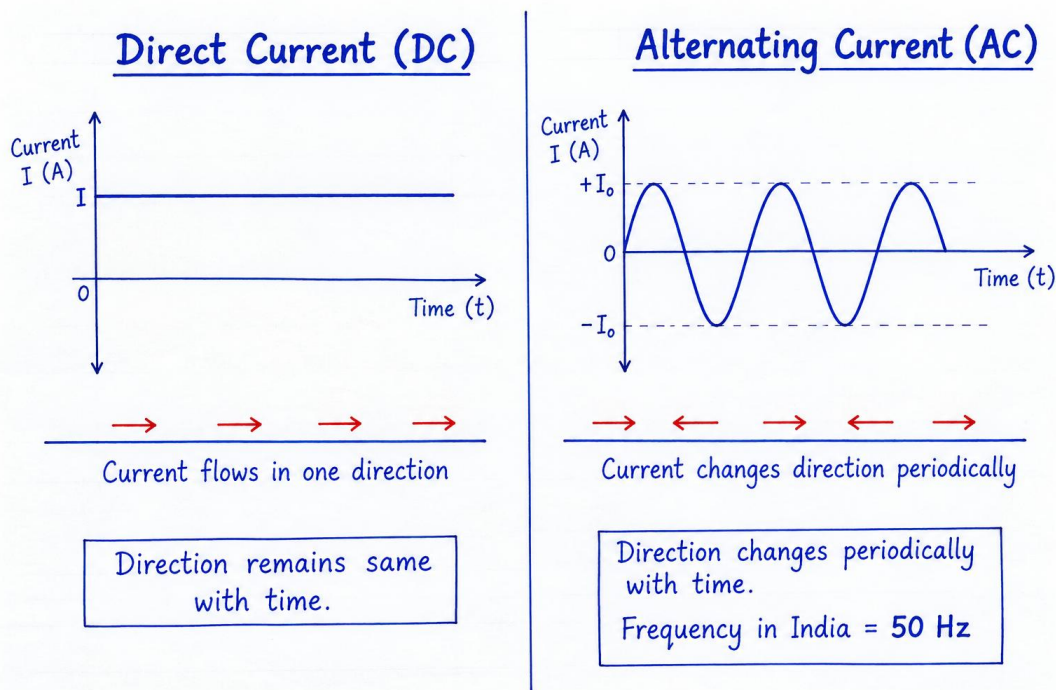
Diagram — Force on a Conductor and Fleming's Left-Hand Rule
Force on a Current-Carrying Conductor & Fleming's Left-Hand Rule


Use the left hand only: Forefinger = Field, Middle finger = Current, Thumb = Force.

5. Direct Current and Alternating Current

Feature	Direct current (DC)	Alternating current (AC)
Direction	Flows in one direction.	Changes direction periodically.
Typical source	Cell / battery.	Mains electricity.
Frequency	0 Hz for steady DC.	50 Hz in India.
Transmission	Not as convenient for conventional long-distance step-up/step-down transmission.	Voltage can be changed conveniently; high-voltage transmission reduces current and I^2R losses.
50 Hz means 50 complete AC cycles each second. Since current reverses twice in one cycle, its direction reverses 100 times per second.		

Diagram — DC and AC



DC stays in one direction; AC reverses direction periodically.

6. Domestic Electric Circuits

- In the standard Class 10 NCERT description, domestic supply is about 220 V, 50 Hz AC.

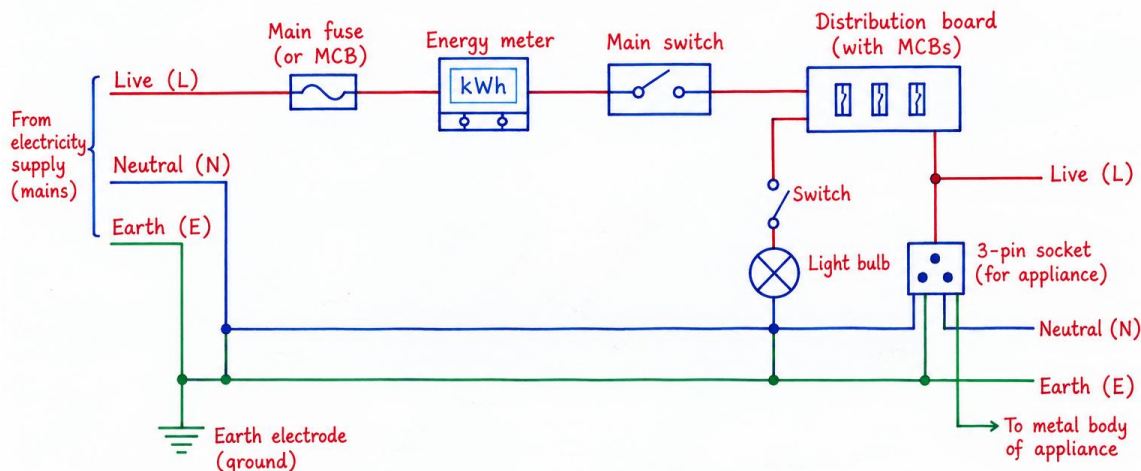
Wire	Function
Live (L)	Carries current from the supply to the appliance; at high potential.
Neutral (N)	Returns current to the supply; approximately zero potential relative to earth.
Earth (E)	Safety wire connected to the metal body of an appliance; provides a low-resistance path to ground during leakage.

6.1 Why appliances are connected in parallel

- Each appliance receives the full supply potential difference.
- Each appliance can be switched independently.
- Failure of one appliance does not stop the others from working.
- Different appliances can draw different currents according to their power ratings.

6.2 Fuse, MCB, short circuit and overloading

- A fuse is connected in the live wire. Its low-melting-point fuse element melts when current exceeds the safe value and breaks the circuit.
- An MCB (Miniature Circuit Breaker) automatically trips when excessive current flows and can normally be reset after the fault is removed.
- Short circuit: live and neutral wires come into direct contact, giving a very low-resistance path and a sudden large current.
- Overloading: too many or high-power appliances draw more current than the circuit is designed to carry.
- Earthing reduces electric-shock risk by giving leakage current a low-resistance path to ground and helping the protective device operate.
- NCERT commonly describes a 5 A circuit for lamps/fans and a 15 A circuit for high-power appliances such as heaters or air conditioners.

Diagram — Domestic Electric Circuit**Domestic Electric Circuit****Notes:**

1. All household appliances are connected in parallel to the mains supply.
2. Fuse (or MCB) is always connected in the live wire (L).
3. Switch is connected in the live wire to cut off the current to the appliance.
4. Earth wire provides a low-resistance path to the ground. If there is a leakage of current to the metal body, it flows through the earth wire and prevents electric shock.

Fuse/MCB and switches are placed in the live wire; appliances are connected in parallel; the earth wire is a safety path.

7. Board Answer Patterns

Marks	Question pattern	Essential answer points
1M	What is a magnetic field?	Region in which a magnetic force can be experienced.
1M	What does crowding of field lines show?	Stronger magnetic field.
2M	State Right-Hand Thumb Rule.	Thumb → current; curled fingers → magnetic-field direction.
2M	How does field around a straight wire change?	Stronger with greater current; weaker with increasing distance.
2M	Why is field inside a solenoid called uniform?	Field lines inside are nearly parallel and equally spaced.
3M	State Fleming's Left-Hand Rule and apply it.	Forefinger = B, middle finger = I, thumb = force; mention reversal logic.
3M	Differentiate AC and DC.	Direction, source/example, frequency; AC in India = 50 Hz.
3M	Explain domestic safety.	Live/neutral/earth functions + fuse/MCB in live + earthing/short circuit.

8. Common Errors to Avoid

- Do not draw magnetic field lines crossing one another.
- Do not reverse the Right-Hand Thumb Rule: thumb gives current, curled fingers give magnetic field.
- Do not use Fleming's right-hand rule here; the summative topic is force on a conductor, so use Fleming's LEFT-hand rule.
- Do not say AC "has no direction"; its direction changes periodically.
- Do not connect a fuse or switch in the neutral wire in a standard domestic-circuit diagram; place it in the live wire.
- Do not treat the earth wire as a normal current-carrying return path; it is primarily a safety path.

9. Last-Minute Recall

Topic	Key point
Field lines	Closed curves; outside a magnet $N \rightarrow S$; never intersect.
Straight wire	Right-hand thumb rule; stronger for larger current, weaker farther away.
Loop / solenoid	More current/turns $\rightarrow$ stronger field; solenoid gives a nearly uniform field inside.
Force	Fleming's left-hand rule; reverse current or field $\rightarrow$ reverse force.
AC / DC	DC one direction; AC reverses; Indian mains frequency = 50 Hz.
Domestic safety	Parallel appliances; fuse/MCB in live wire; earth wire protects against shock.

10. Practice Questions

- Draw magnetic field lines around a bar magnet and write any two properties of magnetic field lines.
- State the Right-Hand Thumb Rule. What happens to the magnetic field if (i) current is increased and (ii) distance from the wire is increased?
- Why is the magnetic field inside a long solenoid considered nearly uniform? How can its strength be increased?
- A conductor carrying current upward is placed in a magnetic field directed from left to right. Use Fleming's Left-Hand Rule to determine the force direction.
- Differentiate DC and AC. State the frequency of domestic AC supply in India.
- Distinguish between overloading and short circuit. Name the devices used to protect a domestic circuit.
- Why are household appliances connected in parallel rather than series?
- What is the purpose of the earth wire? Why are the fuse/MCB and switch placed in the live wire?

CHAPTER 13 — OUR ENVIRONMENT

Chapter Coverage

Board scope: Ecosystem and its components • food chains, food webs and trophic levels • energy flow and the 10% law • decomposers • biological magnification • ozone depletion • biodegradable/non-biodegradable waste • waste-management solutions.

1. Ecosystem — the basic unit

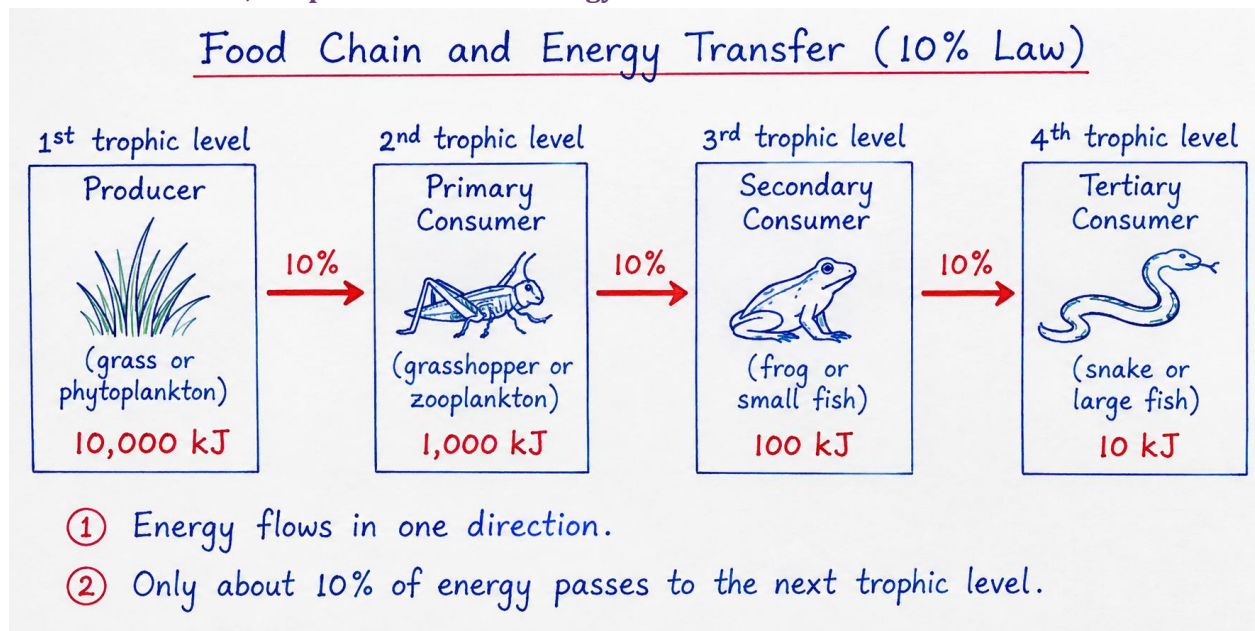
- An ecosystem is a functional unit in which living organisms interact with one another and with the non-living components of their environment.
- Biotic components: producers, consumers and decomposers. Abiotic components: sunlight, air, water, soil, temperature and minerals.
- Natural ecosystems include forests, ponds and lakes. Artificial ecosystems include crop fields and aquariums.
- **Decomposers are not restricted to one trophic level.** They act on dead remains and wastes from organisms at all trophic levels and return nutrients to the environment.

GROUP	ROLE	EXAMPLES
Producers	Make food from inorganic substances using solar energy	Green plants, algae, phytoplankton
Consumers	Obtain food directly or indirectly from producers	Herbivores, carnivores, omnivores
Decomposers	Break down dead organisms and wastes; return nutrients to the environment	Bacteria and fungi

Why decomposers matter: Without decomposers, dead matter would accumulate and nutrients would remain locked in it. Decomposition recycles materials back to the ecosystem.

2. Food Chains, Food Webs and Trophic Levels

- A food chain is a sequence of organisms in which each organism is eaten by the next organism. The arrow points from food to feeder and therefore shows the direction of energy transfer.
- Each feeding step is called a trophic level. Producers occupy the first trophic level, herbivores the second, and carnivores higher levels.
- A food web is a network of interconnected food chains. It gives organisms alternative feeding links and generally makes an ecosystem more stable than a single chain.

Diagram — Food Chain, Trophic Levels and Energy Transfer

Learn the trophic-level names and the direction of the arrows.

2.1 The 10% law of energy transfer

- Only about 10% of the energy available at one trophic level is transferred to the next trophic level. Most of the remaining energy is used in life processes and lost as heat.
- Green plants capture only about 1% of the solar energy** falling on their leaves and convert it into chemical energy stored in food.

Do not confuse: $\approx 1\%$ = solar energy trapped by green plants; $\approx 10\%$ = energy passed from one trophic level to the next.

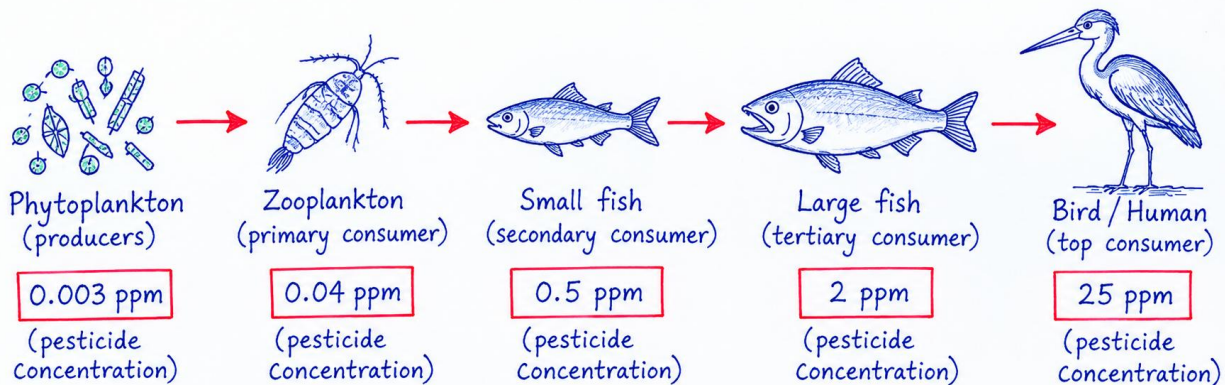
$$10,000 \text{ kJ} \rightarrow 1,000 \text{ kJ} \rightarrow 100 \text{ kJ} \rightarrow 10 \text{ kJ}$$

- Energy flow is unidirectional: Sun $\rightarrow$ producers $\rightarrow$ consumers. Energy is not recycled back to producers.
- Food chains usually contain only a few trophic levels because the usable energy becomes very small at successive levels.

Board calculation: If the fourth trophic level has 10 kJ, the third had about 100 kJ, the second about 1,000 kJ and the producer level about 10,000 kJ.

3. Biological Magnification

- Biological magnification (biomagnification) is the progressive increase in the concentration of a persistent, non-biodegradable pollutant at successive trophic levels of a food chain.
- Pesticides and some toxic chemicals may enter soil or water, pass into producers and then move through consumers without being broken down effectively.
- The top consumer generally accumulates the highest concentration because it feeds on many organisms from the level below.

Diagram — Biological Magnification in an Aquatic Food ChainAquatic Food Chain and Biological Magnification

Non-biodegradable pollutants accumulate and concentration increases at higher trophic levels.

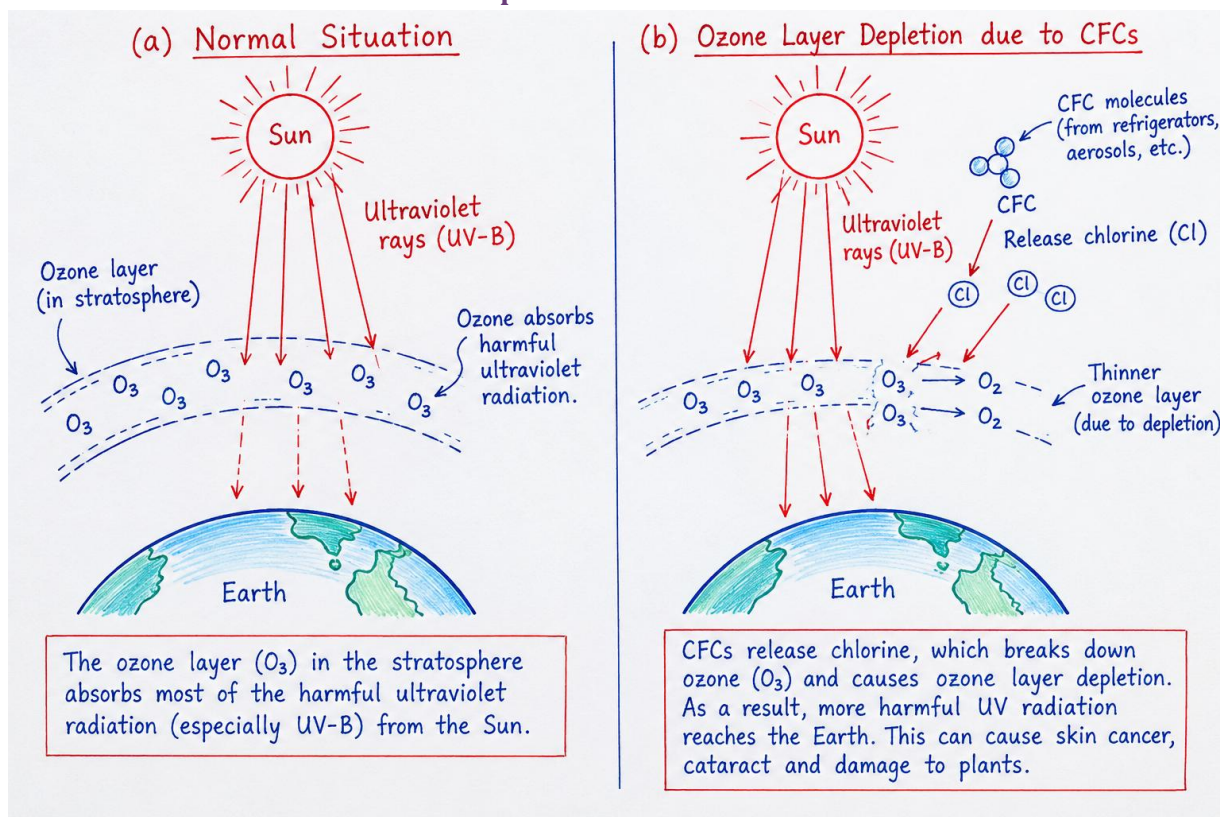
The values are illustrative; the examinable idea is the increasing concentration at higher trophic levels.

Do not confuse: Energy decreases at higher trophic levels, but the concentration of a persistent pollutant may increase.

4. Ozone Layer and Ozone Depletion

- Ozone (O_3) is a form of oxygen present mainly in the stratosphere. It absorbs much of the harmful ultraviolet (UV) radiation from the Sun and protects living organisms.
- Location matters:** stratospheric ozone is protective, but ground-level ozone is harmful/toxic to living organisms.
- Ozone is formed when UV radiation splits oxygen molecules and the free oxygen atoms combine with O_2 .

$$O_2 \xrightarrow{UV} O + O \quad \text{and} \quad O + O_2 \rightarrow O_3$$
- Chlorofluorocarbons (CFCs), formerly used widely in refrigerants and aerosol propellants, can release chlorine in the upper atmosphere. Chlorine promotes the breakdown of ozone.
- Ozone depletion allows more harmful UV radiation to reach Earth, increasing risks such as skin damage/cancer, cataract and damage to plants and other organisms.
- Protection:** reducing the use of CFCs and shifting to ozone-friendly alternatives helps protect the ozone layer.

Diagram — Ozone Protection and Ozone Depletion

The key chain of reasoning is: CFCs → ozone depletion → more UV reaches Earth → biological damage.

5. Waste: Biodegradable and Non-biodegradable

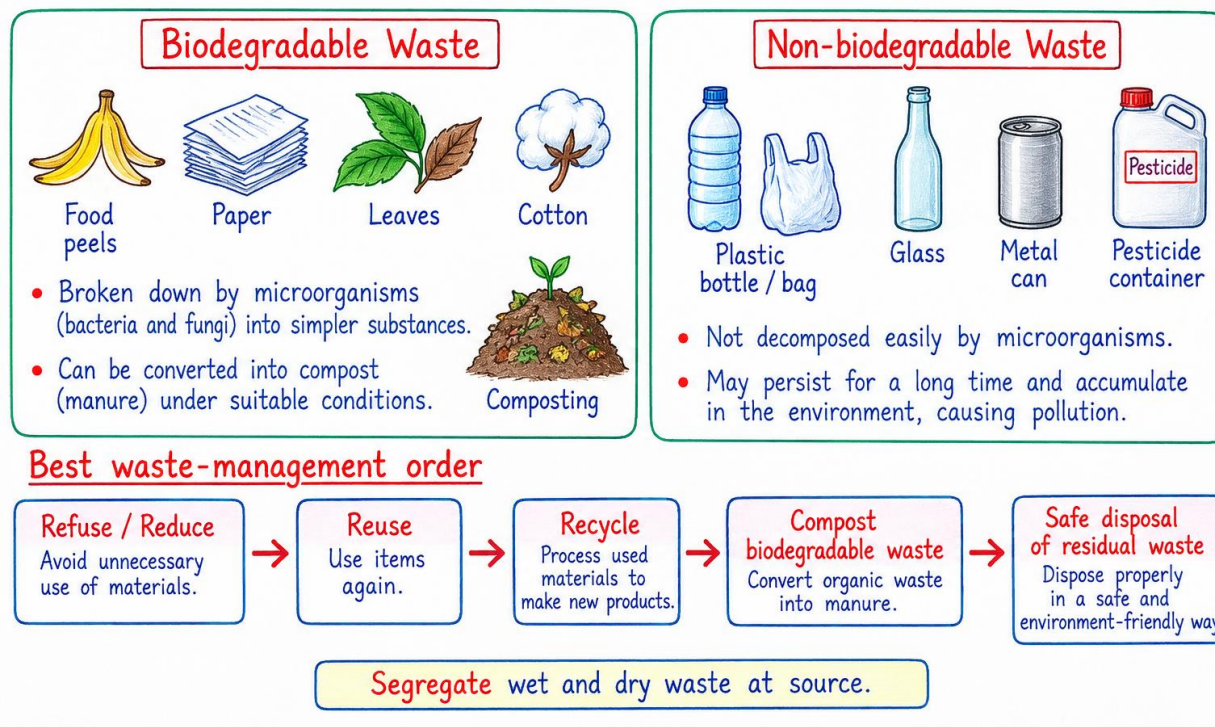
- Biodegradable substances can be decomposed by microorganisms into simpler substances under suitable conditions. Examples: food waste, leaves, paper and cotton.
- Non-biodegradable substances are not decomposed, or are decomposed extremely slowly, by biological processes. Examples: many plastics, glass, metals and persistent synthetic chemicals.

FEATURE	BIODEGRADABLE	NON-BIODEGRADABLE
Breakdown	By microorganisms	Not easily by microorganisms
Persistence	Usually shorter	May remain for a long time
Main management	Composting / biological treatment	Reduce, reuse, recycle and safe disposal

- Even biodegradable waste can create environmental problems if it accumulates: foul smell, disease vectors and oxygen depletion in water may result from uncontrolled decomposition.
- Non-biodegradable waste can persist, accumulate, enter food chains and cause long-term pollution.

5.1 Practical waste-management order

- Refuse or reduce unnecessary consumption and single-use items.
- Reuse products whenever safe and practical.
- Recycle suitable materials such as paper, metal, glass and selected plastics.
- Compost biodegradable kitchen and garden waste.
- Dispose of residual hazardous/non-recyclable waste through appropriate authorised systems; do not dump or burn waste indiscriminately.

Diagram — Waste Types and Practical Management**Waste : Biodegradable vs Non-biodegradable**

Best practice begins with segregation at source and reducing unnecessary waste.

5.2 Environmental problems caused by poor waste handling

- Open dumping of waste can pollute soil and water and may provide breeding sites for disease-carrying organisms.
- Burning mixed waste, especially plastics, is not a safe disposal method because harmful gases and particulates may be released.
- Persistent non-biodegradable chemicals can enter food chains and remain in organisms, creating long-term ecological and health risks.

5.3 Pollution control and special waste

- Sewage:** treat before discharge so organic matter, pathogens and other pollutants do not contaminate water bodies.
- Biomedical waste:** segregate and send through authorised treatment/disposal systems; do not mix it with ordinary household waste.
- E-waste and used batteries:** send to authorised collection/recycling channels because they may contain metals and other hazardous materials.
- Pesticides:** avoid indiscriminate use because persistent chemicals can enter food chains and undergo biological magnification.

6. Reasoning Questions

Why are food chains usually short? Only about 10% of energy is transferred to the next trophic level, so too little usable energy remains after several transfers.

Which trophic level gets maximum pesticide concentration? The highest/top consumer, because persistent pollutants undergo biological magnification.

Why are decomposers essential? They decompose dead matter and recycle nutrients, preventing accumulation of organic remains.

Why is plastic waste a long-term problem? Many plastics are non-biodegradable, persist for long periods and may accumulate in ecosystems.

Why is the ozone layer important? It absorbs harmful ultraviolet radiation and reduces biological damage at Earth's surface.

Why does a food web improve ecosystem stability? Organisms often have alternative feeding links, so loss of one food source does not necessarily break all energy pathways.

7. Board Practice

- [1M] In the chain algae → insect larvae → fish, identify the trophic level of the fish. Answer: secondary consumer.
- [1M] Assertion: 1% of energy is transferred from one trophic level to the next. Reason: about 10% of food energy becomes available to the next level. Answer: Assertion false; Reason true.
- [3M] In phytoplankton → zooplankton → small fish → large fish, the fourth trophic level has 10 kJ. Find producer energy; state whether three more trophic levels are likely; identify where pesticide concentration will be maximum.
- Answer: producer $\approx 10,000$ kJ; three more levels are unlikely because energy falls sharply at each transfer; pesticide concentration is maximum in the large fish because of biological magnification.
- [2M] Differentiate biodegradable and non-biodegradable substances with one example of each.
- [3M] Explain how CFCs can lead to ozone depletion and state two harmful consequences.

8. Common Exam Errors

- Writing food-chain arrows in the wrong direction. Arrow means food/energy passes to the organism that eats it.
- Confusing the two percentages: $\approx 1\%$ refers to solar energy trapped by green plants, while $\approx 10\%$ is transferred from one trophic level to the next.
- Saying energy is recycled. Matter is recycled; energy flows one way and is progressively dissipated.
- Confusing biological magnification with energy transfer: pollutant concentration rises, while available energy falls.
- Calling all waste biodegradable just because it is small or breaks physically. Biodegradability depends on biological decomposition.

9. Final Revision Grid

TOPIC	MUST REMEMBER
Ecosystem	Biotic + abiotic components interacting as a functional unit
Trophic levels	Producer → primary → secondary → tertiary consumer
10% law	$\approx 10\%$ energy reaches the next trophic level
Decomposers	Recycle nutrients from dead matter
Biomagnification	Persistent pollutant concentration increases up the food chain
Ozone	O ₃ absorbs harmful UV; CFCs contribute to depletion

TOPIC	MUST REMEMBER
Waste	Biodegradable vs non-biodegradable; segregate, reduce, reuse, recycle, compost

10. Short-answer Revision

Define a trophic level. Each feeding step or position in a food chain is called a trophic level.

What is the difference between the 1% and 10% figures in energy flow? About 1% of incident solar energy is trapped by green plants; about 10% of the energy at one trophic level passes to the next.

What does an arrow in a food chain show? It points from the organism being eaten to the organism that eats it, showing the direction of energy transfer.

What would happen if decomposers disappeared? Dead matter would accumulate and nutrient recycling would be severely reduced.

Why are top consumers most affected by persistent pesticides? The pesticide concentration increases at successive trophic levels by biological magnification.

Give one difference between a food chain and a food web. A food chain shows one feeding path; a food web links several interconnected feeding paths.

Name two ways to manage biodegradable waste. Segregation followed by composting or other suitable biological treatment.

What is ozone depletion? A reduction in stratospheric ozone concentration, allowing more harmful UV radiation to reach Earth.

Why can ozone be both useful and harmful? Ozone in the stratosphere protects us from harmful UV radiation, whereas ground-level ozone is harmful to living organisms.

Exam writing reminder: For a 3-mark environmental question, write the principle first (10% law / biomagnification / ozone protection), then apply it to the given food chain or situation, and finish with a clear consequence.

CBSE Class X Science 2026–27 • Unit V: Natural Resources — Our Environment

Final Board Revision Checklist

One last scan before you close the book

Chemistry

- ☐ I can balance equations and write correct states/conditions.
- ☐ I can identify reaction types and explain observations.
- ☐ I remember key salts, reactions and carbon-compound conversions.
- ☐ I can compare metals/non-metals and explain reactivity trends.

Biology

- ☐ I can redraw the major labelled diagrams from memory.
- ☐ I can explain life processes in sequence using correct terms.
- ☐ I can distinguish reflex/hormonal/plant responses clearly.
- ☐ I can solve Mendelian ratios and explain sex determination.

Physics

- ☐ I know sign conventions, ray rules and all core formulae.
- ☐ I include units in every numerical and show working clearly.
- ☐ I can draw circuit, magnetic-field and optics diagrams neatly.
- ☐ I can explain vision defects, scattering and domestic-circuit safety.

Environment

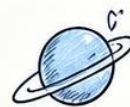
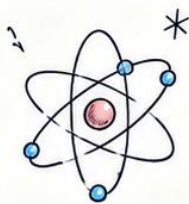
- ☐ I can apply the 10% law and distinguish it from the 1% solar-energy idea.
- ☐ I can explain biomagnification, ozone depletion and waste management.

Last 20-minute routine

- Scan formulae, reactions and diagrams — do not start a new topic.
- Read common-error boxes once.
- Check units, signs, labels and keywords.
- Stop revising in time to enter the exam calm and focused.

**You do not need perfect memory — you need accurate concepts, regular practice and calm execution.
Revise smartly. Write clearly. Trust your preparation.**

Curious
Minds
Brighter
Futures ♡



Thank
you ♡

- ☑ for your hard work
- ☑ for your curiosity
- ☑ for never giving up
- ☑ for believing in a brighter you!



Keep Learning, Keep Growing!

- ☑ Practice daily.
- ☑ Revise smartly.
- ☑ Believe in yourself.



Wishing you success in your board exams ♡

$$E = mc^2$$



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